Lecture notes. Lecture 12

Parametrised quantum circuits
variational ansätze for many-body states, probability models and optimisation

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These notes introduce parametrised quantum circuits for bachelor students of physics, mathematics and computer science. The ground state of many interacting spins is described by a number of amplitudes that grows exponentially with the number of spins. Classical approximations such as mean field, tensor networks and Monte Carlo methods each have known limits. A register of controlled qubits holds such a state physically. A circuit with adjustable gate angles prepares a trial state on the register, the energy is estimated from measured bit strings, and a classical computer changes the angles until the energy is as low as the circuit allows. This variational quantum eigensolver is the original use of parametrised circuits. We follow it from the variational principle through the measurement of energies and gradients to the choice of the circuit, its performance on spin chains and the barren plateaus that limit its training. The same circuits then serve as models of probability distributions, as solvers of binary optimisation problems and as general machine-learning models. All numerical results come from exact state-vector simulations or from simulated measurement records with fixed random seeds, and each result states which of the two it is.

Prerequisites

Qubits and registers of qubits as unit vectors of complex amplitudes, with qubit 0 as the most significant bit of the index. Pauli matrices, Pauli strings and rotations $e^{-i\theta\hat{P}/2}$, the Hadamard and controlled-$\hat{Z}$ gates. The Born rule, measurements in different bases and estimates from a finite number of shots. Hamiltonians as Hermitian matrices, their eigenvalues and ground states, the Schmidt decomposition and the entanglement entropy. Gradient descent and elementary probability. Further physical notions, such as bosons, modes and phase transitions, are explained at their first appearance.

The same objects carry different names in physics, mathematics and computer science. Table 1 collects the names used in these notes.

physics	mathematics	computer science
ansatz $\lvert\psi(\boldsymbol{\theta})\rangle$	parametrised family of unit vectors	model with trainable weights
parametrised circuit $\hat{U}(\boldsymbol{\theta})$	product of matrix exponentials $e^{-i\theta_k \hat{P}_k/2}$	program with continuous parameters
gate angle $\theta = \Omega t$	real parameter	weight
energy $\langle\psi\lvert\hat{\mathcal{H}}\rvert\psi\rangle$	quadratic form, Rayleigh quotient	cost or loss function
variational principle	minimum of the Rayleigh quotient	upper bound of the optimum
measurement setting	common eigenbasis of commuting matrices	readout configuration
shots	independent samples	mini-batch of samples
parameter-shift rule	exact difference formula for trigonometric functions	gradient from two function calls
Born machine	distribution $\lvert\psi_{\mathbf{b}}\rvert^2$	generative model over bit strings
cost Hamiltonian $\hat{\mathcal{H}}_C$	diagonal matrix of a quadratic form	objective function of a QUBO
ground state of $\hat{\mathcal{H}}_C$	minimiser of a quadratic binary form	optimal solution

Table 1: Physics, mathematics and computer-science names for the same objects.

1 The many-body problem and its classical approximations

1.1 The exponential cost of a many-body state

A chain of N interacting spins-1/2, or equivalently a register of N qubits, is described by a state vector with one complex amplitude for every bit string $\mathbf{b} = b_0 b_1 \dots b_{N-1}$,

$$|\psi\rangle = \sum_{\mathbf{b}\in\{0,1\}^N} \psi_{\mathbf{b}} \, |\mathbf{b}\rangle, \qquad \sum_{\mathbf{b}} |\psi_{\mathbf{b}}|^2 = 1. \tag{1.1}$$

The amplitude of $b_0 \dots b_{N-1}$ is stored at the index $i = \sum_q b_q 2^{N-1-q}$, so qubit 0 is the most significant bit. In double precision each amplitude takes 16 bytes and the vector needs $16 \cdot 2^N$ bytes. Ten spins fit into 16 kilobytes, thirty spins need 17 gigabytes, the memory of a large workstation, and fifty spins need about 18 petabytes (Figure 1). The Hamiltonian $\hat{\mathcal{H}}$, the Hermitian matrix of the energy, has 4^N entries, although for local interactions most of them vanish. Finding the ground state, the eigenvector of the lowest eigenvalue of $\hat{\mathcal{H}}$, or following the evolution $e^{-it\hat{\mathcal{H}}}|\psi\rangle$ in time requires repeated products of $\hat{\mathcal{H}}$ with such vectors. The cost of an exact calculation therefore doubles with every added spin, and exact diagonalisation is limited to a few dozen spins.

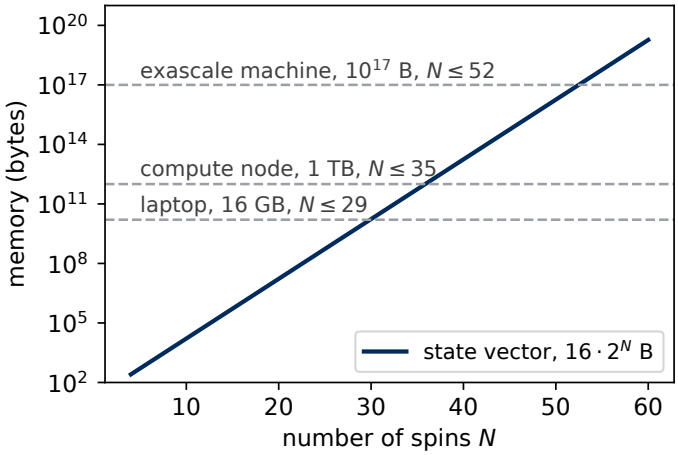


Figure 1: Memory needed to store the state vector of N spins in double precision, $16 \cdot 2^N$ bytes, against N, with the memory of a laptop, of a compute node and of an exascale machine for comparison. Every added spin doubles the memory.

Two spin chains serve as running models in these notes. The transverse-field Ising model (TFIM) on an open chain is

$$\hat{\mathcal{H}}_{\mathrm{TFIM}} = -J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1} - h \sum_{i=0}^{N-1} \hat{X}_i, \tag{1.2}$$

where $\hat{X}_i$ and $\hat{Z}_i$ are the Pauli operators of spin i. The coupling $J > 0$ favours neighbours aligned along z, both in $|0\rangle$ or both in $|1\rangle$, and the transverse field h favours every spin in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, the eigenstate of $\hat{X}$ with eigenvalue $+1$. The two terms do not commute, so the ground state is a superposition of many bit strings. In an infinite chain their competition produces a quantum phase transition at $h = J$ between an ordered phase for $h < J$, in which the spins tend to align along z, and a paramagnetic phase for $h > J$, in which they align with the field. The second model is the XXZ chain,

$$\hat{\mathcal{H}}_{\mathrm{XXZ}} = J \sum_{i=0}^{N-2} (\hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1} + \Delta \, \hat{Z}_i \hat{Z}_{i+1}), \tag{1.3}$$

with the anisotropy Δ. At $\Delta = 1$ it is the antiferromagnetic Heisenberg chain. We use $J = 1$ throughout and measure energies in units of J.

1.2 The variational principle

Every method that avoids the full state vector restricts the search to a family of trial states $|\psi(\boldsymbol{\theta})\rangle$ described by a few parameters $\boldsymbol{\theta}$, called an *ansatz*, and chooses the parameters by minimising the energy. To justify this choice, let $E_0 \le E_1 \le E_2 \le \dots$ be the eigenvalues of $\hat{\mathcal{H}}$ and $|E_m\rangle$ its orthonormal eigenvectors. Any normalised state can be expanded as $|\psi\rangle = \sum_m c_m |E_m\rangle$ with $\sum_m |c_m|^2 = 1$, and its mean energy is

$$\langle\psi|\hat{\mathcal{H}}|\psi\rangle = \sum_m |c_m|^2 E_m \;\ge\; E_0 \sum_m |c_m|^2 = E_0, \tag{1.4}$$

because every E_m in the sum is at least E_0. Equality holds only when all the weight lies on ground states. Every trial state therefore gives an upper bound on the ground-state energy (Figure 2), and minimising the energy over an ansatz gives the best bound that the family allows, which is the Rayleigh–Ritz variational method. The minimising state is the best approximation of the ground state within the family in the sense of the energy, and its quality in other respects must be checked separately.

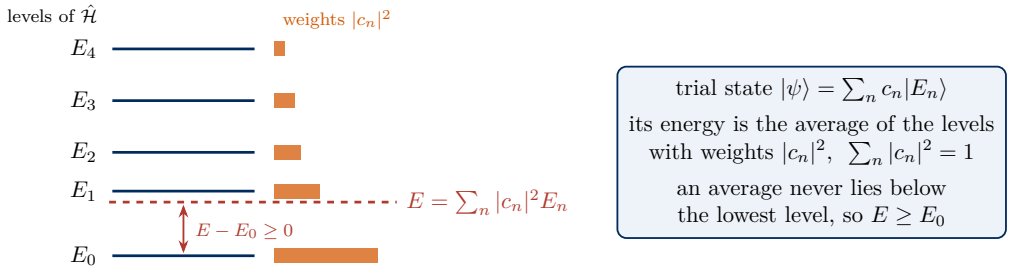


Figure 2: The variational principle (1.4). The energy levels $E_0 \le E_1 \le \dots$ of $\hat{\mathcal{H}}$, and the energy of any trial state as a weighted average of them, which lies at or above E_0. Minimising over an ansatz lowers this upper bound.

Two product states of the TFIM give bounds without any computation. In $|0\dots0\rangle$ every $\hat{Z}_i\hat{Z}_{i+1}$ has the value $+1$ and every $\langle\hat{X}_i\rangle$ vanishes, so the energy is $-J(N-1)$. In $|+\cdots+\rangle$ every $\langle\hat{X}_i\rangle = 1$ and every $\langle\hat{Z}_i\hat{Z}_{i+1}\rangle = 0$, so the energy is $-hN$. For $N = 6$ and $h = J$ these bounds are -5 and -6, both above the exact ground-state energy $E_0 = -7.2962$.

1.3 Mean field for bosons and the Gross–Pitaevskii equation

Mean-field theory, the oldest family of ansätze for interacting particles, is clearest for identical bosons such as the atoms of a Bose–Einstein condensate held in a trap, and its derivation needs the quantum description of such particles.

A single particle of mass m moving along a line is described by a wave function $\phi(x)$, a complex function whose squared modulus $|\phi(x)|^2$ is the probability density of finding the particle at the position x, normalised as $\int dx\, |\phi(x)|^2 = 1$. In a trap potential $V(x)$ its stationary states solve the Schrödinger equation

$$\Big[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\Big]\phi_i(x) = \varepsilon_i\, \phi_i(x), \tag{1.5}$$

where the second derivative is the kinetic energy and $\hbar$ is the reduced Planck constant. The solutions $\phi_0, \phi_1, \dots$ with energies $\varepsilon_0 \le \varepsilon_1 \le \dots$ are orthonormal, $\int dx\, \phi_i^*(x)\phi_j(x) = \delta_{ij}$, and they are called the *modes* of the trap. For the harmonic trap $V(x) = m\omega^2 x^2/2$ the lowest mode is a Gaussian.

Identical bosons cannot be told apart, so a many-boson state is fully specified by the number n_i of bosons in each mode i. The occupation-number state $|n_0, n_1, \dots\rangle$ plays the role of the bit string of qubits, except that each occupation can be any non-negative integer. Second quantisation describes changes of these numbers by two operators per mode. The *creation operator* $\hat{a}_i^\dagger$ adds one boson to mode i and the *annihilation operator* $\hat{a}_i$ removes one,

$$\hat{a}_i|\dots, n_i, \dots\rangle = \sqrt{n_i}\,|\dots, n_i - 1, \dots\rangle, \qquad \hat{a}_i^\dagger|\dots, n_i, \dots\rangle = \sqrt{n_i + 1}\,|\dots, n_i + 1, \dots\rangle, \tag{1.6}$$

and the product $\hat{a}_i^\dagger \hat{a}_i$ counts the bosons in mode i, $\hat{a}_i^\dagger \hat{a}_i |\dots, n_i, \dots\rangle = n_i |\dots, n_i, \dots\rangle$. The factors $\sqrt{n_i}$ make these operators obey the commutation relations

$$[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \qquad [\hat{a}_i, \hat{a}_j] = 0, \tag{1.7}$$

which express that bosons in different modes are independent and that the order in which bosons are added to one mode does not matter. The state without any boson is the vacuum $|\mathrm{vac}\rangle$. The *field operator*

$$\hat{\psi}(x) = \sum_i \phi_i(x)\, \hat{a}_i \tag{1.8}$$

removes one boson at the position x, whichever mode it occupies, and $\hat{\psi}^\dagger(x)\hat{\psi}(x)$ is the operator of the particle density at x. With these operators the Hamiltonian of bosons that interact with strength g when two of them are at the same point reads

$$\hat{\mathcal{H}}_B = \int dx\, \hat{\psi}^\dagger(x) \Big[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) \Big] \hat{\psi}(x) + \frac{g}{2} \int dx\, \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(x)\hat{\psi}(x)\hat{\psi}(x). \tag{1.9}$$

The first term is the single-particle energy (1.5) of every boson, and the second removes two bosons at the same point and puts them back, counting the interaction energy g of every pair at the same position.

The mean-field ansatz puts all N bosons into one normalised mode $\phi(x)$, with its own creation operator $\hat{a}_\phi^\dagger = \int dx\, \phi(x)\hat{\psi}^\dagger(x)$,

$$|\Psi_N\rangle = \frac{(\hat{a}_\phi^\dagger)^N}{\sqrt{N!}} |\mathrm{vac}\rangle. \tag{1.10}$$

Its wave function is the product $\phi(x_1)\phi(x_2)\cdots\phi(x_N)$, so the bosons are uncorrelated. To evaluate the energy (1.9) in this state we expand the field operator in a basis whose first mode is ϕ. Only the component $\phi(x)\hat{a}_\phi$ removes a boson from $|\Psi_N\rangle$, the others annihilate it, and by (1.6)

$$\hat{\psi}(x)|\Psi_N\rangle = \sqrt{N}\, \phi(x)\, |\Psi_{N-1}\rangle, \qquad \hat{\psi}(x)\hat{\psi}(x)|\Psi_N\rangle = \sqrt{N(N-1)}\, \phi(x)^2\, |\Psi_{N-2}\rangle. \tag{1.11}$$

The two expectation values that enter the energy follow by taking the norm of these vectors,

$$\langle \hat{\psi}^\dagger(x)\hat{\psi}(x')\rangle = N\, \phi^*(x)\, \phi(x'), \qquad \langle \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(x)\hat{\psi}(x)\hat{\psi}(x)\rangle = N(N-1)\, |\phi(x)|^4, \tag{1.12}$$

and inserting them into (1.9) turns the energy into a functional of the single mode,

$$E[\phi] = N \int dx\, \phi^*(x) \Big[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) \Big] \phi(x) + \frac{gN(N-1)}{2} \int dx\, |\phi(x)|^4. \tag{1.13}$$

By (1.4) this energy is an upper bound on the ground-state energy of the bosons for every normalised ϕ. We minimise it over ϕ as a function of many variables, one value $\phi(x)$ per point, with the normalisation enforced by a Lagrange multiplier μ. We set to zero the derivative of $E[\phi] - N\mu(\int dx\, |\phi|^2 - 1)$ with respect to $\phi^*(x)$, treating ϕ and ϕ^* as independent variables. With $\partial|\phi|^4/\partial\phi^* = \partial[(\phi^*)^2\phi^2]/\partial\phi^* = 2|\phi|^2\phi$ and after division by N this gives

$$\Big[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) + g(N-1)\, |\phi(x)|^2 \Big] \phi(x) = \mu\, \phi(x). \tag{1.14}$$

The result is the Gross–Pitaevskii equation [1, 2]. It has the form of the Schrödinger equation (1.5) in which the trap potential is supplemented by the mean interaction energy $g(N-1)|\phi(x)|^2$ that one boson feels from the other $N-1$. This potential depends on the solution itself, so the equation is nonlinear and is solved self-consistently. Its eigenvalue μ is the chemical potential, the energy needed to add one boson.

Figure 3 shows the solution in a harmonic trap, obtained by relaxation in imaginary time. Replacing t by $-i\tau$ in the time-dependent Schrödinger equation damps each eigencomponent by $e^{-E\tau}$,

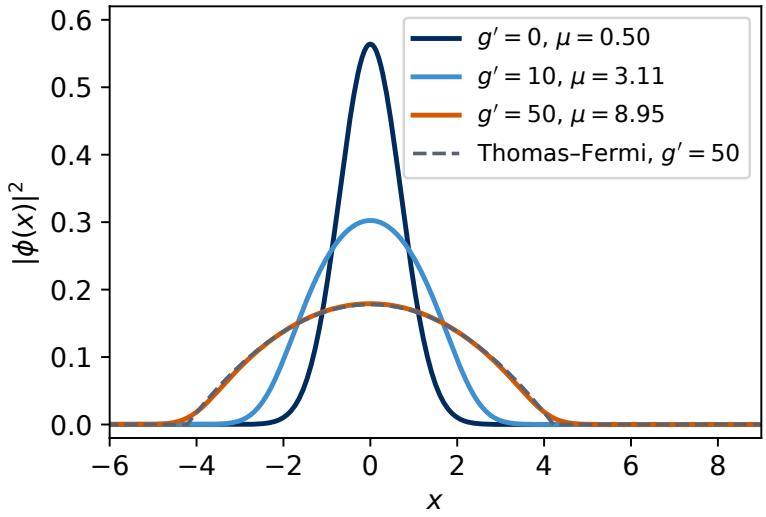


Figure 3: Ground-state density $|\phi(x)|^2$ of the Gross–Pitaevskii equation (1.14) in a harmonic trap, in units $\hbar = m = \omega = 1$, for the interaction strengths $g(N-1) = 0$, 10 and 50. The dashed line is the Thomas–Fermi parabola (1.15) for $g(N-1) = 50$. Repulsion spreads the cloud and lowers its central density.

so repeated small steps followed by normalisation converge to the lowest state. Without interactions the condensate occupies the Gaussian ground state of the oscillator with $\mu = 0.500$. For strong repulsion the kinetic energy is small compared with the interaction, and (1.14) without the kinetic term gives the Thomas–Fermi density

$$|\phi(x)|^2 = \frac{\mu - x^2/2}{g(N-1)} \quad \text{inside the cloud,} \qquad \mu_{\mathrm{TF}} = \Big(\frac{3g(N-1)}{4\sqrt{2}}\Big)^{2/3}, \tag{1.15}$$

an inverted parabola whose normalisation fixes μ_{TF}. At $g(N-1) = 50$ the numerical chemical potential 8.95 is close to $\mu_{\mathrm{TF}} = 8.89$, and the density deviates from the parabola only at the edge of the cloud, where the kinetic energy matters.

Mean field follows the same pattern for any many-body system. A product ansatz without entanglement is chosen, and its energy is computed as a functional of the single-particle state and minimised. The interaction enters as an average field produced by all other particles, and the minimum satisfies a nonlinear self-consistent equation. The approximation is accurate when the true ground state is close to a product state, as for dilute and weakly interacting Bose gases, and it fails where correlations between particles matter.

1.4 Mean field for the Ising chain

For the spin chain the same logic replaces the ground state by a product state in which every spin occupies the same single-qubit state,

$$|\psi_{\mathrm{MF}}(\theta)\rangle = |\phi(\theta)\rangle^{\otimes N}, \qquad |\phi(\theta)\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle = \hat{R}_y(\theta)|0\rangle, \tag{1.16}$$

where $\hat{R}_y(\theta) = e^{-i\theta\hat{Y}/2}$ is a rotation about the y axis of the Bloch sphere. The Bloch vector of every spin lies in the xz plane at the polar angle θ, so $\langle\hat{Z}_i\rangle = \cos\theta$ and $\langle\hat{X}_i\rangle = \sin\theta$. A $\hat{Y}$ component cannot lower the energy of (1.2) and is left out. The state is stored as one angle in place of 2^N amplitudes. In a product state the correlator factorises, $\langle\hat{Z}_i\hat{Z}_{i+1}\rangle = \langle\hat{Z}_i\rangle\langle\hat{Z}_{i+1}\rangle = \cos^2\theta$, and the energy of the open chain becomes

$$E_{\mathrm{MF}}(\theta) = -J(N-1)\cos^2\theta - hN\sin\theta. \tag{1.17}$$

The open chain has $N-1$ bonds and N spins, and the ratio $(N-1)/N$ tends to one for a long chain, where the energy per spin is $E_{\mathrm{MF}}/N = -J\cos^2\theta - h\sin\theta$. The best angle makes the derivative

vanish,

$$\frac{dE_{\mathrm{MF}}}{d\theta} = \cos\theta\,\bigl[2J(N-1)\sin\theta - hN\bigr] = 0. \tag{1.18}$$

This stationarity condition is the self-consistent equation of mean-field theory, as the magnetisation $\zeta = \langle \hat{Z}_i \rangle = \cos\theta$ shows. Each spin feels the transverse field h along x and, from its neighbours, the average longitudinal field $2J(N-1)\zeta/N$ along z, where $2(N-1)/N$ is the average number of neighbours on the open chain. This average field of the neighbours is the molecular field that Weiss introduced for ferromagnets [3]. The Bloch vector points along the total field, so its z component is

$$\zeta = \frac{2J(N-1)\zeta/N}{\sqrt{\bigl(2J(N-1)\zeta/N\bigr)^2 + h^2}}, \tag{1.19}$$

which is (1.18) for $\cos\theta \neq 0$. The magnetisation appears on both sides, since the field that orients a spin is produced by the orientation of the others. Equation (1.19) always has the solution $\zeta = 0$, and for small h a second solution with $\zeta \neq 0$. Solving for $\sin\theta$ gives the minimum of (1.17),

$$\begin{aligned} \sin\theta^* &= \frac{h}{h_c^{\mathrm{MF}}}, & E_{\mathrm{MF}} &= -J(N-1) - \frac{h^2 N^2}{4J(N-1)} && \text{for } h < h_c^{\mathrm{MF}} = \frac{2J(N-1)}{N}, \\ \theta^* &= \frac{\pi}{2}, & E_{\mathrm{MF}} &= -hN && \text{for } h \geq h_c^{\mathrm{MF}}. \end{aligned} \tag{1.20}$$

Figure 4 shows how the minimum of (1.17) moves with the field. Below h_c^{MF} the spins tilt away from z and keep the magnetisation $\zeta = \pm\cos\theta^*$, with two degenerate solutions related by the spin flip $\hat{Z} \to -\hat{Z}$. Above it all spins point along x. In a long chain the mean-field transition lies at $h_c^{\mathrm{MF}} = 2J$, twice the exact critical field $h_c = J$, and for $N = 6$ it lies at $1.667J$ (the ratio $(N-1)/N$ lowers it). The product states $|0\ldots0\rangle$ and $|+\cdots+\rangle$ of Section 1.2 are the members $\theta = 0$ and $\theta = \pi/2$ of this family.

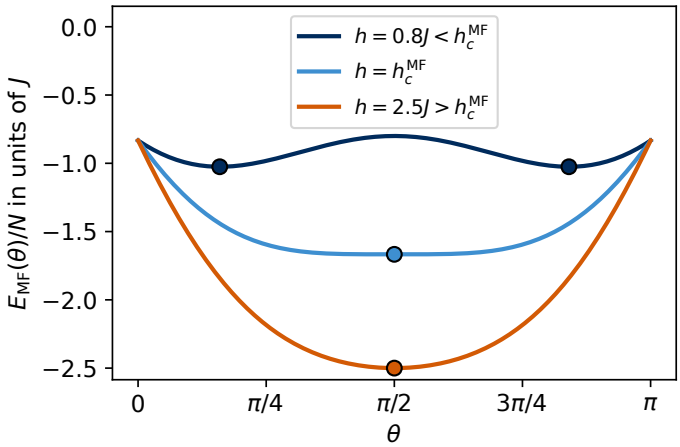


Figure 4: Mean-field energy (1.17) of the six-spin TFIM as a function of the angle θ for several fields h. Below h_c^{MF} the minimum lies at $\sin\theta^* = h/h_c^{\mathrm{MF}}$, above it at $\theta^* = \pi/2$. Look at how the minimum moves with h.

The infinite chain allows an exact comparison, because the TFIM maps onto free fermions and its exact ground-state energy per spin is known [4],

$$e_\infty(h) = -\frac{1}{\pi}\int_0^\pi dk\,\sqrt{J^2 + h^2 - 2Jh\cos k}. \tag{1.21}$$

At $h = J$ the integrand is $2J\sin(k/2)$ and $e_\infty = -4J/\pi \approx -1.2732J$, whereas mean field gives $-J - J/4 = -1.2500J$. Figure 5 compares mean field with the exact results for $N = 6$ and for the infinite chain. Every mean-field energy lies above its exact counterpart, as (1.4) requires, and the difference is largest between the exact critical point and the mean-field transition. At $h = J$ the mean-field energy of the six-spin chain is -6.800 against the exact -7.2962. Far from the transition,

at small and at large h, the exact ground state is close to a product state and mean field becomes accurate. In the range $J < h < 2J$ mean field predicts a magnetisation where the exact infinite chain has none, so it overestimates the order.

In Figure 5(b) the mean-field correlator $\cos^2\theta^*$ vanishes above h_c^{MF}, whereas the exact correlator decreases smoothly and stays positive, because quantum fluctuations correlate neighbouring spins without any magnetisation. A product state has zero half-chain entanglement entropy by construction and describes correlations only through the magnetisation.

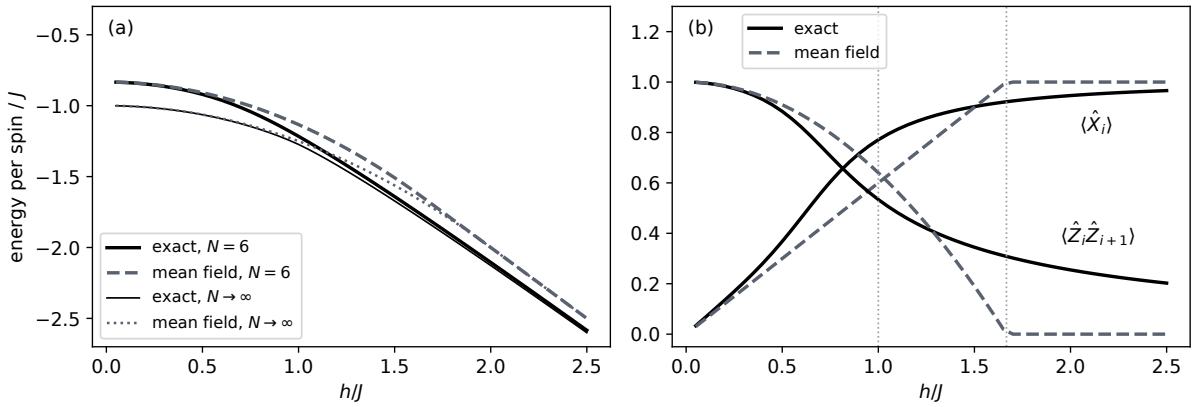


Figure 5: Mean field for the TFIM (1.2). (a) Energy per spin of the mean-field state (1.20) (dashed) and of the exact ground state (solid), for $N = 6$ (blue) and for the infinite chain (1.21) (red). (b) Transverse magnetisation $\langle \hat{X}_i \rangle$ and nearest-neighbour correlator $\langle \hat{Z}_i \hat{Z}_{i+1} \rangle$, averaged over the chain of six spins, exact (solid) and in mean field (dashed). The dotted lines mark the exact critical field $h = J$ and the mean-field transition h_c^{MF} of the six-spin chain. Look at the kink of the mean-field curves at h_c^{MF}, which the exact curves do not have.

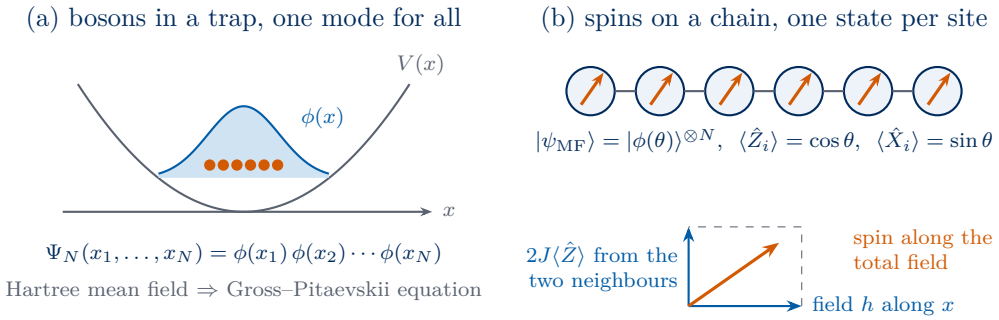


Figure 6: Mean field replaces the interacting chain by independent spins, each in the same state and each feeling the transverse field h and the average field of its neighbours. A Hartree state factorises over particles and a site-factorised state over lattice sites.

The two examples belong to two families of mean-field theory (Figure 6). The Gross–Pitaevskii state (1.10) factorises over particles, since all N bosons occupy one mode, and it is a *Hartree* mean field [5], in which each particle moves in the average potential created by all others. The spin state (1.16) factorises over lattice sites, $|\psi\rangle = \bigotimes_i |\phi_i\rangle$, and each factor is a spin pointing in a definite direction of its Bloch sphere. When all factors are equal, as in (1.16), the product is a spin coherent state. This *site-factorised* mean field is the Weiss mean field of magnetism. For bosons on a lattice the site-factorised idea gives the *Gutzwiller ansatz* [6]

$$|\Psi\rangle = \bigotimes_i \sum_{n=0}^{\infty} f_n^{(i)} |n\rangle_i, \tag{1.22}$$

a product over sites of superpositions of occupation numbers n with amplitudes $f_n^{(i)}$. In the Bose–Hubbard model, in which bosons hop between neighbouring sites with amplitude t_{hop} and repel each other with energy U on the same site, the ground state changes from a superfluid to a Mott

insulator, with a fixed number of bosons on every site, as U/t_{hop} grows [7]. The Gutzwiller ansatz (Figure 7) describes this transition in mean field [8], whereas a single-mode Gross–Pitaevskii state cannot, because it never fixes the number of bosons on a site. Neither family contains entanglement between its factors. In the language of circuits the state (1.16) is a circuit with one $\hat{R}_y(\theta)$ on every qubit and no entangling gate.

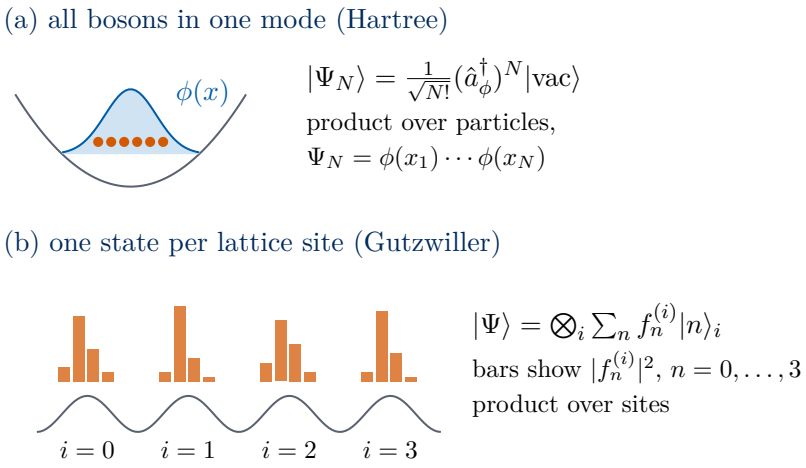


Figure 7: Two kinds of product ansatz. A Hartree state puts all particles into one single-particle mode and factorises over particles, as for the Gross–Pitaevskii equation. A Gutzwiller state (1.22) factorises over lattice sites, each site holding its own superposition of occupation numbers.

1.5 Tensor networks

Tensor networks keep entanglement while storing a number of parameters that grows only polynomially with N. A matrix product state (MPS) writes every amplitude as a product of matrices, one per spin,

$$\psi_{b_0 b_1 \ldots b_{N-1}} = A^{[0]}_{b_0} A^{[1]}_{b_1} \cdots A^{[N-1]}_{b_{N-1}}, \tag{1.23}$$

where $A^{[i]}_{b_i}$ is a matrix of size at most $\chi \times \chi$ chosen by the value of bit b_i, the first and last are a row and a column vector, and χ is the bond dimension (Figure 8). The MPS stores about $2N\chi^2$ numbers. Across any cut of the chain the state has at most χ Schmidt terms, so its entanglement entropy is at most $\log_2 \chi$ bits. A product state is an MPS with $\chi = 1$.

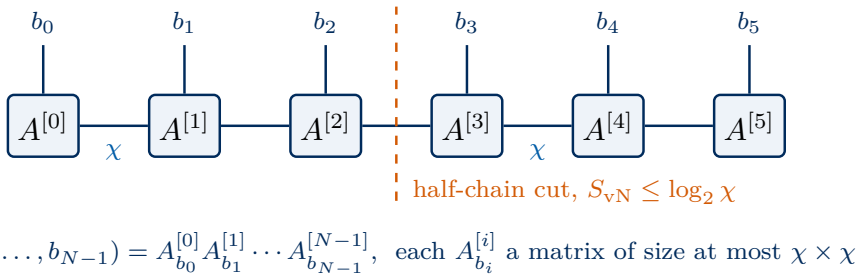


Figure 8: A matrix product state (1.23). Each spin carries one tensor with a physical index b_i and two bond indices of dimension χ, and contracting the bonds gives the amplitude of a bit string. The bond dimension bounds the entanglement across every cut.

Ground states of gapped one-dimensional Hamiltonians with local interactions obey an area law, with an entanglement entropy across a cut that stays bounded as N grows [9, 10]. A modest χ then describes them accurately, and the density-matrix renormalisation group (DMRG) optimises such states variationally [11, 12]. The required χ grows exponentially with the entanglement that must be represented. Entanglement grows with the width of two-dimensional systems and linearly in time after a sudden change of the Hamiltonian, which limits tensor networks in both situations.

1.6 Monte Carlo methods

Monte Carlo methods estimate expectation values from random samples and never store the full state (Figure 9). Quantum Monte Carlo maps the quantum problem onto the sampling of classical configurations with non-negative weights. For fermions and frustrated magnets the weights take both signs, and the statistical error then grows exponentially with the system size and the inverse temperature. This sign problem is NP-hard in general, so an efficient general solution would solve every problem of the complexity class NP efficiently [13]. Variational Monte Carlo samples bit strings from $|\psi_{\mathbf{b}}|^2$ of a trial wave function whose amplitudes are computed by a classical formula, and a neural network can serve as this formula, with its weights trained to minimise the energy [14].

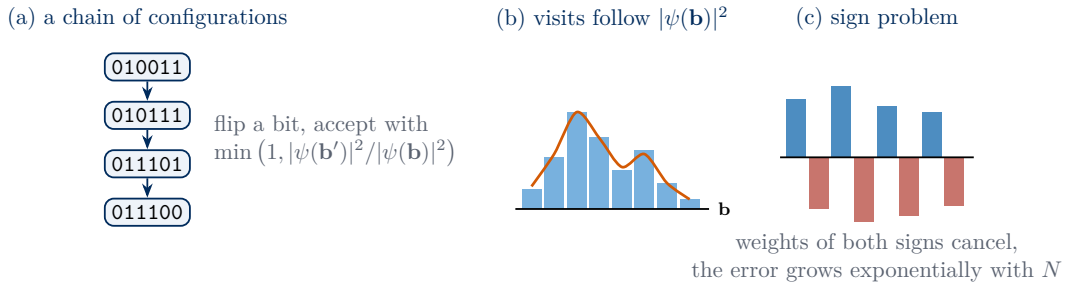


Figure 9: Monte Carlo methods. Configurations are sampled with probabilities given by non-negative weights, and expectation values are estimated as averages over the samples without storing the state. When the weights take both signs, the average is a small difference of large positive and negative contributions, the sign problem.

Each classical method restricts the states it can represent. Mean field excludes entanglement, tensor networks bound it, and Monte Carlo methods are limited by the sign structure of the amplitudes or by the form of the trial wave function.

2 Parametrised circuits as quantum ansätze

2.1 Trial states prepared by a quantum device

Over the last two decades experiments with trapped ions, neutral atoms in optical tweezers and superconducting circuits have reached individual control of tens to hundreds of qubits [15]. Each qubit is prepared in $|0\rangle$, rotated by laser or microwave pulses, coupled to its neighbours by a controlled interaction and measured in the computational basis. A register of N such qubits holds the 2^N amplitudes of (1.1) physically. Such a register can carry a variational ansatz built from the operations of the hardware, with pulse areas and interaction times as parameters optimised by a classical computer and a trial state that is never stored classically. A quantum processor can thus prepare the trial state of a variational calculation itself [16], and the resulting hybrid quantum–classical scheme is the *variational quantum eigensolver* (VQE) [16, 17].

The register starts in $|0\ldots0\rangle$ and a circuit with n adjustable angles $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ acts on it,

$$|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})\,|0\ldots0\rangle, \qquad \hat{U}(\boldsymbol{\theta}) = \hat{W}_n\, e^{-i\theta_n \hat{P}_n/2} \cdots \hat{W}_2\, e^{-i\theta_2 \hat{P}_2/2}\, \hat{W}_1\, e^{-i\theta_1 \hat{P}_1/2}, \tag{2.1}$$

where each $\hat{P}_k$ is a Pauli string, a tensor product of $\hat{I}$, $\hat{X}$, $\hat{Y}$ and $\hat{Z}$ on the qubits, and each $\hat{W}_k$ is a fixed gate without an adjustable angle, such as a controlled-$\hat{Z}$ gate $\widehat{\mathrm{CZ}}$, or the identity. Every parametrised gate is a rotation $\hat{R}_P(\theta) = e^{-i\theta\hat{P}/2}$, and since $\hat{P}^2 = \hat{I}$ its exponential series sums to

$$\hat{R}_P(\theta) = e^{-i\theta\hat{P}/2} = \cos\frac{\theta}{2}\,\hat{I} - i\sin\frac{\theta}{2}\,\hat{P}. \tag{2.2}$$

In the hardware a single-qubit rotation $\hat{R}_x(\theta)$ or $\hat{R}_y(\theta)$ is a resonant laser or microwave pulse of Rabi frequency Ω and duration t, with the pulse area $\theta = \Omega t$. A two-qubit rotation $\hat{R}_{zz}(\theta) = e^{-i\theta\hat{Z}_i\hat{Z}_j/2}$

is an Ising interaction $\frac{J}{2}\hat{Z}_i\hat{Z}_j$ switched on for the time t, with $\theta = Jt$. Changing an angle therefore means changing the length of a pulse or of an interaction.

The ansatz (2.1) is fully quantum, since the trial state exists only on the register, its 2^N amplitudes are never written down and the device returns only bit strings. Different circuit layouts define different ansätze, just as mean field, matrix product states and neural-network wave functions define different classical ansätze. The mean-field state (1.16) is the member with one $\hat{R}_y$ per qubit, and entangling gates such as $\widehat{\mathrm{CZ}}$ or $\hat{R}_{zz}$ take the ansatz beyond product states. Table 1 relates the names used in physics, mathematics and computer science.

2.2 The hybrid loop

The energy $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{\mathcal{H}}|\psi(\boldsymbol{\theta})\rangle$ is a function of n real numbers, and by (1.4) it never falls below E_0. The calculation alternates between the quantum device and a classical computer (Figure 10). The device prepares the trial state and measures it many times, and the classical computer turns the measured bit strings into an estimate of the energy and of its gradient and proposes new angles.

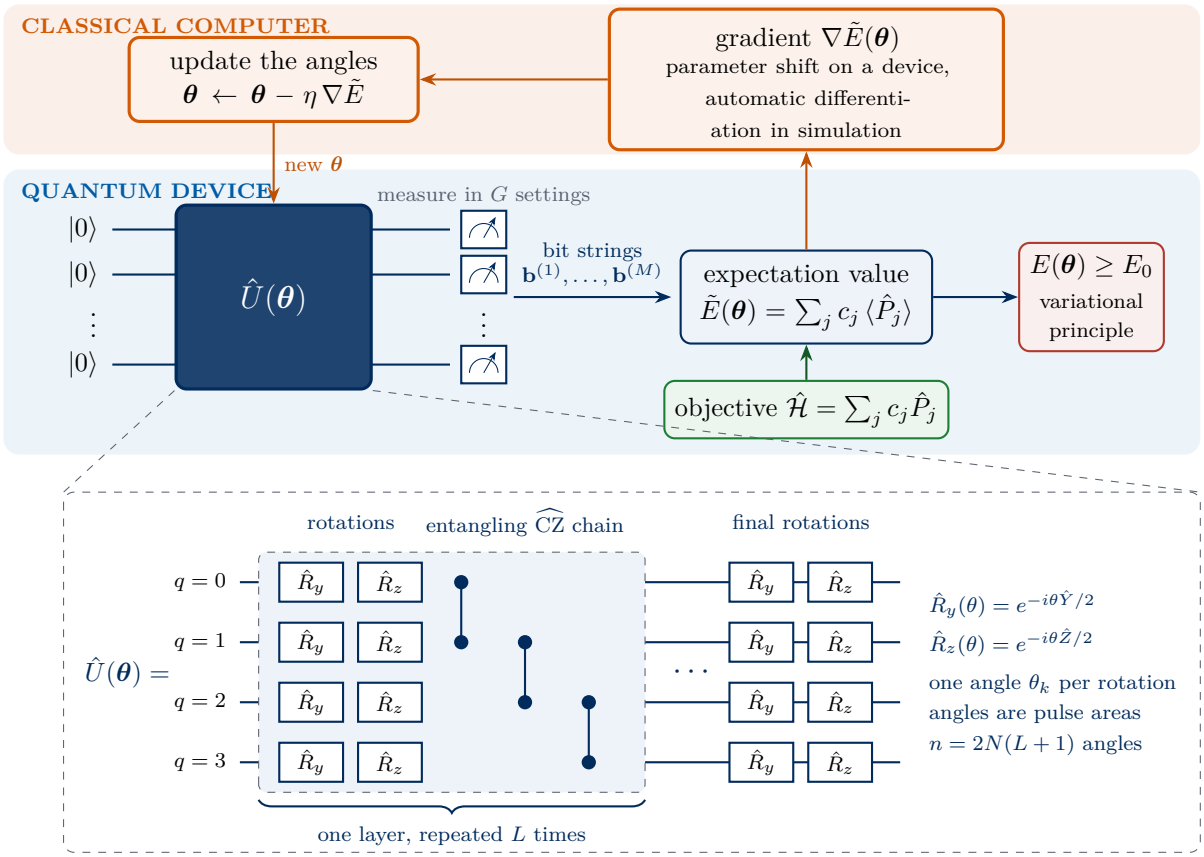


Figure 10: The variational loop. The quantum device prepares $|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|0\ldots0\rangle$, measures it in G settings and returns bit strings. The classical computer estimates the expectation value of the objective $\hat{\mathcal{H}} = \sum_j c_j\hat{P}_j$ and its gradient and updates the angles $\boldsymbol{\theta}$. The dashed frame shows the hardware-efficient ansatz inside $\hat{U}(\boldsymbol{\theta})$, layers of single-qubit rotations with independent angles followed by a chain of $\widehat{\mathrm{CZ}}$ gates.

Variational quantum eigensolver

Given a Hamiltonian $\hat{\mathcal{H}} = \sum_j c_j\hat{P}_j$, a circuit $\hat{U}(\boldsymbol{\theta})$ with n angles, a number M of shots per measurement setting and a step size η, the classical computer chooses initial angles $\boldsymbol{\theta}$, and the algorithm repeats three operations.

1. The device prepares $|\psi(\boldsymbol{\theta})\rangle$, rotates the qubits into each measurement setting of $\hat{\mathcal{H}}$ and records M bit strings per setting.

2. The classical computer forms an estimate $\tilde{E}(\boldsymbol{\theta})$ of the energy from the bit strings and, from further runs at shifted angles, an estimate $\tilde{\mathbf{g}}$ of its gradient.
3. It updates the angles, $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\tilde{\mathbf{g}}$ or by a variant of gradient descent, and sends them to the device, until the energy stops decreasing.

The output is the last energy estimate, an approximate upper bound on E_0 within its statistical error, and the angles, which prepare the trial state again on demand.

2.3 The accuracy of a variational state

The trained state is also used to compute properties other than the energy, such as magnetisations and correlations, so its distance from the ground state matters as well. The *fidelity* $F = |\langle E_0|\psi\rangle|^2 = |c_0|^2$ is the probability that the trial state passes a test for being the ground state, and $1-F$ is the infidelity. Subtracting E_0 from (1.4) gives

$$E - E_0 = \sum_{m\geq 1} |c_m|^2 (E_m - E_0) \;\geq\; \Delta E \sum_{m\geq 1} |c_m|^2 = \Delta E\,(1-F), \tag{2.3}$$

where $\Delta E = E_1 - E_0$ is the gap between the ground state and the first excited state, assuming a non-degenerate ground state. The energy error therefore bounds the infidelity (Figure 11), $1 - F \leq (E - E_0)/\Delta E$. A small energy error guarantees a fidelity close to one only when it is small compared with the gap. For the critical TFIM with $N = 6$ the gap is $\Delta E = 0.482J$, and an energy error of $10^{-3}J$ guarantees an infidelity of at most about 2×10^{-3}. Near a phase transition, or in an ordered phase where the two lowest levels almost coincide, the gap is small and the bound becomes weak. We use $E - E_0$ or the relative error $\delta E = (E - E_0)/|E_0|$, the fidelity F, and the half-chain entanglement entropy S_{vN} to judge trained states. The fidelity and the entropy require the exact ground state and the full trial state, which only a simulation of a small register provides.

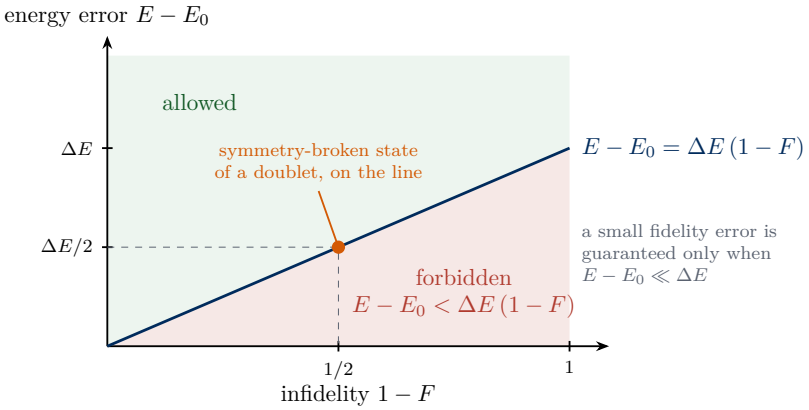


Figure 11: The fidelity bound (2.3). Every unit of weight $1 - F$ outside the ground state raises the energy by at least the gap ΔE, so the energy error bounds the infidelity. A small gap weakens the bound.

3 The energy from bit strings

3.1 Pauli strings and measurement settings

A device returns bit strings, and the energy has to be assembled from them. Every Hamiltonian of N qubits is a real combination of Pauli strings,

$$\hat{\mathcal{H}} = \sum_j c_j \hat{P}_j, \qquad E(\boldsymbol{\theta}) = \sum_j c_j \,\langle \hat{P}_j \rangle_{\boldsymbol{\theta}}, \tag{3.1}$$

where $\langle \hat{P}_j \rangle_{\boldsymbol{\theta}} = \langle \psi(\boldsymbol{\theta}) | \hat{P}_j | \psi(\boldsymbol{\theta}) \rangle$. The TFIM (1.2) has $2N-1$ strings, $N-1$ strings $\hat{Z}_i \hat{Z}_{i+1}$ with coefficient $-J$ and N strings $\hat{X}_i$ with coefficient $-h$, and the XXZ chain (1.3) has $3(N-1)$ strings.

A string of $\hat{Z}$ operators is diagonal in the computational basis. When every qubit is measured and the bit string $\mathbf{b}$ is recorded, $\hat{Z}_i$ takes the value $(-1)^{b_i}$, because $\hat{Z}|0\rangle = |0\rangle$ and $\hat{Z}|1\rangle = -|1\rangle$, and the product $\hat{Z}_i \hat{Z}_{i+1}$ takes the value $(-1)^{b_i + b_{i+1}}$, the parity of the two bits. In general a Pauli string with support S, the set of qubits on which it acts nontrivially, takes the value

$$P(\mathbf{b}) = (-1)^{\sum_{q \in S} b_q} \tag{3.2}$$

after each of its factors has been rotated into $\hat{Z}$. The rotation uses $\hat{H}\hat{X}\hat{H} = \hat{Z}$ for the Hadamard gate $\hat{H}$, so a qubit is measured along x by applying $\hat{H}$ before the readout, and along y by applying $\hat{S}^\dagger$ and then $\hat{H}$, where $\hat{S} = \mathrm{diag}(1, i)$. (The letter $\hat{H}$ denotes the Hadamard gate and $\hat{\mathcal{H}}$ the Hamiltonian.)

One choice of readout basis for every qubit is a *measurement setting*, and one setting measures all strings that carry on each qubit either the identity or the Pauli matrix of that qubit's basis. The TFIM therefore needs $G = 2$ settings, all qubits along z for the bond terms and all qubits along x for the field terms. The XXZ chain needs $G = 3$ settings, all along x, all along y and all along z. Each shot of a setting g gives one value of the part of the energy that this setting measures. For the TFIM

$$e_Z(\mathbf{b}) = -J \sum_{i=0}^{N-2} (-1)^{b_i + b_{i+1}}, \qquad e_X(\mathbf{b}) = -h \sum_{i=0}^{N-1} (-1)^{b_i}, \tag{3.3}$$

where $\mathbf{b}$ in e_X is recorded after a Hadamard gate on every qubit. The energy is the sum of the two averages over the Born distributions of the settings,

$$E(\boldsymbol{\theta}) = \sum_{\mathbf{b}} p_Z(\mathbf{b})\, e_Z(\mathbf{b}) + \sum_{\mathbf{b}} p_X(\mathbf{b})\, e_X(\mathbf{b}), \qquad p_Z(\mathbf{b}) = |\langle \mathbf{b} | \psi \rangle|^2, \quad p_X(\mathbf{b}) = |\langle \mathbf{b} | \hat{H}^{\otimes N} | \psi \rangle|^2. \tag{3.4}$$

3.2 The estimator and its standard error

A device returns M bit strings $\mathbf{b}^{(1)}, \ldots, \mathbf{b}^{(M)}$ in the z setting and M bit strings $\mathbf{b}'^{(1)}, \ldots, \mathbf{b}'^{(M)}$ in the x setting. The estimate of the energy replaces each average in (3.4) by the sample mean (Figure 12),

$$\tilde{E} = \frac{1}{M} \sum_{s=1}^{M} e_Z(\mathbf{b}^{(s)}) + \frac{1}{M} \sum_{s=1}^{M} e_X(\mathbf{b}'^{(s)}). \tag{3.5}$$

Its mean over many repetitions of the experiment is the exact energy. The shots are independent, so the variance of each sample mean is σ_g^2 / M, where

$$\sigma_g^2 = \sum_{\mathbf{b}} p_g(\mathbf{b})\, e_g(\mathbf{b})^2 - \Big(\sum_{\mathbf{b}} p_g(\mathbf{b})\, e_g(\mathbf{b}) \Big)^2 \tag{3.6}$$

is the variance of a single shot of setting $g \in \{Z, X\}$. For a single Pauli string with values ± 1 this is $1 - \langle \hat{P} \rangle^2$, which vanishes only when the state is an eigenstate of $\hat{P}$. The settings are measured independently, so their variances add and the standard error of the energy is

$$\sigma_{\tilde{E}} = \sqrt{\frac{1}{M} \sum_{g=1}^{G} \sigma_g^2}. \tag{3.7}$$

In an experiment σ_g^2 is unknown and is replaced by the sample variance of the recorded values.

As an example we take a six-qubit *hardware-efficient* circuit, named so because it uses only gates that devices apply natively. Each of its four layers rotates every qubit by $\hat{R}_y$ and $\hat{R}_z$ and couples neighbouring qubits by a chain of $\widehat{\mathrm{CZ}}$ gates, which gives $n = 60$ angles. With the critical TFIM as the Hamiltonian and random angles, $M = 1000$ shots per setting give the estimate -0.172 ± 0.107 of the exact energy -0.186. Repeating the whole measurement many times (Figure 13) confirms the scaling

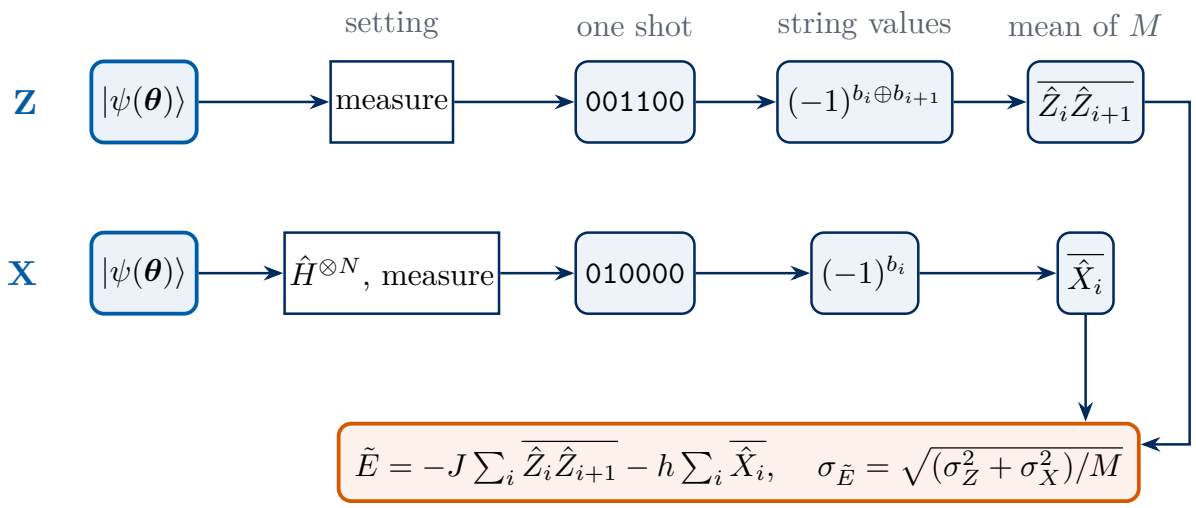


Figure 12: Energy estimate from bit strings for the TFIM. Shots in the z setting give the bond terms as parities of neighbouring bits, shots in the x setting (Hadamard gates before readout) give the field terms, and the two sample means add to $\tilde{E}$ (3.5) with the standard error (3.7).

$M^{-1/2}$ of (3.7) over three orders of magnitude. Every tenfold reduction of the error bar therefore costs a hundred times more shots, and a standard error of $0.01J$ at these angles needs about 1.1×10^5 shots per setting. The single-shot variances depend on the state, so the same target needs a different number of shots at other angles.

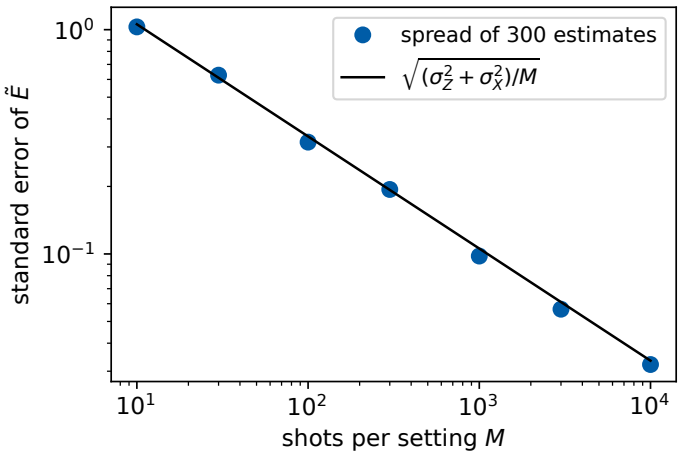


Figure 13: Standard deviation of repeated energy estimates (3.5) against the number M of shots per setting, from simulated measurement records of a six-qubit trial state at fixed random angles (markers), and the prediction (3.7) (line). The slope $-1/2$ on the logarithmic axes is the shot-noise scaling.

The shot budget is the main practical cost of variational algorithms on hardware. Molecular Hamiltonians contain many Pauli strings that fall into many settings, and every setting adds its own record of M shots. Grouping commuting strings into few settings reduces the cost, and the TFIM and XXZ chains with two and three settings are among the cheapest cases.

4 Gradients

4.1 The energy along one angle

Gradient descent needs the derivatives $\partial E/\partial \theta_k$ of the energy with respect to every angle. On a device they have to come from measurements of the same circuit, and their form follows from the dependence

of the energy on a single angle. We split the circuit (2.1) at the rotation that carries θ_k,

$$\hat{U}(\boldsymbol{\theta}) = \hat{A}\, e^{-i\theta_k \hat{P}_k/2}\, \hat{B}, \tag{4.1}$$

where $\hat{B}$ contains all gates before the rotation and $\hat{A}$ all gates after it, and neither depends on θ_k. With $|\phi\rangle = \hat{B}|0\ldots0\rangle$ and the operator $\hat{O} = \hat{A}^\dagger \hat{\mathcal{H}} \hat{A}$, the energy as a function of θ_k alone is

$$E(\theta_k) = \langle\phi|\, e^{i\theta_k \hat{P}_k/2}\, \hat{O}\, e^{-i\theta_k \hat{P}_k/2}\, |\phi\rangle. \tag{4.2}$$

Inserting the rotation (2.2) and multiplying out the four products gives

$$E(\theta_k) = \cos^2\frac{\theta_k}{2}\,\langle\hat{O}\rangle + \sin^2\frac{\theta_k}{2}\,\langle\hat{P}_k\hat{O}\hat{P}_k\rangle + i\sin\frac{\theta_k}{2}\cos\frac{\theta_k}{2}\,\langle[\hat{P}_k,\hat{O}]\rangle, \tag{4.3}$$

with all expectation values in $|\phi\rangle$ and the commutator $[\hat{P}_k,\hat{O}] = \hat{P}_k\hat{O} - \hat{O}\hat{P}_k$. The half-angle identities $\cos^2(\theta/2) = (1+\cos\theta)/2$, $\sin^2(\theta/2) = (1-\cos\theta)/2$ and $2\sin(\theta/2)\cos(\theta/2) = \sin\theta$ turn this into a shifted cosine,

$$E(\theta_k) = a + u\cos\theta_k + v\sin\theta_k = a + b\cos(\theta_k - c), \tag{4.4}$$

where

$$a = \tfrac{1}{2}(\langle\hat{O}\rangle + \langle\hat{P}_k\hat{O}\hat{P}_k\rangle), \qquad u = \tfrac{1}{2}(\langle\hat{O}\rangle - \langle\hat{P}_k\hat{O}\hat{P}_k\rangle), \qquad v = \tfrac{i}{2}\langle[\hat{P}_k,\hat{O}]\rangle, \tag{4.5}$$

the amplitude is $b = \sqrt{u^2+v^2}$ and the phase c satisfies $\cos c = u/b$ and $\sin c = v/b$. All three constants are real, because $i[\hat{P}_k,\hat{O}]$ is Hermitian. The derivation uses only $\hat{P}_k^2 = \hat{I}$, so it holds for every rotation of the circuit and for every Hamiltonian.

Along any single angle the landscape is therefore a cosine of period 2π with one minimum and one maximum, and three energies fix it, $E(0) = a+u$ and $E(\pm\pi/2) = a \pm v$. Figure 14 confirms this for one angle of the six-qubit hardware-efficient circuit of Section 3.2, where the cosine through three points coincides with the energy at all other angles to numerical precision. The fitted constants are $a = -0.582$, $b = 0.563$ and $c = 1.875$. The difficulty of the optimisation comes from the n angles together, since the constants of one angle depend on the values of all others.

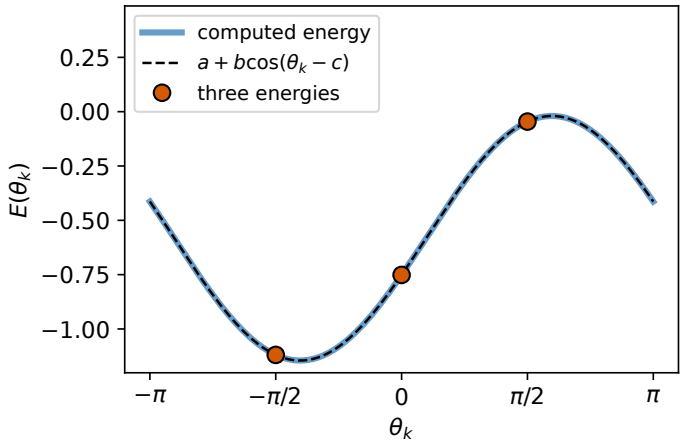


Figure 14: The energy of a six-qubit trial state of the critical TFIM as a function of one angle, with all other angles fixed (dots), and the cosine (4.4) through the energies at $\theta_k = 0$ and $\pm\pi/2$ (line). Exact state-vector values.

4.2 The parameter-shift rule

Differentiating (4.4) gives $\partial E/\partial\theta_k = -u\sin\theta_k + v\cos\theta_k$. The same combination appears in the difference of two energies at angles shifted by $\pm\pi/2$,

$$\begin{aligned} E(\theta_k + \tfrac{\pi}{2}) - E(\theta_k - \tfrac{\pi}{2}) &= u\big[\cos(\theta_k + \tfrac{\pi}{2}) - \cos(\theta_k - \tfrac{\pi}{2})\big] + v\big[\sin(\theta_k + \tfrac{\pi}{2}) - \sin(\theta_k - \tfrac{\pi}{2})\big] \\ &= -2u\sin\theta_k + 2v\cos\theta_k, \end{aligned} \tag{4.6}$$

where $\cos(\theta \pm \pi/2) = \mp \sin\theta$ and $\sin(\theta \pm \pi/2) = \pm \cos\theta$. Hence

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\boldsymbol{\theta} + \frac{\pi}{2}\mathbf{e}_k) - E(\boldsymbol{\theta} - \frac{\pi}{2}\mathbf{e}_k)}{2}, \tag{4.7}$$

where $\mathbf{e}_k$ is the unit vector along angle k. This relation is the parameter-shift rule [18, 19] (Figure 15). The shift is finite, the formula is exact, and both energies are measured with the same circuit on the same device, with one angle changed. For a rotation implemented as a pulse of Rabi frequency Ω and duration $t_k = \theta_k/\Omega$, the two shifted circuits differ from the original only in the pulse duration, $t_k \pm \pi/(2\Omega)$.

Many ansätze use one angle in several gates. When an angle φ enters gates $g = 1, 2, \ldots$ with rotation angles $\theta_g = c_g \varphi$ for constants c_g, the energy depends on φ through all of them, and the chain rule gives

$$\frac{\partial E}{\partial \varphi} = \sum_g c_g \frac{\partial E}{\partial \theta_g}. \tag{4.8}$$

Each term is evaluated with the two-point rule (4.7) by shifting the angle of gate g alone and keeping all other occurrences of φ fixed. Shifting φ itself by $\pm\pi/2$ shifts every gate at once and does not give the derivative, because the energy is then a sum of cosines of different frequencies in φ. The mean-field state (1.16) is an example, with θ in all N gates $\hat{R}_y(\theta)$ and $c_g = 1$, and (4.8) reproduces the derivative (1.18).

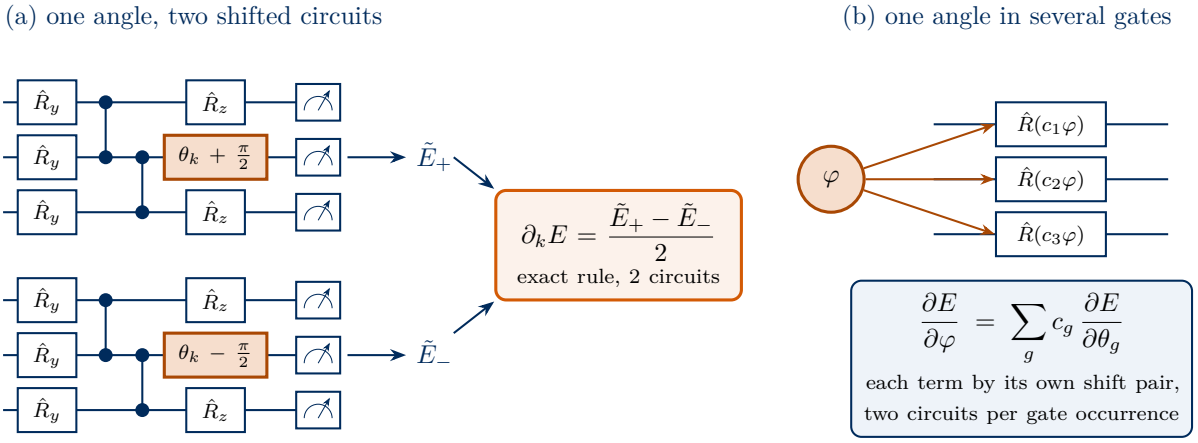


Figure 15: The parameter-shift rule (4.7). Along one angle the energy is a cosine, and the difference of its values at $\theta_k \pm \pi/2$ divided by two is the exact slope at θ_k. For an angle shared by several gates, each gate is shifted separately and the results are summed with the factors c_g of the chain rule (4.8).

4.3 The cost of a gradient on a device

On a device each energy in (4.7) is itself an estimate (3.5). Let n_{rot} be the number of parametrised Pauli rotations of the circuit, the gate occurrences of the angles, which equals n when every angle enters one gate and exceeds n when angles are shared. One gradient then requires $2n_{\mathrm{rot}}$ circuits, each measured in G settings with M shots,

$$\text{shots per gradient} = 2\, n_{\mathrm{rot}}\, G\, M. \tag{4.9}$$

For the six-qubit hardware-efficient circuit of Section 3.2, with $n_{\mathrm{rot}} = n = 60$ and $G = 2$, the gradient needs 120 distinct circuits, and $M = 1000$ shots per setting give 240000 shots for a single gradient. A training run of 600 gradient steps then needs about 1.4×10^8 shots, and the number grows linearly with the number of rotations and with the number of settings.

The statistical error of a gradient component follows from (3.7). The component $\tilde{g}_k = (\tilde{E}_{k,+} - \tilde{E}_{k,-})/2$ combines two independent estimates, so

$$\sigma_{\tilde{g}_k} = \frac{1}{2}\sqrt{\sigma^2_{\tilde{E}_{k,+}} + \sigma^2_{\tilde{E}_{k,-}}} \approx \sqrt{\frac{\sigma^2}{2M}}, \tag{4.10}$$

where $\sigma^2 = \sum_g \sigma_g^2$ is the single-shot variance of the energy, close to its value at the unshifted angles. At the random angles of Section 3.2 with $M = 1000$ this error is about 0.073. Figure 16 compares the estimated components with the exact ones. They scatter within their error bars, and components whose exact value is smaller than about twice the error bar cannot be distinguished from zero with this number of shots. Halving the error bar requires four times as many shots, so small gradients are expensive on hardware.

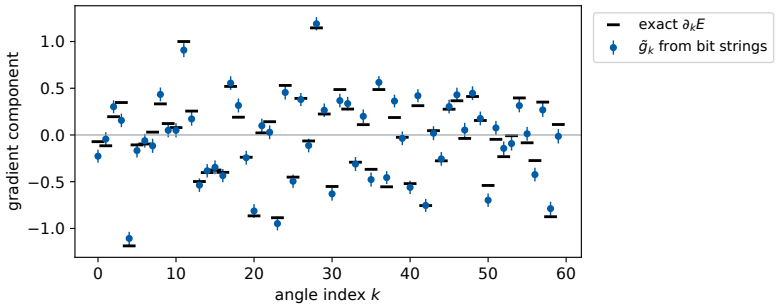


Figure 16: All 60 components of the energy gradient of a six-qubit trial state, estimated with the parameter-shift rule from simulated measurement records with $M = 1000$ shots per setting and shifted circuit (dots with error bars (4.10)), and the exact components (red dashes). Components whose exact value lies within about two error bars of zero are not resolved.

Of the other common gradient estimates, central finite differences $[E(\boldsymbol{\theta} + \epsilon \mathbf{e}_k) - E(\boldsymbol{\theta} - \epsilon \mathbf{e}_k)]/(2\epsilon)$ need the same $2n$ circuits, but the shot noise of the two energies is divided by the small step ϵ, so their error grows as $1/\epsilon$. Simultaneous-perturbation stochastic approximation (SPSA) [20] shifts all angles at once along a random direction with entries ± 1 and needs two energies per step for any n. Its estimate, averaged over the directions, equals the gradient up to corrections of second order in the size of the shift.

4.4 Automatic differentiation in simulation

A simulation holds the state vector and can differentiate the program that computes the energy. Automatic differentiation applies the chain rule to every elementary operation of that program. Its reverse mode [21] runs the program forwards, stores the intermediate states and propagates derivatives backwards through the gates, so that the whole gradient costs a few energy evaluations for any n. We use the implementation in the JAX library [22]. For the circuit with 60 angles, the reverse-mode gradient and the parameter-shift gradient (4.7), evaluated from exact energies, agree to numerical precision in all components. In a simulation the parameter-shift rule costs $2n$ circuit evaluations, so reverse mode is preferred there. A device has no access to intermediate states, and there the parameter-shift rule is the exact method. All training runs in these notes use exact gradients from reverse-mode differentiation, which correspond to the limit $M \to \infty$ of a device and show what an ansatz can reach.

4.5 Training

With the gradient available, the angles are updated by gradient descent,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \, \nabla E(\boldsymbol{\theta}), \tag{4.11}$$

with a step size η. We use Adam [23], a variant of gradient descent that scales the step of each angle with running averages of its past gradients and of their squares. Figure 17 shows a training run of the six-qubit hardware-efficient circuit on the critical TFIM, started from small angles near the product state $|0 \ldots 0\rangle$. The energy starts at -4.45, close to the value -5 of $|0 \ldots 0\rangle$, falls below the first excited level E_1 within the first 17 steps and then approaches E_0 slowly. After 600 steps the relative energy error is of order 1×10^{-3} and the fidelity with the exact ground state is 0.998, consistent with the bound (2.3). On a device the same run would cost the shots counted in (4.9).

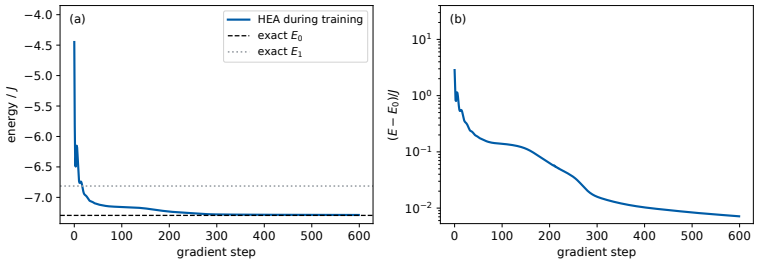


Figure 17: Energy of the six-qubit trial state of the critical TFIM during 600 Adam steps with exact gradients. The dashed and dotted lines mark the exact E_0 and E_1, and the inset shows the energy error $E - E_0$ on a logarithmic scale.

5 Circuit ansätze

5.1 Requirements and symmetries

The circuit layout fixes the family of states that VQE can reach, so the choice of the ansatz decides the quality of the result. A useful ansatz meets four requirements. It contains a good approximation of the target state, a property called *expressibility*, and its angles are trainable by gradient descent from a reasonable starting point. It uses the native gates and the qubit connectivity of the hardware, because every additional gate adds errors, and it respects the symmetries of the Hamiltonian, which keeps the search inside the symmetry sector of the ground state.

Both running models have symmetries. Every term of the TFIM (1.2) commutes with the *parity*

$$\hat{\Pi} = \prod_{i=0}^{N-1} \hat{X}_i, \tag{5.1}$$

which flips every spin, $|b_0 \dots b_{N-1}\rangle \to |\bar{b}_0 \dots \bar{b}_{N-1}\rangle$ with $\bar{b} = 1 - b$. This holds because $\hat{X}_i \hat{X}_{i+1}$ anticommutes with each of $\hat{Z}_i$ and $\hat{Z}_{i+1}$ and therefore commutes with their product. Eigenstates of $\hat{\mathcal{H}}_{\mathrm{TFIM}}$ thus have a definite parity ± 1. In the XXZ chain (1.3) the operator $\hat{X}\hat{X} + \hat{Y}\hat{Y} = 2(|01\rangle\langle 10| + |10\rangle\langle 01|)$ exchanges a 0 and a 1 between neighbours and $\hat{Z}\hat{Z}$ is diagonal. Neither term changes the number of bits equal to one, so the Hamiltonian conserves the total magnetisation

$$\hat{Z}_{\mathrm{tot}} = \sum_{i=0}^{N-1} \hat{Z}_i. \tag{5.2}$$

For even N the Heisenberg ground state lies in the sector $\hat{Z}_{\mathrm{tot}} = 0$, spanned by the bit strings with equal numbers of zeros and ones.

We compare three ansätze that weigh these requirements differently. In quantum chemistry the standard choice is the unitary coupled-cluster ansatz, which builds the trial state from excitations of electrons out of a reference configuration [16, 24].

5.2 The hardware-efficient ansatz

The hardware-efficient ansatz (HEA) is built only from gates that a device applies natively, without reference to the Hamiltonian [25]. It starts from $|0 \dots 0\rangle$. Each of its L layers applies $\hat{R}_y$ and then $\hat{R}_z$ with independent angles to every qubit, followed by a chain of $\widehat{\mathrm{CZ}}$ gates on the neighbouring pairs $(0,1), (1,2), \dots, (N-2, N-1)$, and a final rotation block closes the circuit (Figure 18),

$$|\psi_{\mathrm{HEA}}(\boldsymbol{\theta})\rangle = \hat{R}^{(L)}\, \hat{C}_Z\, \hat{R}^{(L-1)} \cdots \hat{C}_Z\, \hat{R}^{(0)} |0 \dots 0\rangle, \qquad \hat{R}^{(\ell)} = \bigotimes_{q=0}^{N-1} \hat{R}_z(\theta^z_{\ell q})\, \hat{R}_y(\theta^y_{\ell q}), \tag{5.3}$$

where $\hat{C}_Z = \prod_{q=0}^{N-2} \widehat{\mathrm{CZ}}_{q,q+1}$. The controlled-$\hat{Z}$ gate $\widehat{\mathrm{CZ}} = \mathrm{diag}(1, 1, 1, -1)$ changes the sign of the amplitude when both qubits are in $|1\rangle$, so all $\widehat{\mathrm{CZ}}$ gates commute and the chain multiplies the amplitude

of $\mathbf{b}$ by $(-1)^{\sum_q b_q b_{q+1}}$. The circuit has

$$n = 2N(L+1) \tag{5.4}$$

angles and $(N-1)L$ two-qubit gates. One block $\hat{R}_z\hat{R}_y$ takes $|0\rangle$ to any single-qubit state, so the first block alone prepares every product state, including the mean-field state (1.16), and the $\widehat{\mathrm{CZ}}$ layers create entanglement. With growing depth the HEA approaches any state of the register. Its layers carry no information about the Hamiltonian, and its states mix the parity sectors of the TFIM and the magnetisation sectors of the XXZ chain.

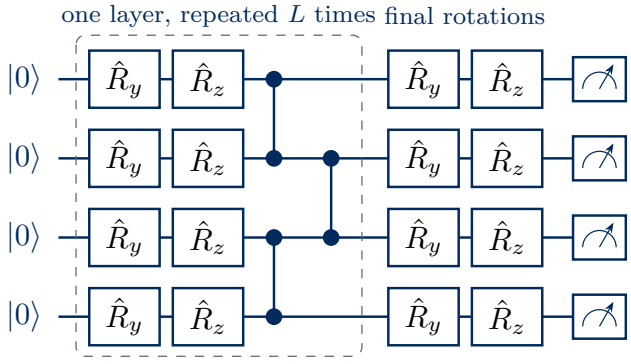


Figure 18: The hardware-efficient ansatz (5.3). One layer (dashed box) applies $\hat{R}_y$ and $\hat{R}_z$ to every qubit and a chain of $\widehat{\mathrm{CZ}}$ gates on neighbouring pairs, and is repeated L times before a final rotation block.

5.3 The Hamiltonian-variational ansatz

The Hamiltonian-variational ansatz (HVA) builds every layer from the evolution operators of the terms of the Hamiltonian [26, 27]. For the TFIM it starts in $|+\rangle^{\otimes N}$, the ground state of the field term alone, and applies L layers (Figure 19),

$$|\psi_{\mathrm{HVA}}(\boldsymbol{\gamma},\boldsymbol{\beta})\rangle = \hat{V}_L \cdots \hat{V}_2\hat{V}_1|+\rangle^{\otimes N}, \qquad \hat{V}_\ell = e^{-i\beta_\ell \sum_i \hat{X}_i}\, e^{-i\gamma_\ell \sum_i \hat{Z}_i\hat{Z}_{i+1}}. \tag{5.5}$$

All bonds of a layer share the angle γ_ℓ and all qubits share β_ℓ, so the circuit has only $n = 2L$ angles. In gates, $e^{-i\gamma_\ell \hat{Z}_i\hat{Z}_{i+1}} = \hat{R}_{zz}(2\gamma_\ell)$ on every bond and $e^{-i\beta_\ell \hat{X}_i} = \hat{R}_x(2\beta_\ell)$ on every qubit. A layer therefore contains $N-1$ two-qubit gates, as many as a layer of the HEA, and the circuit has $n_{\mathrm{rot}} = L(2N-1)$ parametrised rotations. Both exponents commute with the parity (5.1), and $|+\rangle^{\otimes N}$ has $\hat{\Pi} = +1$, so every state of the ansatz is even, like the ground state of the TFIM.

The structure of (5.5) follows from approximating the evolution under a sum of non-commuting terms by alternating short evolutions under each term, $e^{-i(\hat{A}+\hat{B})t} \approx (e^{-i\hat{A}t/m}e^{-i\hat{B}t/m})^m$, a Trotter decomposition that becomes exact as $m \to \infty$ [28]. We write the two terms of (1.2) as $\hat{\mathcal{H}}^{ZZ} = -J\sum_i \hat{Z}_i\hat{Z}_{i+1}$ and $\hat{\mathcal{H}}^{X} = -h\sum_i \hat{X}_i$. Each layer $\hat{V}_\ell$ has the form of one such step of evolution under $s_\ell\hat{\mathcal{H}}^{ZZ} + (1-s_\ell)\hat{\mathcal{H}}^{X}$, a Hamiltonian that interpolates between the field term at $s=0$ and the full model at $s=1$. The adiabatic theorem states that a system prepared in the ground state of a Hamiltonian stays in the instantaneous ground state when the Hamiltonian changes slowly enough, on a time scale set by the inverse of the gap to the first excited state [29]. A slow sweep from the field term to the full Hamiltonian, discretised into layers, therefore ends near the ground state, and we start the training from the angles of such a sweep,

$$\gamma_\ell = -J\tau\, s_\ell, \qquad \beta_\ell = -h\tau\,(1-s_\ell), \qquad s_\ell = \frac{\ell - 1/2}{L}, \tag{5.6}$$

with a time step $\tau = 0.8/J$ and small random offsets, so that $\hat{V}_\ell = e^{-i\tau(1-s_\ell)\hat{\mathcal{H}}^X}e^{-i\tau s_\ell\hat{\mathcal{H}}^{ZZ}}$.

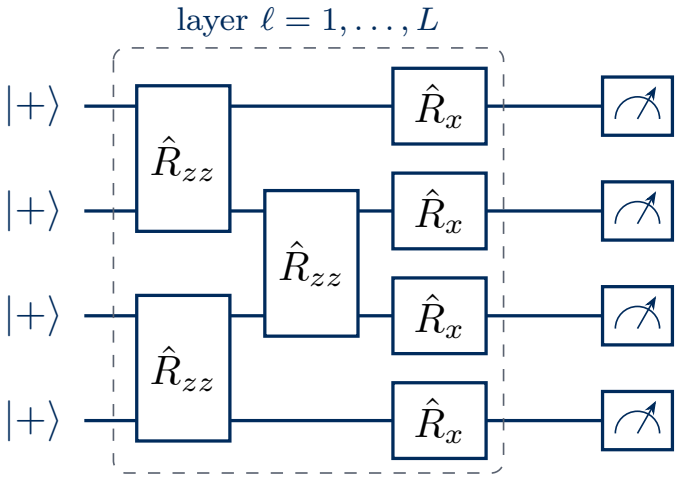


Figure 19: The Hamiltonian-variational ansatz (5.5) for the TFIM. The register starts in $|+\rangle^{\otimes N}$, and each layer applies $\hat{R}_{zz}(2\gamma_\ell)$ on all bonds and $\hat{R}_x(2\beta_\ell)$ on all qubits, with one shared angle per kind of gate and layer.

5.4 A symmetry-preserving ansatz for the XXZ chain

For the Heisenberg chain we use an ansatz that never leaves the sector $\hat{Z}_{\text{tot}} = 0$ of its ground state. It starts from the product of singlets $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ on the pairs $(0,1), (2,3), \ldots$, which has $\hat{Z}_{\text{tot}} = 0$ and is the ground state of (1.3) with every second bond removed. Each layer applies the two-qubit gate

$$\hat{G}(\theta_{xy}, \theta_{zz}) = e^{-i\theta_{xy}(\hat{X}\hat{X}+\hat{Y}\hat{Y})/2}\, e^{-i\theta_{zz}\hat{Z}\hat{Z}/2} \tag{5.7}$$

first on the even bonds $(0,1), (2,3), \ldots$ and then on the odd bonds $(1,2), (3,4), \ldots$, every gate with its own two angles (Figure 20). The two exponents commute, and in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ the gate is

$$G = \begin{pmatrix} e^{-i\theta_{zz}/2} & 0 & 0 & 0 \\ 0 & e^{i\theta_{zz}/2}\cos\theta_{xy} & -ie^{i\theta_{zz}/2}\sin\theta_{xy} & 0 \\ 0 & -ie^{i\theta_{zz}/2}\sin\theta_{xy} & e^{i\theta_{zz}/2}\cos\theta_{xy} & 0 \\ 0 & 0 & 0 & e^{-i\theta_{zz}/2} \end{pmatrix}. \tag{5.8}$$

It rotates $|01\rangle$ and $|10\rangle$ into each other and only adds phases to $|00\rangle$ and $|11\rangle$. No gate changes the number of ones, so every state of this symmetry-preserving ansatz (SPA) has $\hat{Z}_{\text{tot}} = 0$. The circuit has $n = 2(N-1)L$ angles and $(N-1)L$ two-qubit gates. Each gate (5.7) is the evolution of two qubits coupled by an anisotropic exchange interaction, and written as rotations it is $\hat{R}_{xx}(\theta_{xy})\hat{R}_{yy}(\theta_{xy})\hat{R}_{zz}(\theta_{zz})$, so $n_{\text{rot}} = 3(N-1)L$. We count it as one two-qubit gate, although a device whose native two-qubit gate is $\widehat{\text{CZ}}$ compiles it into several.

5.5 Entanglement capacity

A circuit can create only a limited amount of entanglement, and this bounds the states it reaches. We cut the chain into a left half A and a right half B. A state with χ Schmidt terms across the cut has an entanglement entropy of at most $\log_2 \chi$ bits. Gates inside one half leave the Schmidt coefficients unchanged. A gate across the cut can be written as a sum of r products,

$$\hat{U} = \sum_{k=1}^{r} \hat{A}_k \otimes \hat{B}_k, \tag{5.9}$$

with $\hat{A}_k$ acting on A and $\hat{B}_k$ on B, and applied to a state with χ Schmidt terms it produces at most $r\chi$ terms. For the controlled-$\hat{Z}$ gate $r = 2$, since $\widehat{\text{CZ}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{Z}$, and the same holds for $\hat{R}_{zz}(\theta) = \cos\frac{\theta}{2}\,\hat{I}\otimes\hat{I} - i\sin\frac{\theta}{2}\,\hat{Z}\otimes\hat{Z}$. One layer of the HEA or of the HVA contains one such gate

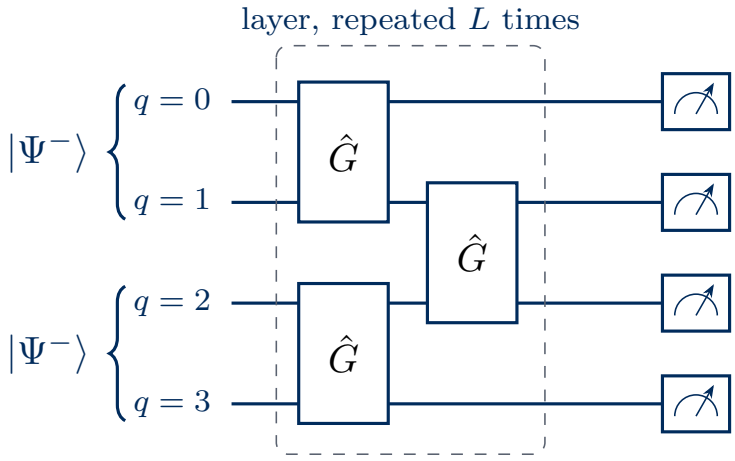


Figure 20: The symmetry-preserving ansatz for the XXZ chain. The register starts in a product of singlets (brackets), and each layer applies the gate $\hat{G}$ of (5.7) on the even and then on the odd bonds, each gate with its own angles $(\theta_{xy}, \theta_{zz})$. No gate changes the number of ones.

across the middle cut. After L layers the number of Schmidt terms is therefore at most 2^L, and the half-chain entropy obeys (Figure 21)

$$S_{\mathrm{vN}} \leq \min\left(L,\ N/2\right) \text{ bits}, \tag{5.10}$$

where $N/2$ bits is the maximum for $N/2$ qubits on each side. The gate (5.7) has $r = 4$ and adds at most two bits per layer. A ground state with an entropy of one bit across the middle needs at least one layer, and a volume-law state, with an entropy proportional to N, needs a depth that grows with N. The bound is a necessary condition only, since correlations between distant spins also need enough layers for the gates to connect them.

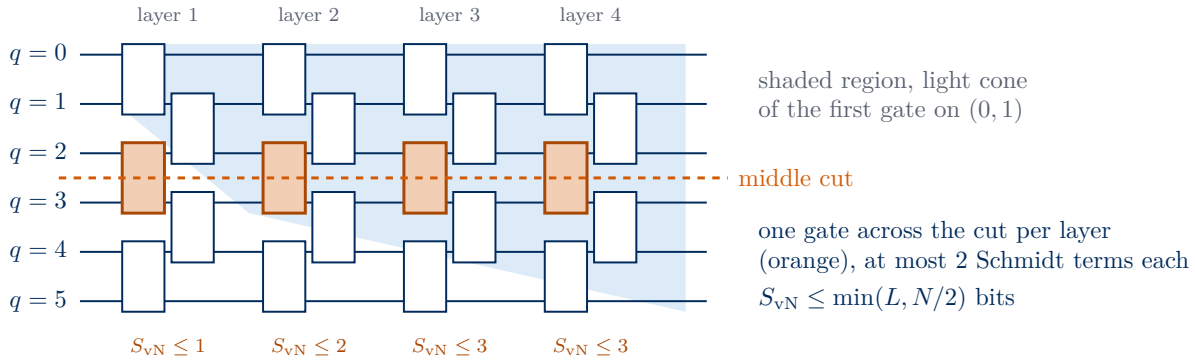


Figure 21: Entanglement capacity of a brickwork circuit. Gates inside one half leave the entanglement across the middle cut unchanged, and each gate across the cut with operator Schmidt rank $r = 2$ at most doubles the number of Schmidt terms, so L layers give at most L bits (5.10).

6 The variational quantum eigensolver for spin chains

6.1 Models, references and training

We test the three ansätze on chains of $N = 6$ qubits, small enough for exact diagonalisation of the 64×64 Hamiltonians, which gives the exact ground state, the energies E_0 and E_1 and the gap ΔE. The half-chain entanglement entropy follows from the Schmidt decomposition across the cut between qubits 2 and 3. The 64 amplitudes are reshaped into an 8×8 matrix with rows labelled by $b_0 b_1 b_2$

and columns by $b_3 b_4 b_5$, its singular values s_k are the Schmidt coefficients, and

$$S_{\mathrm{vN}} = -\sum_k s_k^2 \log_2 s_k^2 \tag{6.1}$$

in bits. A product state across the cut has $S_{\mathrm{vN}} = 0$, and one Bell pair shared between the halves contributes one bit. Of the three models studied, the critical TFIM at $h = J$ has the gap $\Delta E = 0.4821J$ and a ground-state entropy of 0.473 bits. The ordered TFIM at $h = 0.5J$ has a much smaller gap, $0.0235J$, and an entropy of almost one bit, 0.964. The Heisenberg chain, the XXZ chain at $\Delta = 1$, has its ground state in the sector $\hat{Z}_{\mathrm{tot}} = 0$ with an entropy of 1.026 bits. In both TFIM cases the ground state is even and the first excited state odd under $\hat{\Pi}$.

All three ansätze get $L = 6$ layers and therefore the same number $(N-1)L = 30$ of two-qubit gates, with $n = 84$ angles for the HEA, 12 for the HVA and 60 for the SPA. The HEA and the SPA start from small random angles near their reference states $|0\ldots0\rangle$ and the singlet product, and the HVA starts from the discretised sweep (5.6). Each run takes 600 Adam steps with exact gradients, and 4 independent starts, called restarts, are trained for every ansatz and model. As a user of VQE would, we keep the restart with the lowest final energy. The fidelity and the entropy are recorded at every step to judge the result, which on a device would require additional measurements.

6.2 The critical Ising chain

Figure 22 follows the training on the critical TFIM. Both ansätze lower the energy error by three to four orders of magnitude, and the best HVA restart, with 12 angles, ends below the best HEA restart, with 84 angles. The restarts of each ansatz end at different energies. The HEA starts from a product state and builds the entanglement gradually, so its entropy approaches the exact value from below. The HVA starts from the sweep (5.6), overshoots the exact entropy during the first tens of steps and then settles near it. For both best states the entropy error $\delta S = |S_{\mathrm{vN}} - S_{\mathrm{vN}}^{\mathrm{exact}}|$ is one to two orders of magnitude larger than the relative energy error (Table 2).

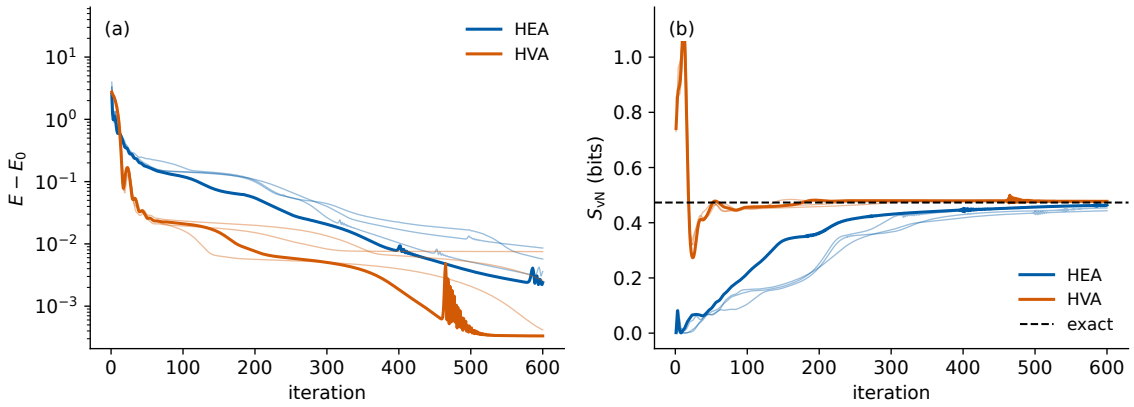


Figure 22: VQE on the critical TFIM, $N = 6$, $h = J$, with the HEA and the HVA at $L = 6$. (a) Energy error $E - E_0$ against the training step on a logarithmic scale. (b) Half-chain entropy S_{vN} against the step, with the exact value dashed. Thick lines are the restart with the lowest final energy, thin lines the other restarts. Exact state-vector simulation with exact gradients.

6.3 Energy errors and entanglement errors

The two errors differ in general, because they depend differently on a small admixture to the ground state. We write a trial state as the ground state with an admixture of amplitude ϵ of a normalised state $|\phi\rangle$ orthogonal to it,

$$|\psi\rangle = \frac{|\psi_0\rangle + \epsilon|\phi\rangle}{\sqrt{1+\epsilon^2}}, \qquad 1 - F = \frac{\epsilon^2}{1+\epsilon^2}, \qquad E - E_0 = \frac{\epsilon^2\big(\langle\phi|\hat{\mathcal{H}}|\phi\rangle - E_0\big)}{1+\epsilon^2}. \tag{6.2}$$

The cross terms of the energy, $\epsilon\langle\psi_0|\hat{\mathcal{H}}|\phi\rangle + \text{c.c.}$, vanish because $\hat{\mathcal{H}}|\psi_0\rangle = E_0|\psi_0\rangle$ and $\langle\psi_0|\phi\rangle = 0$, so the energy error is of second order in ϵ, like the infidelity. The ground state is a stationary point of the energy, which is why the variational principle works so well for the energy itself. The entropy has no such stationary point at $|\psi_0\rangle$. The reduced density matrix of the half chain changes at first order in ϵ, through the cross terms $\epsilon(|\psi_0\rangle\langle\phi| + |\phi\rangle\langle\psi_0|)$ traced over the other half, and so does its entropy,

$$S_{\text{vN}}(\psi) - S_{\text{vN}}(\psi_0) = \epsilon\, S_1 + O(\epsilon^2), \tag{6.3}$$

with a coefficient S_1 that depends on $|\phi\rangle$. The entropy error thus scales as $\sqrt{1-F}$, whereas the energy error scales as $1-F$, and a state with a relative energy error of 10^{-4} can carry a visibly wrong entanglement. Figure 23 confirms both scalings with every state recorded during the training runs of this section, 14400 states in total. All points of panel (a) lie above the bound (2.3) and follow a slope of one, and the entropy errors in panel (b) follow the slope one half of $\sqrt{1-F}$. At small infidelity the entropy is therefore the more sensitive test of the trained state.

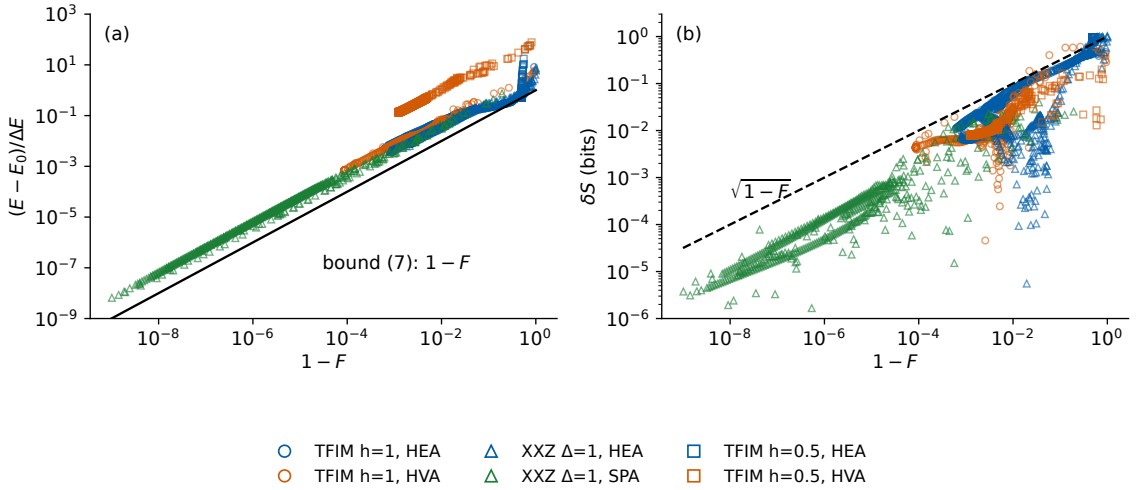


Figure 23: Errors of all states recorded during the VQE runs on the three chains. (a) Energy error in units of the gap, $(E-E_0)/\Delta E$, against the infidelity $1-F$, with the bound (2.3) as a solid line. (b) Entropy error δS against $1-F$, with the line $\sqrt{1-F}$ dashed. Look at the slopes, one in (a) and one half in (b).

6.4 The Heisenberg chain

On the Heisenberg chain we compare the HEA with the SPA (Figure 24). The SPA prepares the ground state to a relative energy error of 1.3×10^{-9}, and its entropy stays at the exact value from the first steps. It starts from the singlet product, which already carries one bit across the middle cut and lies in the sector $\hat{Z}_{\text{tot}} = 0$, so the circuit only redistributes singlet correlations along the chain. The HEA has to build this structure from $|0\ldots0\rangle$ and to remove the weight outside the sector. Its entropy rises to the exact value only after about a hundred steps, and its best state ends at $\delta E = 7.3 \times 10^{-4}$ and $\delta S = 7.6 \times 10^{-3}$ bits.

6.5 The ordered phase and symmetry breaking

At $h = 0.5J$ the exact ground state is close to the even superposition of the two ordered configurations,

$$|\psi_0\rangle \approx \frac{|0\ldots0\rangle + |1\ldots1\rangle}{\sqrt{2}}, \tag{6.4}$$

dressed by spin flips that the field creates, and the first excited state $|\psi_1\rangle$ is close to the odd superposition. A superposition of two macroscopically different configurations is called a cat state, and across any cut it carries about one bit of entropy. The two levels form a doublet separated by the

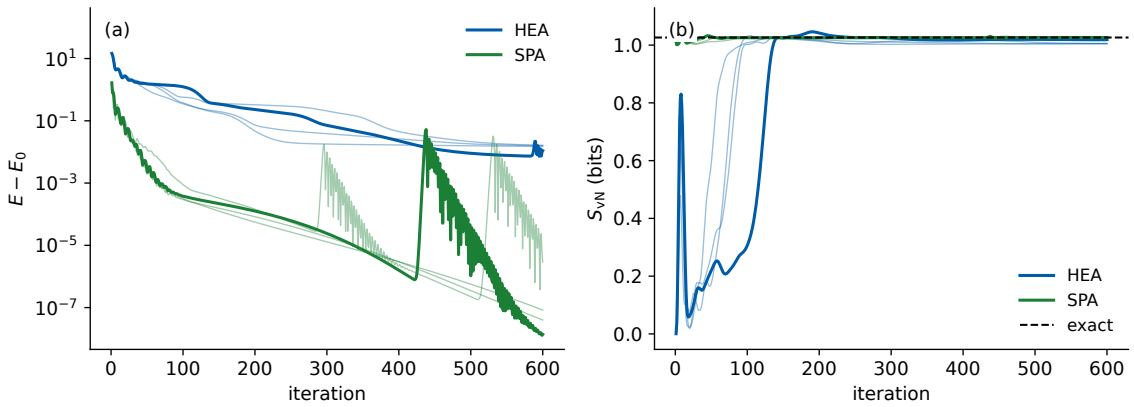


Figure 24: VQE on the Heisenberg chain, $N = 6$, with the HEA and the SPA at $L = 6$, shown as in Figure 22. (a) Energy error. (b) Half-chain entropy with the exact value dashed. The short spikes of the SPA late in training are single large Adam steps, after which the energy returns to its previous level.

small gap ΔE, which closes exponentially with N. A trial state within the doublet, $|\psi\rangle = a|\psi_0\rangle + b|\psi_1\rangle$ with $|a|^2 + |b|^2 = 1$, has the parity and the energy

$$\langle\hat{\Pi}\rangle = |a|^2 - |b|^2, \qquad E - E_0 = |b|^2 \Delta E = \frac{1 - \langle\hat{\Pi}\rangle}{2}\,\Delta E, \tag{6.5}$$

where the second form uses $|b|^2 = (1 - \langle\hat{\Pi}\rangle)/2$. The symmetry-broken state (Figure 25) $(|\psi_0\rangle + |\psi_1\rangle)/\sqrt{2} \approx |0\ldots0\rangle$, dressed by spin flips, has $\langle\hat{\Pi}\rangle = 0$, fidelity $F = 1/2$ and an energy only $\Delta E/2$ above E_0. It is close to a product state and carries almost no entanglement. A circuit that reaches this state has a relative energy error of order $\Delta E/(2|E_0|)$, small on the scale of the coupling, while its fidelity and entropy are far from the exact values.

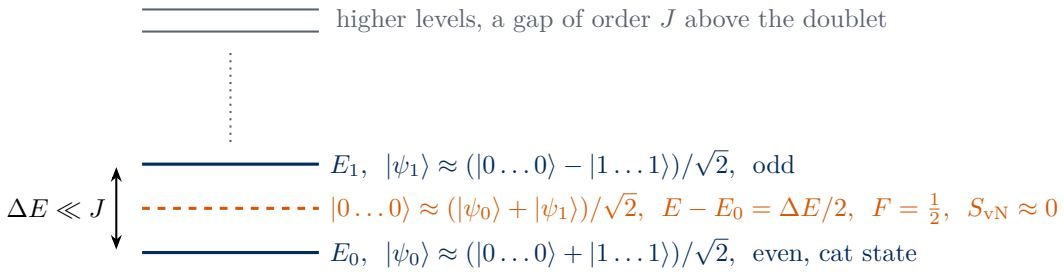


Figure 25: The ground doublet of the ordered TFIM. The even cat state $|\psi_0\rangle$ and the odd state $|\psi_1\rangle$ are split by the small gap ΔE. Their equal superposition is a symmetry-broken branch close to $|0\ldots0\rangle$, with $F = 1/2$, no entanglement and an energy only $\Delta E/2$ above E_0 (6.5).

Figure 26 shows that every HEA restart descends into the symmetry-broken branch near $|0\ldots0\rangle$, from which it starts, and stays there with $F \approx 1/2$, almost no entanglement and a parity close to zero. Its weight lies almost entirely within the doublet, and its energy error agrees with $\Delta E/2$, as (6.5) predicts (Table 2). Turning this state into the cat state would lower the energy by only $\Delta E/2$ and requires a coherent superposition with $|1\ldots1\rangle$, which differs in all six spins. The HVA keeps the parity at $+1$ at every step and cannot enter the symmetry-broken branch. Its best state reaches a fidelity close to one and the exact entropy of almost one bit.

In the computational basis the exact ground state and the HVA state return 000000 and 111111 in nearly equal shares, about 0.40 each, whereas the HEA state returns 000000 with probability 0.81 and 111111 almost never (Figure 27). Measurements along z alone cannot distinguish the cat state from an equal mixture of the two branches, and the parity $\langle\hat{\Pi}\rangle$, measured in the all-x setting, decides between them.

The HEA fails through two properties together, its missing parity symmetry and its start near $|0\ldots0\rangle$, which lies in one branch. The HVA needs enough depth, because the cat state (6.4) correlates

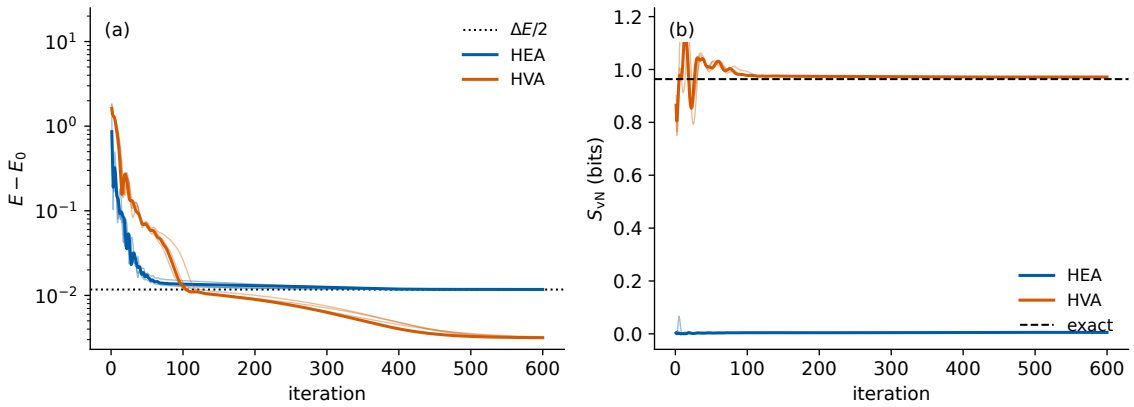


Figure 26: VQE on the ordered TFIM, $N = 6$, $h = 0.5J$, shown as in Figure 22. (a) Energy error, with the dotted line at $\Delta E/2$ of (6.5). (b) Half-chain entropy, with the exact value of almost one bit dashed. The HEA settles at the dotted line with zero entropy.

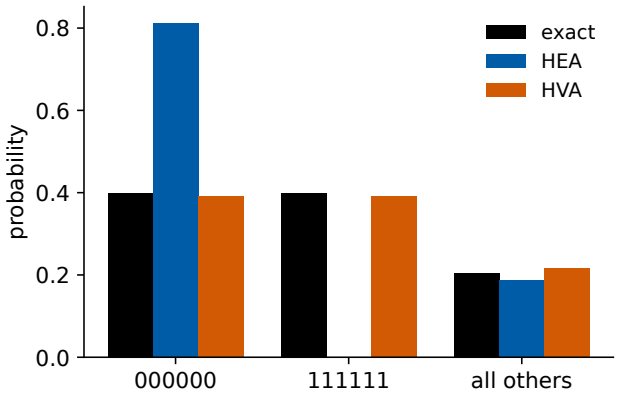


Figure 27: Probabilities of the bit strings 000000 and 111111 and of all other bit strings together in the exact ground state of the ordered TFIM and in the best trained HEA and HVA states, computed from the state vectors.

the two end spins, $N - 1$ bonds apart, and each layer of (5.5) extends the region influenced by a spin by one bond in each direction. On the ordered chain the HVA infidelity falls from about 4.8×10^{-1} at one layer to 4.1×10^{-3} at five, while the HEA stays near $1 - F \approx 1/2$ at every depth. The entanglement bound (5.10) allows one bit already at $L = 1$, so the required depth comes from the range of the correlations.

6.6 Excited states

Variational quantum deflation (VQD) finds excited states with the same circuits [30]. After a VQE run has produced an approximate ground state $|\tilde{\psi}_0\rangle$, a second run minimises

$$C(\boldsymbol{\theta}) = E(\boldsymbol{\theta}) + \lambda\, |\langle \tilde{\psi}_0 | \psi(\boldsymbol{\theta}) \rangle|^2, \tag{6.6}$$

where the penalty weight λ raises the cost of any overlap with the ground state. If $|\tilde{\psi}_0\rangle$ were exact, the cost of a trial state $\sum_m c_m |E_m\rangle$ would be $\sum_m |c_m|^2 E_m + \lambda |c_0|^2$, the energy of a Hamiltonian whose ground level is moved up (Figure 28) to $E_0 + \lambda$. For $\lambda > \Delta E$ its minimum is E_1, reached at the first excited state. On a device the overlap is estimated by applying the trial circuit followed by the inverse of the ground-state circuit and counting the outcome $0 \ldots 0$, since

$$|\langle 0 \ldots 0 | \hat{U}(\tilde{\boldsymbol{\theta}}_0)^\dagger \hat{U}(\boldsymbol{\theta}) | 0 \ldots 0 \rangle|^2 = |\langle \tilde{\psi}_0 | \psi(\boldsymbol{\theta}) \rangle|^2. \tag{6.7}$$

We apply VQD to the critical TFIM with the HEA, its best trained ground state as $|\tilde{\psi}_0\rangle$ and $\lambda = 2J$, larger than the gap. The first excited state is odd under $\hat{\Pi}$, so the HVA, confined to even

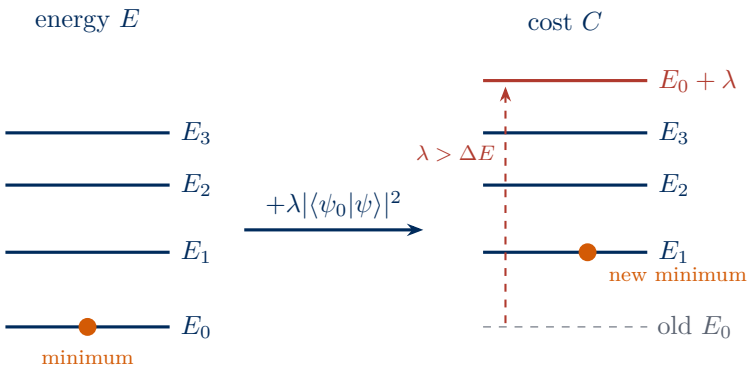


Figure 28: Variational quantum deflation (6.6). The overlap penalty $\lambda|\langle\tilde{\psi}_0|\psi\rangle|^2$ lifts the ground level to $E_0 + \lambda$, so for $\lambda > \Delta E$ the lowest level of the penalised cost is the first excited state E_1.

states, cannot reach it. Figure 29 shows that the energy first drops below E_1 and stays on a plateau where the trial state still overlaps with $|\tilde{\psi}_0\rangle$, since by (1.4) an energy below E_1 requires weight on the ground state. Once this overlap is removed, the energy settles on E_1 with an infidelity below 10^{-3}, and the gap obtained as the difference of the VQD and VQE energies agrees with the exact gap to within $10^{-3}J$. Errors of $|\tilde{\psi}_0\rangle$ propagate into the excited state, because the penalty acts on the approximate ground state and leaves a small admixture of the exact one unpenalised.

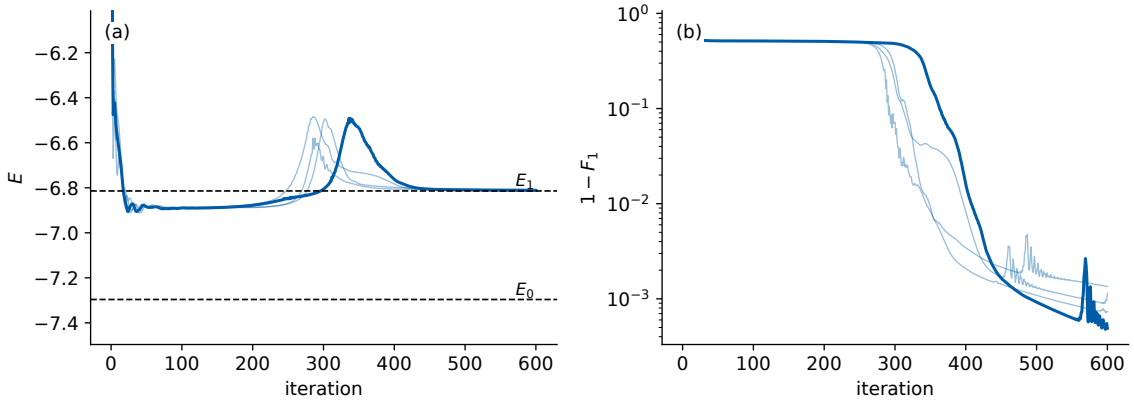


Figure 29: Variational quantum deflation (6.6) for the first excited state of the critical TFIM with the HEA. (a) Energy against the training step, with the exact E_0 and E_1 dashed. (b) Infidelity $1 - F_1$ with the exact first excited state, $F_1 = |\langle\psi_1|\psi\rangle|^2$. Exact state-vector simulation.

6.7 Comparison of the ansätze

Table 2 collects the best restart of every run together with the hardware cost of one gradient by (4.9), $2n_{\mathrm{rot}}$ shifted circuits each measured in G settings, which gives $2n_{\mathrm{rot}}G$ circuit runs of M shots each. In our runs (four restarts, 600 Adam steps), at equal depth and equal two-qubit gate count, the ansätze built for their Hamiltonian need fewer angles and reach better states. The SPA prepares the Heisenberg ground state to numerical precision, and the HVA, with 12 angles, reaches the TFIM ground state at both fields. The HEA, with 84 angles, ends at a lower fidelity on every model and at $F \approx 1/2$ on the ordered chain. The cost of a hardware gradient counts gate occurrences, so the HVA with only 12 free angles still needs 264 circuits per gradient. These advantages are tied to the model, since the HVA is defined by one Hamiltonian and the SPA reaches only states with $\hat{Z}_{\mathrm{tot}} = 0$.

6.8 Phase diagram of the Ising chain

A ground state changes character across a phase diagram, and an ansatz that works in one region can fail in another. We train the HEA and the HVA on the TFIM with $N = 8$ qubits at 18 fields between

model	ansatz	n	n_{rot}	2-qubit gates	circuits	δE	F	δS (bits)
TFIM, $h = J$	HEA	84	84	30	336	3.4×10^{-4}	0.9994	1.1×10^{-2}
TFIM, $h = J$	HVA	12	66	30	264	4.6×10^{-5}	0.9999	4.4×10^{-3}
TFIM, $h = 0.5J$	HEA	84	84	30	336	2.1×10^{-3}	0.5093	9.6×10^{-1}
TFIM, $h = 0.5J$	HVA	12	66	30	264	5.7×10^{-4}	0.9988	8.0×10^{-3}
Heisenberg	HEA	84	84	30	504	7.3×10^{-4}	0.9992	7.6×10^{-3}
Heisenberg	SPA	60	90	30	540	1.3×10^{-9}	1.0000	4.1×10^{-6}

Table 2: Comparison of the ansätze on chains of six qubits with $L = 6$ layers. The columns give the number of angles n, the number of parametrised rotations n_{rot}, the two-qubit gate count, the number $2n_{\mathrm{rot}}G$ of circuit runs (shifted circuits times settings) for one parameter-shift gradient with $G = 2$ settings for the TFIM and $G = 3$ for the Heisenberg chain, and for the best of four restarts the relative energy error δE, the fidelity F and the entropy error δS. Exact state-vector simulation with exact gradients.

$h = 0.1J$ and $h = 2J$, with a denser grid near the critical field, and compare every trained state with exact diagonalisation. Both ansätze have $L = 8$ layers and 56 two-qubit gates, with $n = 144$ angles for the HEA and 16 for the HVA, and every field is trained independently from the starts used above. Besides the energy per site and the entropy (6.1) we record two magnetisations. The transverse magnetisation

$$m_x = \frac{1}{N} \sum_{i=0}^{N-1} \langle \hat{X}_i \rangle \tag{6.8}$$

measures the alignment with the field. The order along z is harder to measure, because $\hat{\Pi} \hat{Z}_i \hat{\Pi} = -\hat{Z}_i$ implies $\langle \hat{Z}_i \rangle = 0$ in every state of definite parity, so the magnetisation vanishes in the exact ground state of a finite chain on both sides of the transition. We therefore use the parity-invariant order parameter

$$m_z = \sqrt{\Big\langle \Big(\frac{1}{N} \sum_{i=0}^{N-1} \hat{Z}_i \Big)^2 \Big\rangle}, \tag{6.9}$$

which tends to one in the ordered phase and falls to a value of order $1/\sqrt{N}$ in the paramagnet. It takes nearly the same value in the cat state (6.4) and in either of its symmetry-broken branches, since $\big(\sum_i \hat{Z}_i \big)^2$ is diagonal and has no matrix elements between the two branches.

In Figure 30 the energy per site and the transverse magnetisation of both ansätze follow the exact curves on the scale of the figure, and the relative energy error stays below a few per cent at every field. The other panels separate the two ansätze. Deep in the ordered phase the HEA has the smallest energy errors of the whole scan, 1.3×10^{-6} at $h = 0.2J$, while its fidelity is 0.500 and its entropy zero. It sits in a symmetry-broken branch, and its energy error shrinks with h together with the splitting of the doublet, in line with (6.5). Its order parameter agrees with the exact one, because m_z takes nearly the same value in a branch and in the cat state. Its fidelity stays near 1/2 up to $h = 0.6J$, rises through a crossover, remains below 0.99 up to $h = 0.85J$ and exceeds 0.99 from $h = 0.9J$ upward.

The HVA keeps even parity at every field and follows the exact entropy through the transition. It reaches $F > 0.99$ at every field from $h = 0.6J$ upward. For $h \leq 0.4J$ its fidelity lies between 0.92 and 0.96, its entropy overshoots the exact one bit, and its energy error exceeds that of the trapped HEA. The same eight-layer circuit reaches $F > 0.99$ from $h = 0.6J$, so the deficit is confined to the deep ordered phase, where the circuit must build the long-range correlations of the cat state. Near the critical point the HVA reaches fidelities above 0.99 and the exact entropy, whereas the entropy error of the HEA grows as the field approaches the critical point from the ordered side. A phase diagram drawn from the energy alone would therefore hide the failure of the HEA in the ordered phase, which the entropy and the fidelity reveal.

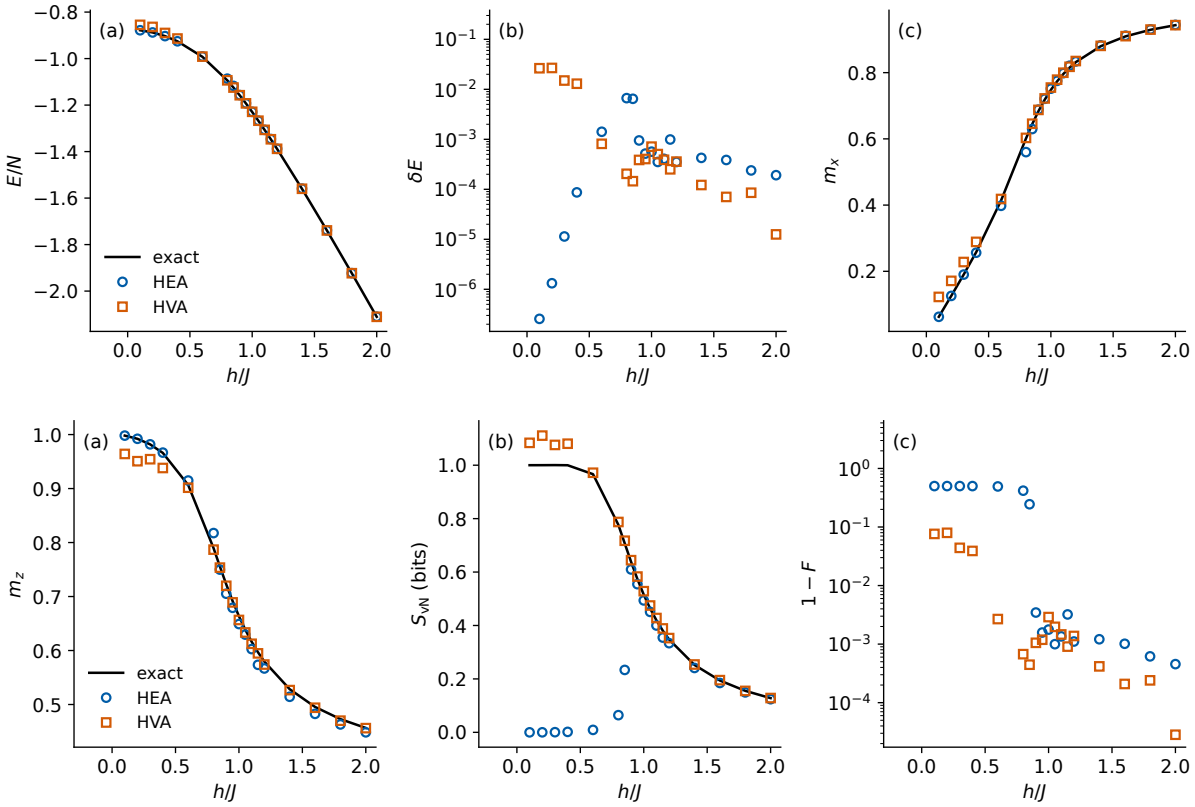


Figure 30: Variational phase diagram of the TFIM with $N = 8$ against the field h/J. Top row, the energy per site, the relative energy error δE and the transverse magnetisation (6.8). Bottom row, the order parameter (6.9), the half-chain entropy and the infidelity $1 - F$. Black lines are exact diagonalisation, blue circles the HEA and orange squares the HVA, each the best of two restarts. Exact state-vector simulation. Compare δE with $1 - F$ in the ordered phase, where the ranking of the two ansätze by energy is the reverse of their ranking by fidelity.

6.9 The XXZ chain in a transverse field

To test the ansätze on ground states of very different character, we add a magnetic field along x to the XXZ chain,

$$\hat{\mathcal{H}} = J \sum_{i=0}^{N-2} \left(\hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1} + \Delta\, \hat{Z}_i \hat{Z}_{i+1} \right) - h_x \sum_{i=0}^{N-1} \hat{X}_i, \tag{6.10}$$

on an open chain of $N = 8$ qubits with $J = 1$. The field term flips single bits and breaks the conservation of $\hat{Z}_{\text{tot}}$, while (6.10) still commutes with the parity (5.1), because $\hat{\Pi}$ commutes with each of $\hat{X}\hat{X}$, $\hat{Y}\hat{Y}$, $\hat{Z}\hat{Z}$ and $\hat{X}$. Every eigenstate therefore has parity ± 1. A transverse field is chosen because it mixes the magnetisation sectors, so the ground state builds its entanglement from superpositions across them. A longitudinal field $-h_z \sum_i \hat{Z}_i$ would commute with $\hat{Z}_{\text{tot}}$ and only move the ground state from one sector to another in steps.

On eight sites the phases of the infinite chain appear as smooth crossovers, recognisable in the bit strings that dominate the ground state. For $\Delta < -1$ the coupling favours aligned spins along z, a ferromagnet dominated by 00000000 and 11111111. For $-1 < \Delta < 1$ and small h_x the exchange term dominates and the chain is in the critical XY region, with weight spread over many bit strings with nearly equal numbers of zeros and ones and correlations that decay as powers of the distance. For $\Delta > 1$ the coupling favours anti-aligned neighbours, the Néel antiferromagnet dominated by 01010101 and 10101010. A large h_x turns the spins towards $+x$. At the largest field of the grid, $h_x = 2.5J$, the transverse magnetisation reaches 0.85 for $\Delta < 0$ and stays at or below 0.5 for $\Delta > 0$, so the fully polarised regime on the antiferromagnetic side lies beyond the grid. Since $\langle \hat{Z}_i \rangle = 0$ in every parity

eigenstate, the two kinds of order are measured by m_z of (6.9) and by the staggered magnetisation

$$m_{\mathrm{stag}} = \sqrt{\left\langle \left(\frac{1}{N} \sum_{i=0}^{N-1} (-1)^i \hat{Z}_i \right)^2 \right\rangle}, \tag{6.11}$$

which equals one in a Néel state and zero in a ferromagnet. Figure 31 summarises the four regimes.

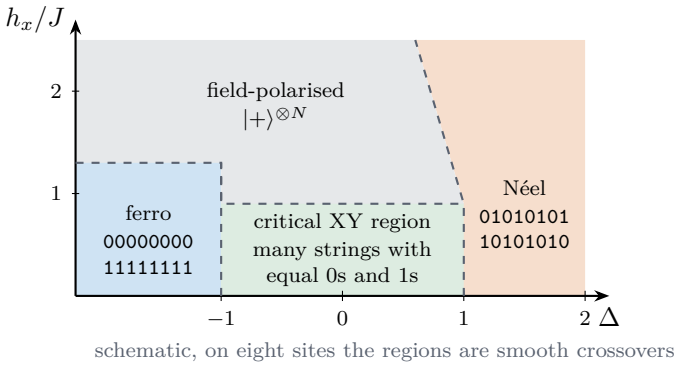


Figure 31: The regimes of the XXZ chain in a transverse field (6.10) with their dominant bit strings, ferromagnet ($\Delta < -1$), critical XY region ($-1 < \Delta < 1$), Néel antiferromagnet ($\Delta > 1$) and the field-polarised state at large h_x. Use it as a legend for the maps of Figure 32.

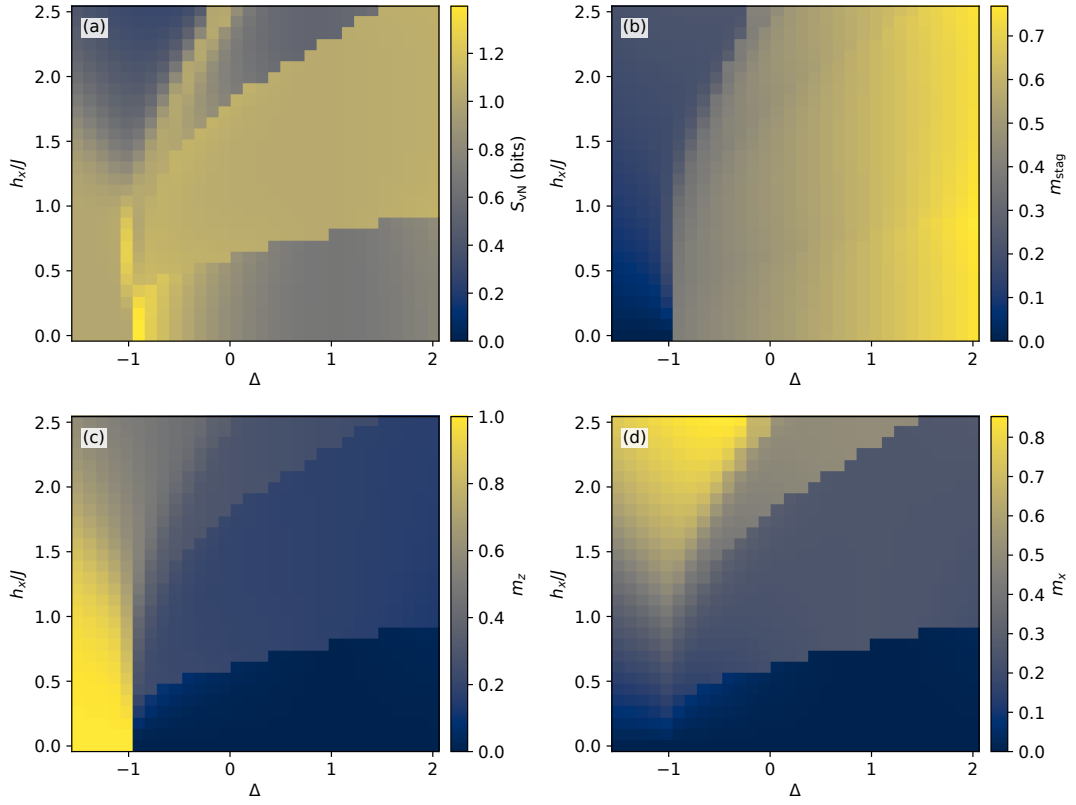


Figure 32: Exact ground state of the XXZ chain in a transverse field (6.10), $N = 8$, on a 30×30 grid of the anisotropy Δ and the field h_x/J. (a) Half-chain entropy, (b) staggered magnetisation (6.11), (c) order parameter m_z (6.9), (d) transverse magnetisation (6.8). Look for the ferromagnet in the lower left corner and for the steps on the antiferromagnetic side.

The exact maps (Figure 32) come from a Lanczos diagonalisation [31] that applies the Hamiltonian to a vector without storing its matrix. In the ferromagnetic corner $m_z \approx 1$ and the entropy is one bit, as for a cat state, with the even and odd levels nearly degenerate. The staggered magnetisation

grows with Δ from the XY region near $\Delta = 0$ and stays below one at $\Delta = 2$, because quantum fluctuations reduce the Néel order of eight spins. Towards large h_x on the ferromagnetic side the transverse magnetisation rises to 0.85. On the antiferromagnetic side the field changes the ground state in steps, at crossings between the lowest even and odd levels, which cover a wedge of odd ground states on 44% of the grid. They are finite-size level crossings of the eight-site chain, whose positions move with N.

We train two ansätze with $L = 6$ layers and 42 two-qubit gates each on a 20×20 grid. The HEA has $n = 112$ angles. The second circuit is built from the terms of (6.10) with an independent angle per gate, and we call it the *HVA with independent angles*. A layer applies the SPA gate (5.7) on the even and then on the odd bonds and finally a field rotation $\hat{R}_x$ on every qubit, $n = 132$ angles in total. Without the field rotations it is the SPA, which conserves $\hat{Z}_{\mathrm{tot}}$, and the field rotations reduce the conserved quantity to the parity. The circuit therefore stays in the parity sector of its initial state. Any product of $N/2$ singlets has the parity $(-1)^{N/2} = +1$, so we train it from the singlet product (even) and from the same product with the first pair replaced by the triplet $(|01\rangle + |10\rangle)/\sqrt{2}$, that is with $\hat{Z}_0$ applied to it (odd), and keep the lower energy. Both ansätze get two starts per grid point and 300 Adam steps.

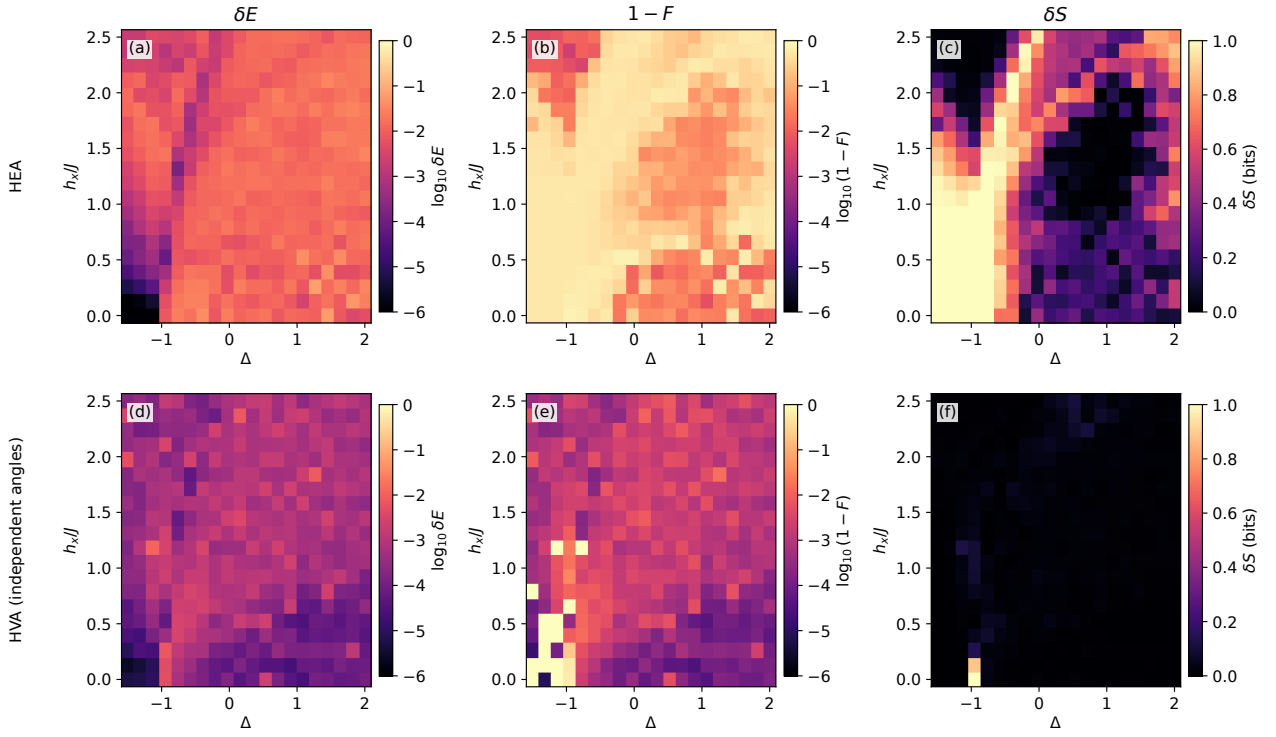


Figure 33: Errors of the trained states on the (Δ, h_x) grid for the HEA (top row) and the HVA with independent angles (bottom row). Left, $\log_{10}$ of the relative energy error δE. Middle, $\log_{10}$ of the infidelity $1 - F$. Right, the entropy error δS in bits. Dark colours mark small errors. Exact state-vector simulation with exact gradients.

The HVA with independent angles reaches $F > 0.99$ at 94% of the grid points (Figure 33), also in the odd wedge, because it is trained in both sectors. It fails in the ferromagnetic corner and along $\Delta \approx -1$, where the even and odd levels are closer than its energy error and the lower trained energy can belong to the wrong sector. The fidelity with the chosen exact state then drops to zero while the energy and the entropy stay correct. Near $\Delta = -1$ at small h_x the circuit also fails within the correct sector, with fidelities between 0.4 and 0.8 and entropy errors above one bit. The HEA reaches $F > 0.99$ at only 5% of the points. Deep in the ferromagnet it has the smallest energy errors of the map together with $F = 1/2$ and an entropy error of one bit, the symmetry-broken branch of Section 6.5. Elsewhere its energy error stays near 10^{-2} and its entropy lies below the exact value, so with this budget it builds too little of the entanglement of the critical and antiferromagnetic ground

states. At 34 of the 400 points the HEA has $\delta E < 10^{-3}$ and $F < 0.6$, so a small energy error does not certify the trained state.

7 Barren plateaus

7.1 Gradients at random angles

On a device every gradient component is estimated from a finite number of shots, and by (4.10) it is useful only when it exceeds the shot noise. We estimate how large a gradient component typically is at the start of a training run, when the angles are drawn at random, and how this size changes with the number of qubits. Along a single angle every cost of the form $C(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{O}|\psi(\boldsymbol{\theta})\rangle$, with a Hermitian observable $\hat{O}$, is a cosine (4.4), $a + b\cos(\theta_k - c)$, with constants that depend on the other angles. Its derivative $-b\sin(\theta_k - c)$ averages to zero over θ_k for any values of the other angles, so the mean gradient over uniformly random angles vanishes for every circuit. The typical size of a component is therefore measured by its variance,

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \big\langle (\partial_k C)^2 \big\rangle_{\boldsymbol{\theta}}, \tag{7.1}$$

where $\langle\cdot\rangle_{\boldsymbol{\theta}}$ is the average over the uniform distribution of the angles and $\partial_k C = \partial C/\partial\theta_k$. When this variance decays exponentially with N, the landscape is flat almost everywhere, and this flatness is called a *barren plateau* [32] (Figure 34).

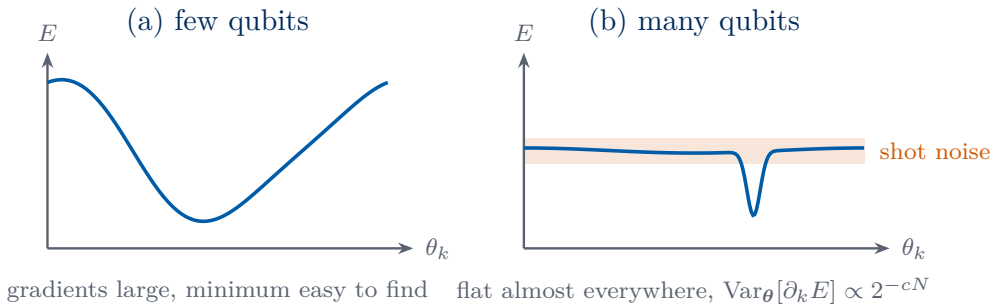


Figure 34: A barren plateau. For a small register the cost landscape has slopes that the shot noise does not hide. For a large register and a deep random circuit the landscape is flat almost everywhere, with the variance of the gradient decaying exponentially with the number of qubits, and the minimum lies in a narrow region.

7.2 Concentration in random states

A circuit with many layers of randomly chosen gates behaves, for averages of expectation values and of their squares, like a unitary drawn uniformly from all unitaries on the $d = 2^N$ dimensional state space. The uniform distribution over unitaries is the Haar distribution, and a set of circuits that reproduces its averages up to second order is a unitary 2-design [33]. A Haar-random unitary applied to $|0\ldots0\rangle$ gives a random state.

In a random state every expectation value is close to its average over all states. For a Pauli string $\hat{P} \neq \hat{I}$ the average of $\langle\hat{P}\rangle$ vanishes and its variance is

$$\mathrm{Var}\langle\hat{P}\rangle = \frac{1}{d+1} \approx 2^{-N}. \tag{7.2}$$

Figure 35 confirms (7.2) with sampled random states. For $\hat{Z}_0$ the scaling is easy to see, since $\langle\hat{Z}_0\rangle$ is the total probability of the bit strings with $b_0 = 0$ minus that of the strings with $b_0 = 1$, two sums of $d/2$ random weights that each fluctuate around one half by about $2^{-N/2}$. Two random states are also nearly orthogonal, $\langle|\langle 0\ldots0|\psi\rangle|^2\rangle = 1/d$, so the probability of any fixed bit string is of order 2^{-N}. A

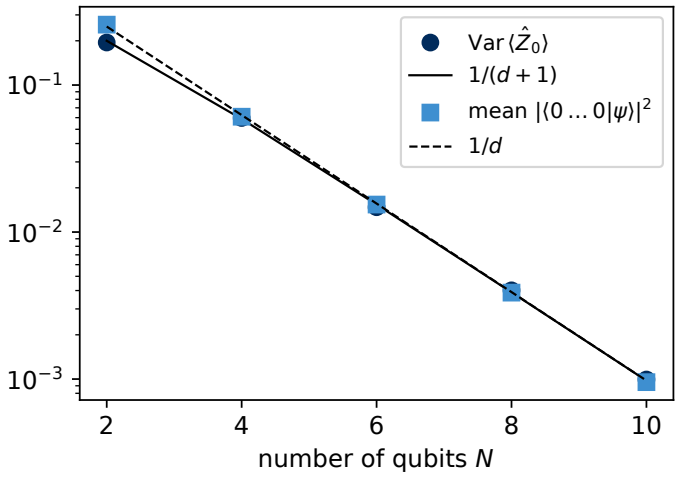


Figure 35: Concentration in random states of N qubits. The variance of $\langle \hat{Z}_0 \rangle$ and the mean overlap $|\langle 0 \ldots 0|\psi\rangle|^2$ over random states (markers) follow $1/(d+1)$ and $1/d$ with $d = 2^N$ (lines) (7.2). Sampled state vectors.

cost that is an expectation value in a nearly random state therefore carries almost no information about the angles.

The gradient inherits this concentration when the differentiated rotation acts first on $|0 \ldots 0\rangle$ and a Haar-random unitary $\hat{W}$ represents all gates after it. The gradient variance of the cost $C = \langle \hat{O} \rangle$ is then [32]

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \frac{\mathrm{Tr}(\hat{O}_0^2)}{2(d^2-1)}, \tag{7.3}$$

where $\hat{O}_0 = \hat{O} - \mathrm{Tr}(\hat{O})\hat{I}/d$ is the traceless part of $\hat{O}$. The formula follows from the Haar average $\langle \mathrm{Tr}(\hat{W}^\dagger \hat{A} \hat{W} \hat{B})^2 \rangle_{\hat{W}} = \mathrm{Tr}(\hat{A}^2)\mathrm{Tr}(\hat{B}^2)/(d^2-1)$ for traceless Hermitian $\hat{A}$ and $\hat{B}$, with $\hat{A} = \hat{O}_0$ and $\hat{B}$ the traceless operator produced by differentiating the rotation, which has $\mathrm{Tr}(\hat{B}^2) = 1/2$. The decay is therefore set by $\mathrm{Tr}(\hat{O}_0^2)$, which we compare for three costs,

$$C_G(\boldsymbol{\theta}) = 1 - |\langle 0 \ldots 0|\psi(\boldsymbol{\theta})\rangle|^2, \qquad C_L(\boldsymbol{\theta}) = \frac{1}{N}\sum_{q=0}^{N-1} \frac{1 - \langle \hat{Z}_q \rangle}{2}, \qquad E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta})|\hat{\mathcal{H}}_{\mathrm{TFIM}}|\psi(\boldsymbol{\theta})\rangle. \tag{7.4}$$

The *global cost* C_G is the probability that a measured bit string differs from $0 \ldots 0$ and depends on all qubits together. The *local cost* C_L is the probability that a single qubit, averaged over the qubits, reads one. Both vanish only in $|0 \ldots 0\rangle$, so they have the same minimum, whereas C_L is a sum of single-qubit terms. The energy is a sum of terms on one or two neighbouring qubits, so it is a local cost as well. For the global cost $\hat{O} = \hat{I} - |0 \ldots 0\rangle\langle 0 \ldots 0|$ and $\mathrm{Tr}(\hat{O}_0^2) = 1 - 1/d$, so its variance $1/[2d(d+1)] \approx 2^{-2N-1}$ decays as 4^{-N}. A sum of Pauli strings $\sum_j c_j \hat{P}_j$ has $\mathrm{Tr}(\hat{O}_0^2) = d\sum_j c_j^2$, so the energy has Var $\approx [(N-1)J^2 + Nh^2]/2^{N+1}$ and the local cost Var $\approx 2^{-N}/(8N)$. For a circuit that acts as a 2-design every cost decays exponentially with N, the global one twice as fast.

7.3 Global and local costs

Figure 36 shows the variance of the derivative with respect to the first angle of the HEA, estimated from 300 random angle vectors for $N = 2, \ldots, 10$. The global cost decays exponentially at depth $L = N$, with a fitted decay Var $\propto 2^{-cN}$ with $c = 1.96$, close to the 2-design value two. A shallow circuit gives the global cost no protection, since at $L = 2$ it decays at a similar rate, $c = 1.81$. The global cost is the probability of a single bit string out of 2^N, exponentially small for almost all angles, whatever the depth. This dependence on the cost was proved for layered circuits of local 2-designs [34], and the HEA shows the same behaviour.

The local cost at $L = 2$ decays slowly because of a light cone. The differentiated angle rotates qubit 0 in the first block, and each $\widehat{\mathrm{CZ}}$ layer passes its influence on by one qubit. After two layers it

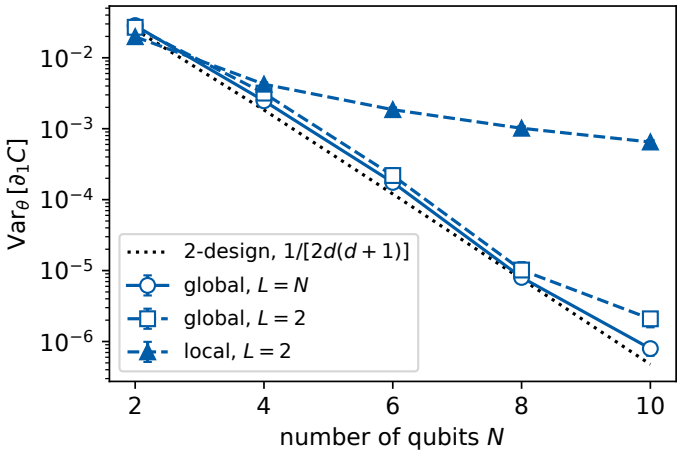


Figure 36: Variance of the derivative of the global cost C_G and the local cost C_L (7.4) with respect to the first angle of the HEA, over 300 uniformly random angle vectors, against the number of qubits, for the depths $L = N$ and $L = 2$. The dotted line is the 2-design value $1/[2d(d+1)]$ of the global cost. Exact gradients from the state vector.

reaches only qubits 0, 1 and 2, so only $\langle \hat{Z}_0 \rangle$, $\langle \hat{Z}_1 \rangle$ and $\langle \hat{Z}_2 \rangle$ depend on it, and these three expectation values depend only on the angles of the first few qubits. For $N \geq 5$ the distribution of their derivative no longer changes with N, and the prefactor $1/N$ of C_L gives

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_1 C_L] = \frac{\mathrm{const}}{N^2} \qquad (L = 2,\ N \geq 5), \tag{7.5}$$

a power law in N. At $N = 10$ the local gradient variance is 311 times larger than the global one of the same circuit. For local costs and depths growing as $\log N$ the variance decays at most polynomially [34], whereas circuits deep enough to form 2-designs bring back the exponential decay (7.3) for local costs as well.

7.4 The energy gradient of structured ansätze

For the energy of the TFIM, the cost of VQE, the gradient variance of the HEA at depth $L = N$ decays far more slowly than the 2-design value at the sizes studied here and lies more than an order of magnitude above it at $N = 10$ (Figure 37). Each term of the energy is affected only by the gates inside its light cone, and at depth N the circuit has only just connected the two ends of the chain and is still far from a random unitary. Once the circuit is deep enough to form a 2-design, the decay approaches (7.3).

The gradient variance of the HVA grows with N. Each of its angles multiplies a sum over all bonds or all qubits, so one angle changes all $2N - 1$ terms of the energy together, and the derivative collects contributions from the whole chain. The HVA also never approaches a random unitary. Its gates commute with the parity and generate only a small family of unitaries, whose dimension grows polynomially with N, so the states it prepares fill a small part of the 2^N dimensional space. Mild or absent barren plateaus for this ansatz were reported in Ref. [27]. In general the gradient variance is set by the dimension of the set of operators generated by the gates of a circuit, its dynamical Lie algebra [35, 36].

7.5 Shots needed to resolve a gradient

A barren plateau is a problem of measurement cost, because a component estimated by the parameter-shift rule from two energies with M shots per setting has the statistical variance $\sigma^2/(2M)$ by (4.10), where σ^2 is the single-shot variance of the energy. The typical gradient becomes visible above the

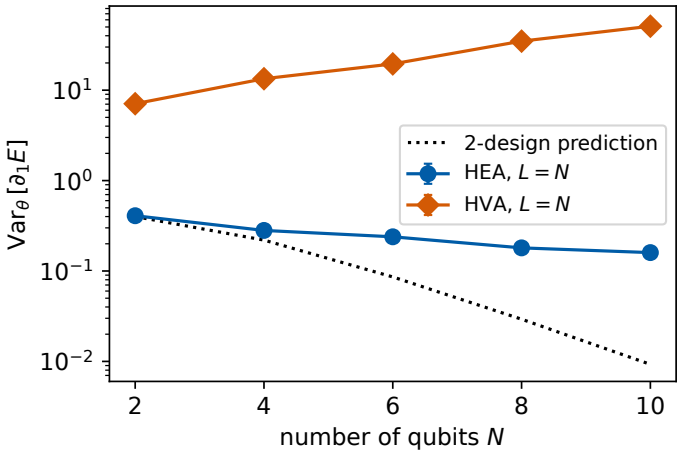


Figure 37: Variance of the derivative of the TFIM energy at $h = J$ over random angles, for the HEA with respect to its first angle and for the HVA with respect to γ_1, both at depth $L = N$, against the number of qubits. The dotted line is the 2-design value of (7.3). Exact gradients.

shot noise when this variance equals (7.1), which requires

$$M^\star = \frac{\sigma^2}{2\ \mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E]} \tag{7.6}$$

shots per setting and per shifted circuit. In a random state $\sigma^2 = d\sum_j c_j^2/(d+1) \approx \sum_j c_j^2$, and with (7.3) the 2-design value is $M^\star = d - 1 = 2^N - 1$, so every added qubit doubles the number of shots for one gradient component. For an angle shared by gates with factors c_g, the chain rule (4.8) measures each term separately and their variances add with the weights c_g^2, so the numerator of (7.6) becomes $\sigma^2 \sum_g c_g^2$.

At $N = 10$ (Figure 38) a typical energy gradient of the deep HEA is resolved with about 58 shots per setting, while a 2-design would need about 1023. For the 2-design the count reaches about 10^6 at $N = 20$ and 10^{15} at $N = 50$, for a single component of a single gradient. The HVA angle γ_1 is resolved with about 6.7 shots per setting in each of its 18 shifted circuits at $N = 10$. A better optimiser cannot remove a barren plateau, because every optimiser on a device has to resolve the gradient, or differences of energies, above the shot noise, and (7.6) sets the cost of doing so.

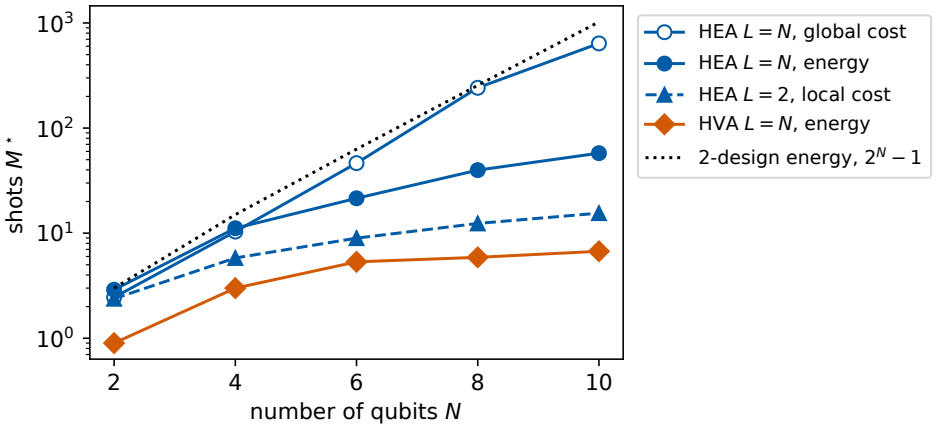


Figure 38: Shots per setting and per shifted circuit needed to resolve a typical energy gradient (7.6) at random angles, against the number of qubits, for the HEA at depth $L = N$ with the global cost and with the energy, for the HEA at $L = 2$ with the local cost, for the HVA angle γ_1 with the energy, and the 2-design value $M^\star = 2^N - 1$ (dotted). Measured gradient variances from exact state vectors.

7.6 Noise and mitigation

Noise on a device produces a second kind of plateau. Depolarising noise after each layer replaces part of the state by the maximally mixed state, on which every traceless observable has expectation value zero, so the whole cost landscape contracts towards a constant by a factor that decays exponentially with the depth [37]. For circuits whose depth grows linearly with N the gradient then vanishes exponentially in N, independently of the random-unitary argument and also for structured ansätze. The remedies within the circuit are a smaller depth and lower error rates.

The known ways of keeping the gradient measurable change the cost, the circuit or the starting point. Local costs decay at most polynomially for circuits of logarithmic depth [34]. Shallow circuits keep the light cone of each gate small and limit the effect of noise. Structured ansätze that respect the symmetries of the problem, such as the HVA, prepare states in a small part of the state space and keep large gradients [27, 35]. Initial angles chosen so that the circuit starts as the identity [38], and circuits trained layer by layer [39], start the training away from the random region. All of these measures restrict the set of states that the circuit explores. A circuit that can prepare almost any state has exponentially small gradients, so expressibility and trainability have to be balanced [40].

8 Circuits as probability models

8.1 Generative modelling

Up to this point a circuit prepared a state whose energy was of interest. Measured in the computational basis, the same circuit produces random bit strings whose distribution can itself be the object of interest, which turns the circuit into a machine-learning model.

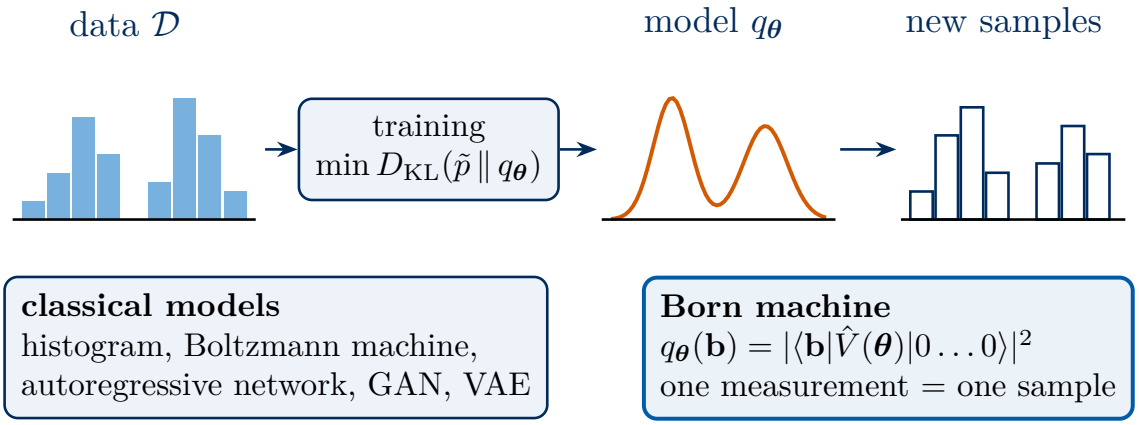


Figure 39: Generative modelling. Samples from an unknown distribution p form the data set $\mathcal{D}$, a model $q_{\boldsymbol{\theta}}$ is trained on them, and the trained model produces new samples that resemble the data.

We consider an unknown distribution p over a space of outcomes and a data set

$$\mathcal{D} = \{x^{(1)}, x^{(2)}, \ldots, x^{(|\mathcal{D}|)}\} \tag{8.1}$$

of $|\mathcal{D}|$ independent samples drawn from it. The learner sees the samples and never the distribution p itself. A *generative model* is a family of distributions $q_{\boldsymbol{\theta}}$ labelled by parameters $\boldsymbol{\theta}$, from which new samples can be drawn (Figure 39). Training adjusts $\boldsymbol{\theta}$ so that $q_{\boldsymbol{\theta}}$ comes close to p in a measure of distance between distributions, and a trained model produces new samples that resemble the data without copying them. Among the classical families, a histogram of the data can be sampled by picking a bin with the stored frequency. A Boltzmann machine assigns the probability $e^{-E_{\boldsymbol{\theta}}(x)}/Z$ through an energy function with trainable couplings and a normalisation Z. An autoregressive network writes the probability of a vector as a product of conditional probabilities, one component after another, and generative adversarial networks and variational autoencoders transform random noise into samples

with a neural network. The quantum model of this section addresses the same task and differs in how the probabilities arise.

8.2 Data as bit strings

A register of N qubits returns bit strings $\mathbf{b} = b_0 b_1 \dots b_{N-1}$, so the data must be written as bit strings before a circuit can model them. Discrete data such as spin configurations or binary images are bit strings already. A continuous variable $x \in [x_{\min}, x_{\max}]$ is first discretised by cutting the interval into 2^N bins of equal width

$$w = \frac{x_{\max} - x_{\min}}{2^N} \tag{8.2}$$

and number them from left to right by the bin index

$$k(x) = \left\lfloor \frac{x - x_{\min}}{w} \right\rfloor \in \{0, 1, \dots, 2^N - 1\}, \tag{8.3}$$

where $\lfloor \cdot \rfloor$ rounds down to an integer and the right end $x = x_{\max}$ is assigned to the last bin. The index written in binary,

$$k = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q}, \tag{8.4}$$

is the bit string $\mathbf{b}$, with b_0 the most significant bit, the same ordering as the amplitudes of the register. Every value of x in bin k therefore corresponds to the computational basis state $|\mathbf{b}\rangle$, and every measured bit string corresponds back to a bin. When a value of x is needed, we take the centre of the bin,

$$x_k = x_{\min} + \left(k + \frac{1}{2}\right) w. \tag{8.5}$$

For $N = 3$ and the interval $[-4, 4]$ the width is $w = 1$, and there are eight bins labelled $000, 001, \dots, 111$ (Figure 40). The value $x = -0.37$ lies in the bin $[-1, 0)$ with index $k = \lfloor 3.63 \rfloor = 3$, and $3 = 0 \cdot 4 + 1 \cdot 2 + 1 \cdot 1$ is the bit string 011. The value belongs to the basis state $|011\rangle$, and a measurement that returns 011 is read as the bin centre $x_3 = -0.5$. The discretisation loses the position of x inside its bin, an error of at most $w/2$. Each added qubit doubles the number of bins and halves w, so the resolution improves exponentially with the number of qubits.

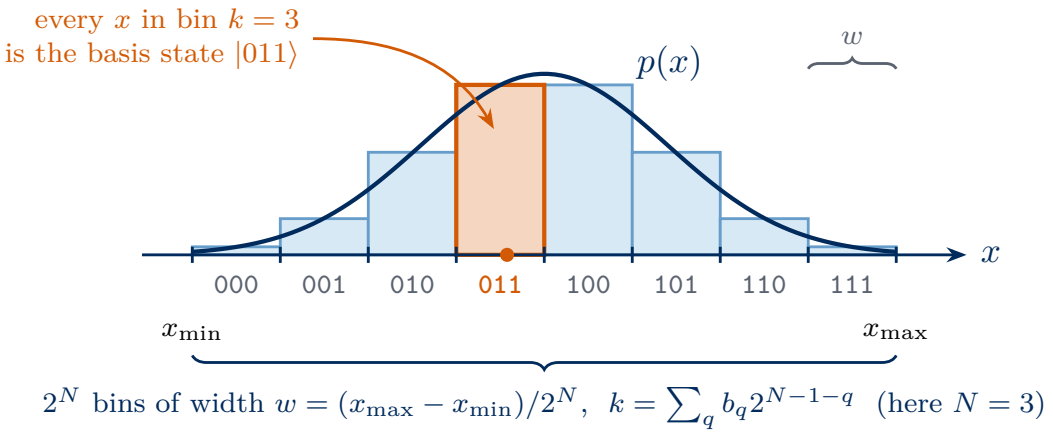


Figure 40: Continuous data as bit strings. The interval $[x_{\min}, x_{\max}]$ is cut into 2^N bins, here $N = 3$, and the binary form (8.4) of the bin index is the bit string of a computational basis state. A value in bin 3 corresponds to $|011\rangle$, and the bin probabilities of the distribution $p(x)$ become a distribution over bit strings.

The data define the empirical distribution over bit strings,

$$\tilde{p}(\mathbf{b}) = \frac{n(\mathbf{b})}{|\mathcal{D}|}, \tag{8.6}$$

where $n(\mathbf{b})$ counts how often the bit string $\mathbf{b}$ occurs in the data set, and the tilde marks an estimate obtained from samples. As a running example we take a mixture of two Gaussian peaks on $x \in [-4, 4]$,

$$p(x) \propto 0.4\, e^{-(x+1.6)^2/(2\sigma^2)} + 0.6\, e^{-(x-1.6)^2/(2\sigma^2)}, \qquad \sigma = 0.6, \tag{8.7}$$

discretised on $N = 6$ qubits into 64 bins of width $w = 0.125$, and a data set of $|\mathcal{D}| = 3000$ values drawn from it (Figure 41). The bin probabilities $p(\mathbf{b})$ are known only because we chose the target, and they serve to judge how well a trained model generalises beyond the finite data. The two peaks make the bits strongly correlated. The left peak lies around bin 19, bit string 010011, and the right peak around bin 44, bit string 101100, so the leading bits of a value near one peak are the complements of those near the other.

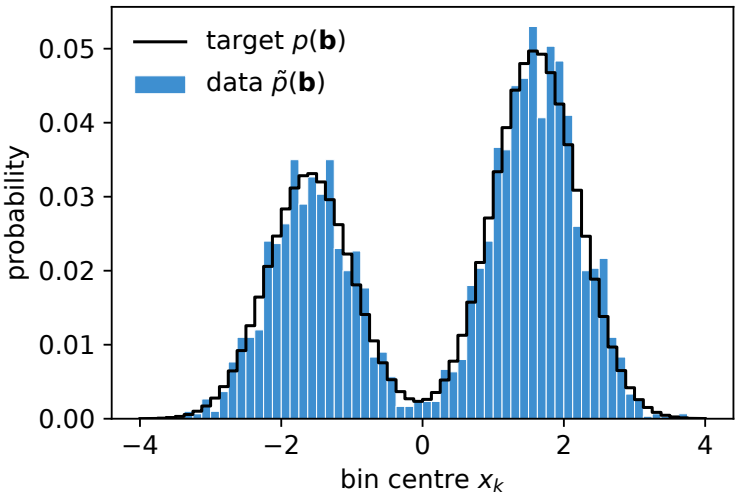


Figure 41: The two-peak target (8.7) on 64 bins (line) and the empirical distribution (8.6) of 3000 samples (bars). Some bins in the tails received no sample. The total variation distance (8.13) between the two is 0.065.

8.3 The Born machine

A circuit $\hat{V}(\boldsymbol{\theta})$ acts on the register prepared in $|0\rangle^{\otimes N}$ and produces the state $|\psi(\boldsymbol{\theta})\rangle = \hat{V}(\boldsymbol{\theta})|0\rangle^{\otimes N}$. A measurement of every qubit in the computational basis returns the bit string $\mathbf{b}$ with the Born-rule probability

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \left|\langle \mathbf{b}|\hat{V}(\boldsymbol{\theta})|0\rangle^{\otimes N}\right|^2. \tag{8.8}$$

These probabilities are non-negative and sum to one for every $\boldsymbol{\theta}$, because the state is normalised, so (8.8) is a family of distributions over the 2^N bit strings (Figure 42). A model whose probabilities are squared amplitudes is called a *Born machine*. The idea was introduced with matrix product states as the source of the amplitudes [41] and compared with Boltzmann machines in Ref. [42]. With a parametrised circuit as the source, the model is the quantum circuit Born machine [43, 44], whose sampling hardness and training were studied in Ref. [45].

Sampling needs no further computation, since one run of the circuit followed by a measurement of all qubits draws one bit string from $q_{\boldsymbol{\theta}}$ and, through (8.5), one value of x. On a device $q_{\boldsymbol{\theta}}(\mathbf{b})$ is accessible only as a frequency $\tilde{q}(\mathbf{b}) = n(\mathbf{b})/M$ over M shots, and a bit string of small probability may not appear at all. A simulation stores all 2^N amplitudes and gives the exact $q_{\boldsymbol{\theta}}$. The circuit is the same object as in VQE, where the bit strings serve only to estimate an energy and the goal is the state that minimises it. In a Born machine the distribution of the bit strings is the goal, and the samples themselves are the output.

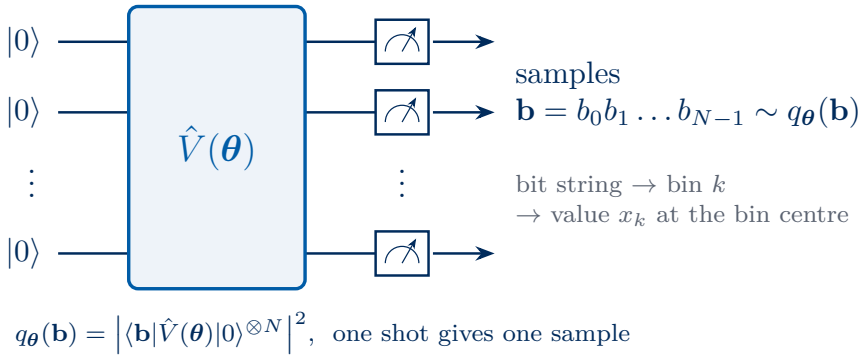


Figure 42: A quantum circuit Born machine. A parametrised circuit $\hat{V}(\boldsymbol{\theta})$ acts on $|0\rangle^{\otimes N}$, and each measurement of all qubits returns one sample $\mathbf{b} \sim q_{\boldsymbol{\theta}}(\mathbf{b})$ of the model (8.8).

8.4 The loss function

Training needs a number that measures how far $q_{\boldsymbol{\theta}}$ is from the data. The Kullback–Leibler (KL) divergence [46] of q from p is

$$D_{\mathrm{KL}}(p \,\|\, q_{\boldsymbol{\theta}}) = \sum_{\mathbf{b}} p(\mathbf{b}) \, \ln \frac{p(\mathbf{b})}{q_{\boldsymbol{\theta}}(\mathbf{b})}, \tag{8.9}$$

with the convention $0 \ln 0 = 0$, measured in nats because of the natural logarithm. It is non-negative and vanishes only for $q_{\boldsymbol{\theta}} = p$, which follows from $\ln u \leq u - 1$ applied to $u = q_{\boldsymbol{\theta}}/p$. It is asymmetric in its arguments, and the weight $p(\mathbf{b})$ makes it most sensitive to the bit strings that are frequent under p.

With the empirical distribution (8.6) in place of p, the KL divergence becomes the negative log-likelihood of the data, the standard loss of statistics. Splitting the logarithm gives

$$D_{\mathrm{KL}}(\tilde{p} \,\|\, q_{\boldsymbol{\theta}}) = \underbrace{\sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln \tilde{p}(\mathbf{b})}_{-H(\tilde{p})} \; \underbrace{- \sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln q_{\boldsymbol{\theta}}(\mathbf{b})}_{\text{depends on } \boldsymbol{\theta}} . \tag{8.10}$$

The first sum is minus the Shannon entropy $H(\tilde{p})$ of the data, in nats, which does not depend on $\boldsymbol{\theta}$. In the second sum each bit string occurs with the weight $n(\mathbf{b})/|\mathcal{D}|$, so the sum over bit strings is a sum over the data points,

$$-\sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln q_{\boldsymbol{\theta}}(\mathbf{b}) = -\frac{1}{|\mathcal{D}|} \sum_{\mathbf{b}} n(\mathbf{b}) \ln q_{\boldsymbol{\theta}}(\mathbf{b}) = -\frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \ln q_{\boldsymbol{\theta}}(\mathbf{b}^{(i)}) \equiv \mathcal{L}_{\mathrm{NLL}}(\boldsymbol{\theta}), \tag{8.11}$$

where $\mathbf{b}^{(i)}$ is the bit string of the data point $x^{(i)}$. Therefore

$$D_{\mathrm{KL}}(\tilde{p} \,\|\, q_{\boldsymbol{\theta}}) = \mathcal{L}_{\mathrm{NLL}}(\boldsymbol{\theta}) - H(\tilde{p}), \tag{8.12}$$

and minimising the KL divergence to the data and maximising the likelihood of the data are the same task with the same gradient.

The loss (8.11) contains $\ln q_{\boldsymbol{\theta}}(\mathbf{b})$ for every bit string in the data. If the model gives a data bit string a vanishing probability, the loss diverges, so training starts from a model that covers all bit strings, for example with every qubit near $|+\rangle$, where $q_{\boldsymbol{\theta}}$ is close to the uniform distribution 2^{-N}. In a simulation the exact $q_{\boldsymbol{\theta}}$ is available and the loss and its gradient follow by automatic differentiation. On a device the loss is costly, because it needs the probability of each data bit string, which can only be estimated from shot frequencies, and a zero frequency makes the estimate of $\ln q$ infinite. The gradient follows from the parameter-shift rule (4.7), since $q_{\boldsymbol{\theta}}(\mathbf{b})$ is the expectation value of the projector $|\mathbf{b}\rangle\langle\mathbf{b}|$ and depends on each angle as a cosine.

A sampled model is judged by a quantity that stays finite when some bins receive no shot. The total variation distance

$$d_{\mathrm{TV}}(p, q) = \frac{1}{2} \sum_{\mathbf{b}} |p(\mathbf{b}) - q(\mathbf{b})| \tag{8.13}$$

is the largest difference between the probabilities that the two distributions assign to any set of bit strings. It lies between 0 and 1 and can be computed directly from shot frequencies.

8.5 Product circuits and entangling circuits

The circuit we use alternates rotation blocks $\hat{B}_\ell = \bigotimes_q \hat{R}_z(\phi_{\ell q})\hat{R}_y(\vartheta_{\ell q})$ with chains of $\widehat{\mathrm{CZ}}$ gates,

$$\hat{V}(\boldsymbol{\theta}) = \hat{B}_L\,\hat{C}_Z \cdots \hat{B}_1\,\hat{C}_Z\,\hat{B}_0, \tag{8.14}$$

the layout of the HEA (5.3) with $n = 2N(L+1)$ angles. Without entangling gates, $L = 0$, the circuit prepares a product state. Each qubit q ends in $\cos(\vartheta_q/2)|0\rangle + e^{i\phi_q}\sin(\vartheta_q/2)|1\rangle$ up to a global phase, the amplitude of a bit string is a product of one-qubit amplitudes, and the Born probability factorises,

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \prod_{q=0}^{N-1} q_q(b_q), \qquad q_q(0) = \cos^2\frac{\vartheta_q}{2}, \quad q_q(1) = \sin^2\frac{\vartheta_q}{2}. \tag{8.15}$$

Every bit is an independent biased coin, and the phases ϕ_q have no effect on the outcomes. The best product model of a distribution p follows from (8.9), because the logarithm of a product splits into a sum over qubits. Summing $p(\mathbf{b})$ over all bits except b_q gives the marginal $p_q(b_q)$, the probability that bit q takes the value b_q,

$$D_{\mathrm{KL}}\Big(p \,\Big\|\, \prod_q q_q\Big) = -H(p) - \sum_{q=0}^{N-1} \sum_{b_q \in \{0,1\}} p_q(b_q)\ln q_q(b_q). \tag{8.16}$$

Each inner sum is smallest for $q_q = p_q$, by the non-negativity of the KL divergence between two one-bit distributions. The best product model is therefore the product of the marginals, and no training of the angles goes below the value of (8.16) at $q_q = p_q$. For the two-peak target the marginals mix the bits of the two peaks, and their product places probability on bit strings that combine the leading bits of one peak with the trailing bits of the other, between and outside the peaks. For the empirical distribution $\tilde{p}$ of the data set, which is the distribution the training sees, the product bound is 0.500 nats.

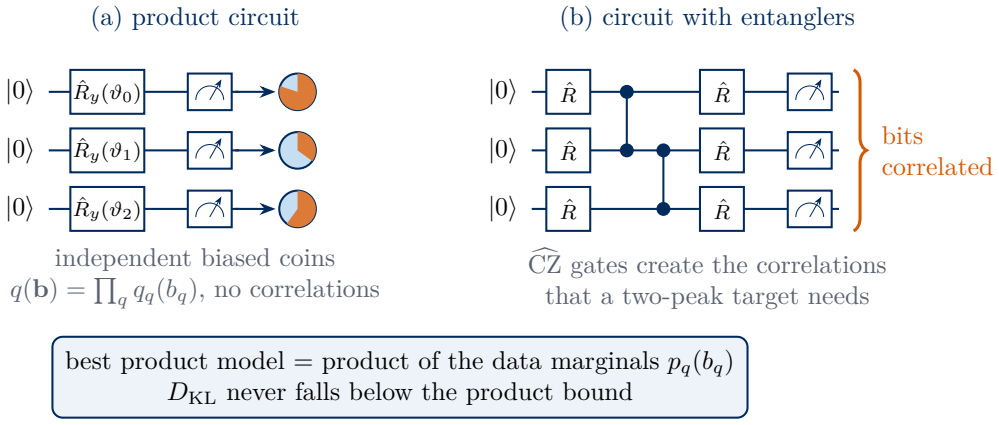


Figure 43: Product and entangling circuits as probability models. Without entangling gates every bit is an independent biased coin and the model is a product of one-bit distributions (8.15). Entangling gates between rotation layers correlate the bits, which a two-peak target requires.

Entangling layers make the amplitudes, and therefore the bits, correlated (Figure 43). We train circuits of depth $L = 0, \ldots, 5$ by minimising the KL divergence to the data with exact probabilities and automatic differentiation, from three random starts near the uniform distribution each, with Adam (Figure 44). The product circuit converges to the bound (8.16). One entangling layer lowers the KL divergence from 0.5 to below 0.1, and both divergences fall quickly up to $L = 2$. Beyond it the loss on the data keeps decreasing by an order of magnitude, while the divergence from the target

stays near 0.0137. The deeper circuits start to reproduce the statistical fluctuations of the 3000 data values, including empty bins, which are not part of p. Fitting such fluctuations is called overfitting, and the divergence from a target that the training never sees detects it. In practice p is unknown, and a part of the data set held back from training plays the role of the target.

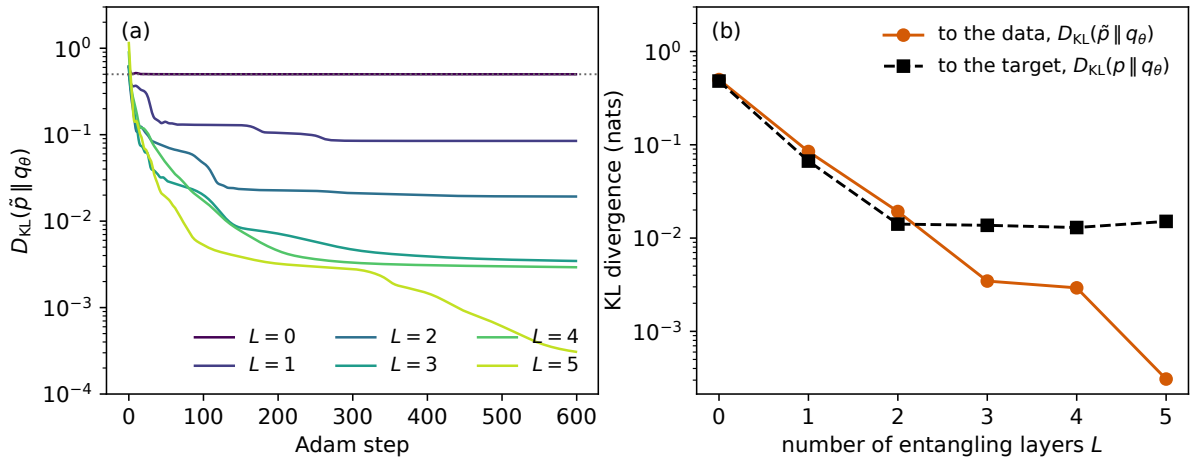


Figure 44: Born machines of depth $L = 0, \ldots, 5$ trained on the two-peak data. (a) KL divergence to the data against the Adam step for the best start at each depth, with the product bound (8.16) dotted. (b) Final KL divergence against the number of entangling layers, to the data (circles) and to the exact target (squares). Exact probabilities from the state vector.

The saturation comes from the finite data set. With a second data set ten times larger, whose empirical distribution lies at $d_{\mathrm{TV}} = 0.015$ from the target, the divergence of the trained circuits from the target drops to 0.0019 at $L = 3$ and 0.0014 at $L = 5$, whereas the $L = 2$ circuit improves less, because its error comes mostly from the limited circuit. In Figure 45 the product model spreads probability over the whole interval and misses both peaks, one entangling layer places both peaks at the right positions with distorted shapes, and three layers follow the target in the peaks and in the tails.

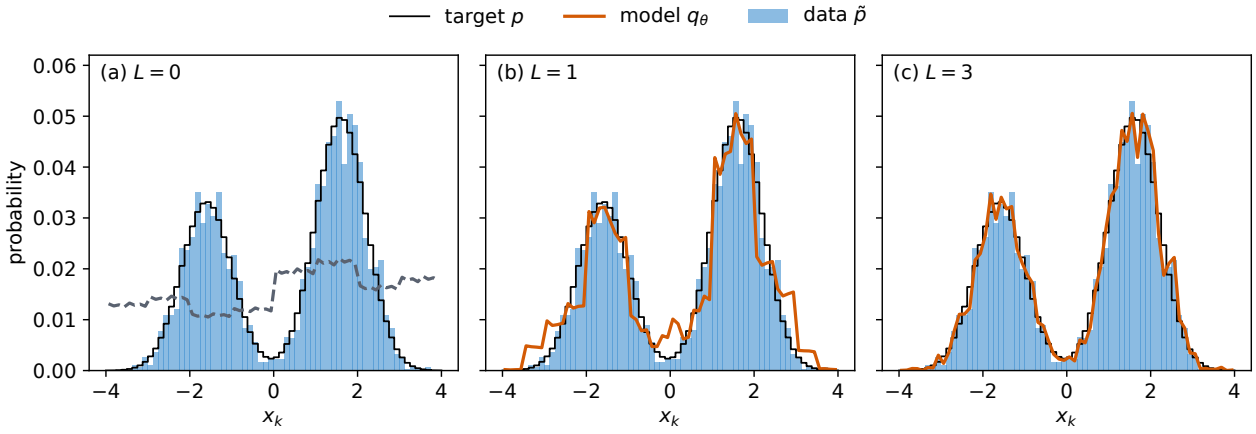


Figure 45: Trained model distributions $q_{\boldsymbol{\theta}}$ (coloured lines) against the bin centres x_k, with the exact target p (black steps) and the empirical distribution $\tilde{p}$ of the data (bars), for (a) the product circuit, $L = 0$, (b) one entangling layer and (c) three entangling layers. Exact model probabilities.

On a device the trained model is used and tested through its samples. We measure the $L = 3$ model 10000 times in simulation and convert the bit strings to bin frequencies and values of x (Figure 46). The sampled histogram reproduces the two peaks with their weights and widths, and its total variation distance from the target, 0.051, is close to the value 0.055 of the exact model probabilities. The KL divergence from samples would be infinite, since 2 bins with nonzero target probability received no shot.

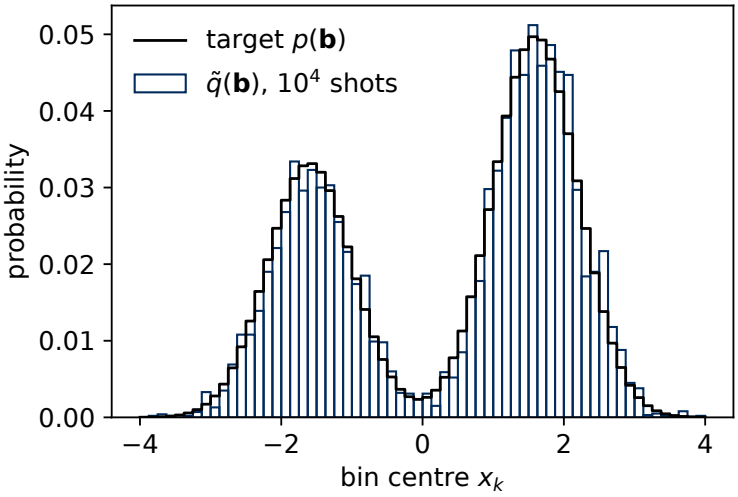


Figure 46: Frequencies of 10000 bit strings sampled from the trained $L = 3$ Born machine, read as bin centres (8.5) (bars), and the exact target (line). Simulated measurement record.

8.6 Trainability and ancilla qubits

The Born machine meets the same balance of expressibility and trainability as VQE. Deep circuits with every angle trainable are expressive, but their gradients vanish as in Section 7. One remedy fixes the entangling part of each layer and trains only single-qubit rotations, which keeps the number of trainable angles small and the gradients of the rotations measurable. A pure state of N qubits also limits the model, since every distribution it represents is a squared amplitude. Ancilla qubits that take part in the circuit but are not measured lift this restriction. Ignoring their outcomes sums the Born probabilities over their bits $\mathbf{a}$,

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \sum_{\mathbf{a}} \left|\langle \mathbf{b}, \mathbf{a}|\psi(\boldsymbol{\theta})\rangle\right|^2 = \langle \mathbf{b}|\hat{\rho}_{\text{meas}}|\mathbf{b}\rangle, \tag{8.17}$$

where $\hat{\rho}_{\text{meas}}$ is the reduced density matrix of the measured qubits (Figure 47). The measured qubits thus follow a mixture of Born distributions, at the cost of extra qubits.

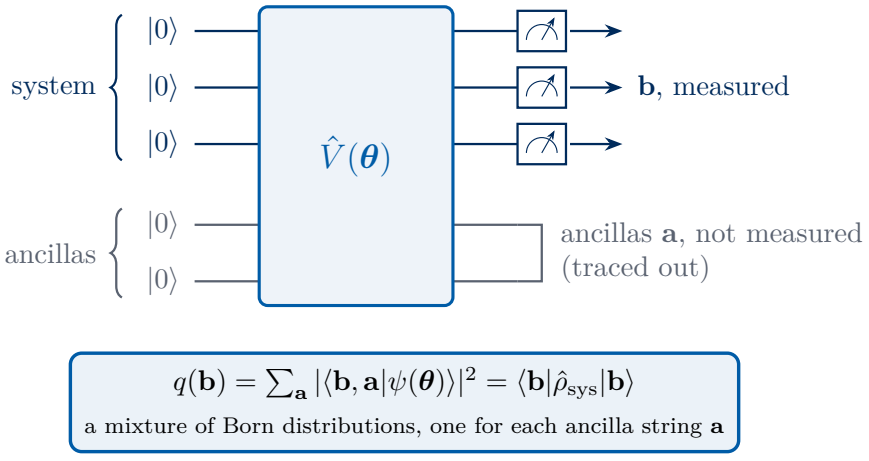


Figure 47: Ancilla qubits in a Born machine. The circuit acts on measured and ancilla qubits, and only the measured qubits are read out. Summing over the unread ancilla bits (8.17) makes the model a mixture of Born distributions.

The quantum scrambling Born machine [47] is one construction of this kind. Each of its layers applies trainable rotations of every qubit around a fixed, non-trainable entangling unitary that spreads correlations over the whole register. Ref. [47] considers three choices of this unitary, a Haar-random unitary, a finite-depth brickwork of random two-qubit gates, and evolution for a fixed time under a spin-chain Hamiltonian with nearest-neighbour couplings. Once the fixed unitary creates entanglement

close to that of a random state, the trained models reach similar accuracy for all three choices, and only single-qubit angles need to be optimised. A variant also trains the couplings and fields of the spin-chain Hamiltonian. Figure 48 reproduces a published result for a two-dimensional distribution, in which two registers encode the two variables x and y by the binning of Section 8.2 and ancilla qubits are traced out as in (8.17).

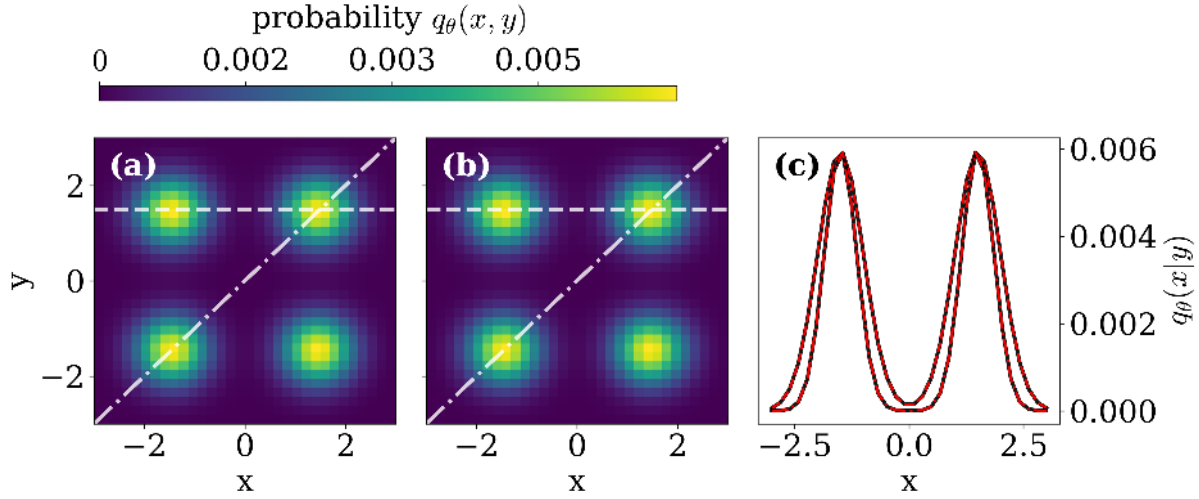


Figure 48: Published result of the quantum scrambling Born machine, reproduced from Ref. [47] and not computed for these notes. (a) Target distribution $p(x, y)$, a mixture of four Gaussian peaks on a grid of bins. (b) Distribution learned by the variant with trainable Hamiltonian parameters. (c) Slices along the dashed and dash-dotted lines of (a) and (b), target in black and learned model in red. All four peaks are reproduced with their weights.

9 Combinatorial optimisation with QUBO and QAOA

9.1 Quadratic unconstrained binary optimisation

Many decisions are choices between two options made for many items at once. A delivery is assigned to one of two vans, a component to one of two chips, a vertex of a network to one of two groups. When the cost of a choice depends on pairs of items, the task is a quadratic unconstrained binary optimisation (QUBO) problem. It has N binary variables $x_i \in \{0, 1\}$, collected in $\mathbf{x} = (x_0, \ldots, x_{N-1})$, and a real $N \times N$ matrix Q, and it asks for the vector that minimises the quadratic cost (Figure 49)

$$C(\mathbf{x}) = \mathbf{x}^T Q\, \mathbf{x} = \sum_{i,j} Q_{ij}\, x_i x_j. \tag{9.1}$$

A binary variable satisfies $x_i^2 = x_i$, so the diagonal elements Q_{ii} act as linear costs, and every quadratic function of binary variables has the form (9.1) up to a constant. The word unconstrained means that all 2^N vectors are allowed, and constraints are added to the cost as penalty terms. Each variable is stored in one qubit, so N is also the number of qubits.

Checking every vector costs 2^N evaluations of C, about 10^{15} for $N = 50$. QUBO is NP-hard, so no known algorithm solves every instance in a time that grows polynomially with N, and practical solvers for large instances are heuristics. A greedy method flips one variable at a time as long as the cost decreases. Simulated annealing [48] also accepts some cost increases, with a probability that decreases during the run, so that it can leave local minima. For the small instances below, exhaustive search gives the exact answer against which the quantum algorithm is compared.

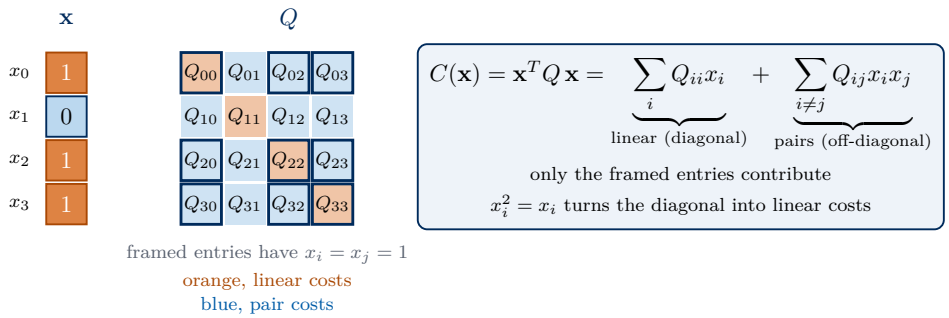


Figure 49: A QUBO problem. Each of the N binary variables is a yes-or-no choice, the matrix Q assigns linear costs on its diagonal and pair costs off the diagonal, and the answer is the bit string with the lowest cost (9.1) among 2^N candidates.

9.2 MaxCut

A graph consists of N vertices and a set E of edges, each joining a pair of vertices (i, j). We may think of the vertices as people and the edges as conflicts between them, and we must split the people into two rooms so that as many conflicting pairs as possible end up in different rooms. A split of the vertices into two groups is called a *cut*, because a line drawn between the groups cuts every edge that crosses it. We label the groups by one binary variable per vertex, $x_i = 0$ for the first group and $x_i = 1$ for the second, so a cut is a bit string $x_0 x_1 \ldots x_{N-1}$. The *size* of the cut is the number of edges whose two ends carry different labels, and the maximum-cut problem (MaxCut) asks for the bit string with the largest cut size.

On a square, four vertices $0, 1, 2, 3$ are joined in a ring by the edges $(0, 1)$, $(1, 2)$, $(2, 3)$ and $(3, 0)$ (Figure 50). The bit string 0000 puts all vertices in one group and cuts no edge. The string 1000 moves vertex 0 to the other group and cuts its two edges $(0, 1)$ and $(3, 0)$, a cut of size two. The string 1010 alternates the groups around the ring, every edge joins vertices of different groups, and all four edges are cut. No string can do better, so the maximum cut of the square is four, reached by 1010 and by its complement 0101. Exchanging the two groups never changes a cut, so optimal bit strings always come in complementary pairs. Around a triangle, a cycle of three vertices, the labels cannot alternate, and at most two of its three edges can be cut. Every cycle of odd length leaves at least one edge uncut.

For a square the answer is evident, whereas a general graph has 2^N candidates to compare. MaxCut is one of the problems shown to be NP-complete in its decision form in Ref. [49], and no algorithm is known that finds the optimum of every graph in a time that grows polynomially with N. Approximate algorithms are therefore judged by the *approximation ratio*, the cut they find divided by the maximum cut. The Goemans–Williamson algorithm, based on semidefinite programming, runs in polynomial time and guarantees an expected ratio of at least 0.878 on every graph [50]. MaxCut appears in the partitioning of networks and circuits, in clustering, where the edges carry the dissimilarities of items, and in physics as the ground-state problem of an Ising antiferromagnet, in which a cut edge is a satisfied bond.

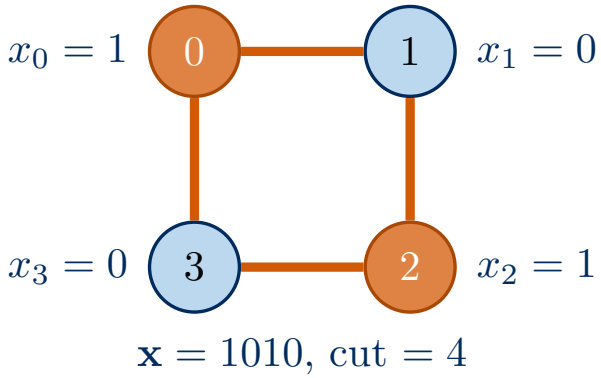
An edge (i, j) is cut when exactly one of x_i, x_j equals one, and the expression $x_i + x_j - 2x_i x_j$ equals one in that case and zero otherwise. The cut size is therefore

$$\mathrm{cut}(\mathbf{x}) = \sum_{(i,j)\in E} (x_i + x_j - 2x_i x_j), \tag{9.2}$$

and maximising the cut is minimising $C(\mathbf{x}) = -\mathrm{cut}(\mathbf{x})$, a QUBO with the matrix

$$Q_{ii} = -d_i, \qquad Q_{ij} = Q_{ji} = \begin{cases} 1 & (i, j) \in E \\ 0 & \text{otherwise} \end{cases} \quad (i \neq j), \tag{9.3}$$

where the degree d_i is the number of edges at vertex i. Each edge contributes $Q_{ij} + Q_{ji} = 2$ to the product term, and the linear terms x_i collect on the diagonal.



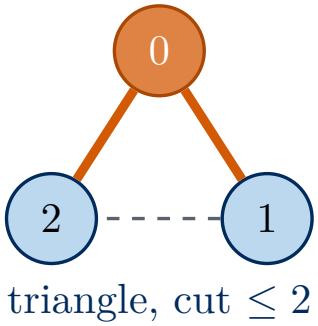


Figure 50: MaxCut on a square. The bit string 1010 puts vertices 0 and 2 in one group and vertices 1 and 3 in the other, and all four edges (highlighted) join vertices of different groups. Its complement 0101 gives the same cut.

9.3 The Ising Hamiltonian of a QUBO

A qubit measured in the computational basis gives the bit b_i, and its operator $\hat{Z}_i$ has the eigenvalue $z_i = 1 - 2b_i = \pm 1$. We identify each binary variable with the bit of one qubit, $x_i = b_i$, so that

$$x_i = \frac{1 - z_i}{2}. \tag{9.4}$$

Inserting (9.4) into (9.1) gives $C = \frac{1}{4}\sum_{i,j} Q_{ij}(1 - z_i - z_j + z_i z_j)$. The diagonal terms $i = j$ contain $z_i^2 = 1$ and add a constant, and collecting the constant, linear and quadratic terms gives

$$C = \underbrace{\frac{1}{4}\sum_{i,j} Q_{ij} + \frac{1}{4}\sum_{i} Q_{ii}}_{\text{constant } c_0} + \sum_i \underbrace{\Big(-\frac{1}{4}\sum_j (Q_{ij} + Q_{ji})\Big)}_{\text{field } h_i} z_i + \sum_{i<j} \underbrace{\frac{Q_{ij} + Q_{ji}}{4}}_{\text{coupling } J_{ij}} z_i z_j. \tag{9.5}$$

The result is the energy of classical Ising spins $z_i = \pm 1$ with couplings J_{ij}, fields h_i and a constant c_0. Replacing each spin by its Pauli operator gives the cost Hamiltonian

$$\hat{\mathcal{H}}_C = \sum_{i<j} J_{ij}\, \hat{Z}_i \hat{Z}_j + \sum_i h_i\, \hat{Z}_i + c_0\, \hat{I}. \tag{9.6}$$

It contains only $\hat{Z}$ operators and is therefore diagonal in the computational basis,

$$\hat{\mathcal{H}}_C |\mathbf{b}\rangle = C(\mathbf{b})\, |\mathbf{b}\rangle. \tag{9.7}$$

Every basis state $|\mathbf{b}\rangle$ is a candidate solution with its cost as the energy, and the ground state of $\hat{\mathcal{H}}_C$ is the optimal bit string (Figure 51). A degenerate ground level holds all optimal bit strings. For MaxCut the fields vanish, since $\sum_j (Q_{ij} + Q_{ji}) = -2d_i + 2d_i = 0$, and (9.5) becomes

$$\hat{\mathcal{H}}_C = \frac{1}{2} \sum_{(i,j)\in E} \hat{Z}_i \hat{Z}_j - \frac{|E|}{2}\, \hat{I}, \tag{9.8}$$

so a cut edge lowers the energy by one relative to an uncut edge. For the square, $\hat{\mathcal{H}}_C$ has the couplings $J_{ij} = 1/2$ on the four edges and $c_0 = -2$, and $|1010\rangle$ has the energy $4 \cdot (-\frac{1}{2}) - 2 = -4$, minus the maximum cut.

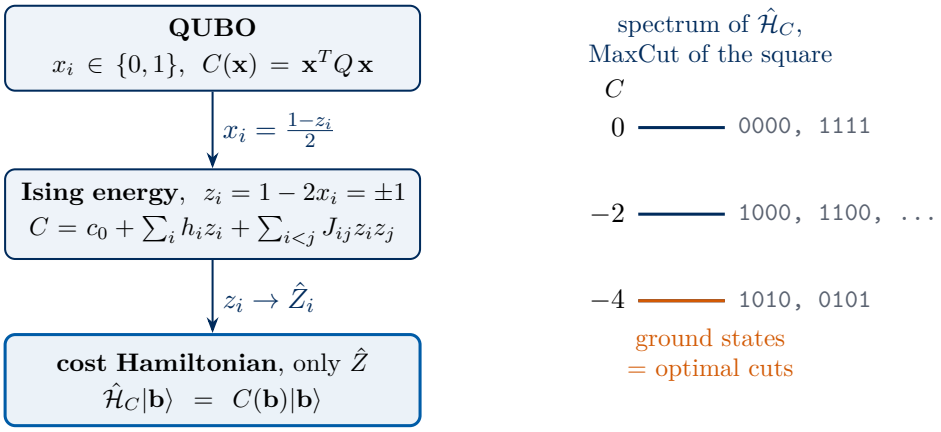


Figure 51: From a QUBO to a diagonal Ising Hamiltonian. Each binary variable becomes the bit of one qubit through $x_i = (1 - z_i)/2$, the quadratic cost becomes couplings J_{ij}, fields h_i and a constant, and every computational basis state is a candidate solution whose energy is its cost.

9.4 Answers from measured bit strings

A quantum algorithm for a QUBO prepares some state $|\psi\rangle = \sum_{\mathbf{b}} \psi_{\mathbf{b}} |\mathbf{b}\rangle$ and measures all qubits in the computational basis. Each shot returns one bit string $\mathbf{b}$ with probability $|\psi_{\mathbf{b}}|^2$, which is one candidate solution, and its cost $C(\mathbf{b})$ is evaluated on a classical computer in a time polynomial in N. From M shots we keep the best bit string. If the optimal bit strings together carry the probability P_{opt}, at least one of M independent shots is optimal with probability

$$P_{\text{hit}}(M) = 1 - (1 - P_{\text{opt}})^M, \tag{9.9}$$

and the number of shots for a chance of 95 per cent is $M_{95} = \lceil \ln 0.05 / \ln(1 - P_{\text{opt}}) \rceil$, about $3/P_{\text{opt}}$ for small P_{opt}. The state is useful when its weight on optimal or near-optimal bit strings is much larger than the uniform weight 2^{-N} of random guessing, which is what the uniform superposition $|+\rangle^{\otimes N}$ gives.

In VQE the same circuits and measurements give the ground-state energy, a number estimated as an average over many shots. Here the answer is one sampled bit string, whose cost is known exactly as soon as it is measured, and the average energy serves only as the training signal for the circuit.

As an example we take a graph of $N = 8$ vertices, a ring in which each vertex is also joined to the opposite vertex, with 12 edges and three neighbours per vertex (Figure 52). It contains cycles of odd length, such as 0–1–2–3–4–0, so no division cuts all edges. Exhaustive search over the 256 bit strings gives the maximum cut 10, reached by 8 bit strings in complementary pairs. A uniformly random bit string cuts each edge with probability one half, six edges on average, and is optimal with probability $8/256 = 1/32$, so about 95 random shots are needed for a 95 per cent chance of an optimum.

9.5 The QAOA circuit

The quantum approximate optimisation algorithm (QAOA) [51] prepares a state concentrated on low-cost bit strings with a circuit of p layers, where p plays the role of the depth L (Figure 53). It starts in $|+\rangle^{\otimes N}$, the uniform superposition of all candidates, and alternates two evolutions. The cost layer is the evolution under the cost Hamiltonian for a time γ_ℓ, and the mixing layer is the evolution under a transverse field for a time β_ℓ,

$$\hat{U}_C(\gamma_\ell) = e^{-i\gamma_\ell \hat{\mathcal{H}}_C}, \qquad \hat{U}_B(\beta_\ell) = e^{-i\beta_\ell \sum_i \hat{X}_i} = \bigotimes_{i=0}^{N-1} \hat{R}_x(2\beta_\ell). \tag{9.10}$$

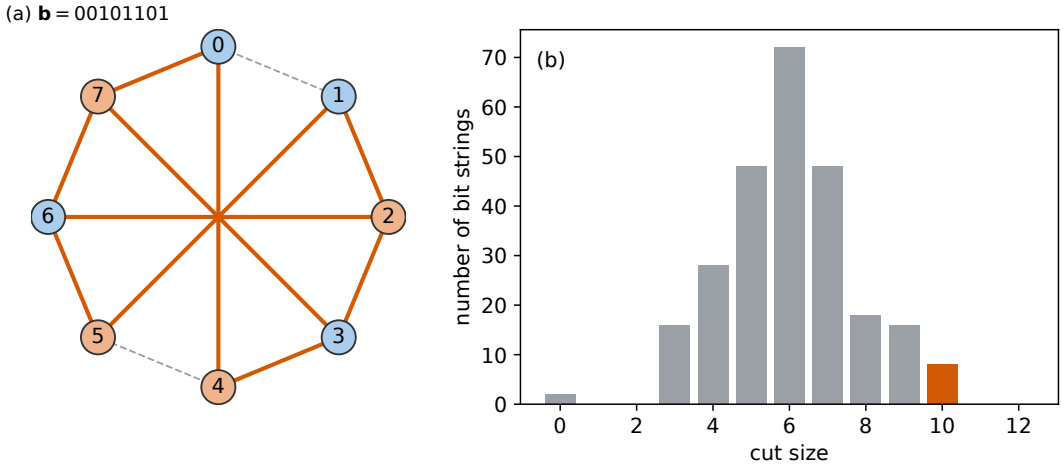


Figure 52: The MaxCut example. (a) The graph of eight vertices with the vertices coloured by the optimal bit string 00101101, blue for $b_i = 0$ and orange for $b_i = 1$, the ten cut edges solid and the two uncut edges dashed. (b) Number of bit strings for each cut size, with the eight optimal strings in orange. The optimum lies in the tail of the distribution.

The QAOA state with $2p$ angles $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$ is

$$|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = \hat{U}_B(\beta_p)\hat{U}_C(\gamma_p) \cdots \hat{U}_B(\beta_1)\hat{U}_C(\gamma_1)\, |+\rangle^{\otimes N}. \tag{9.11}$$

The cost layer is diagonal and multiplies the amplitude of each bit string by the phase $e^{-i\gamma_\ell C(\mathbf{b})}$, so bit strings of different cost acquire different phases. The mixing layer rotates every qubit and makes the amplitudes of bit strings that differ by single bit flips interfere. The cost layer alone leaves all probabilities unchanged, and the mixing layer alone leaves $|+\rangle^{\otimes N}$ unchanged, since it is an eigenstate of $\sum_i \hat{X}_i$. Only the alternation moves probability, and the angles are chosen so that the interference collects it on low-cost bit strings. The circuit has the structure of the HVA (5.5), with the cost Hamiltonian in place of the Ising couplings of the chain.

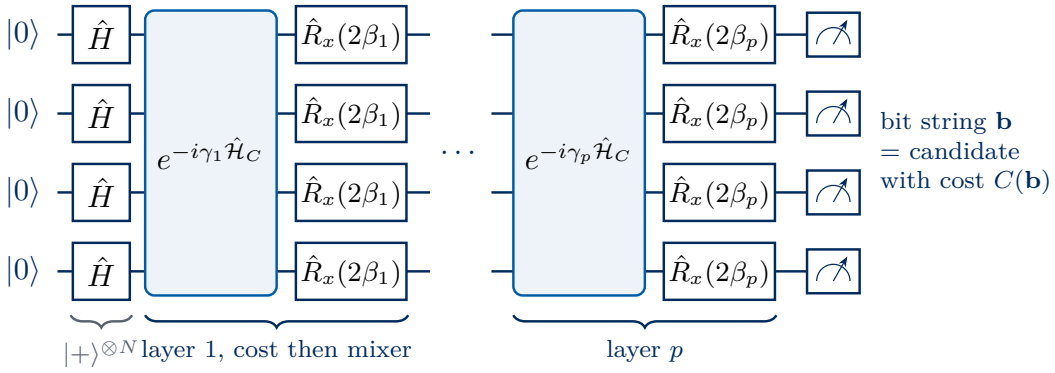


Figure 53: The QAOA circuit (9.11). Hadamard gates prepare $|+\rangle^{\otimes N}$, and each of the p layers applies the cost layer, compiled into $\hat{R}_{zz}$ and $\hat{R}_z$ gates, and the mixing layer of $\hat{R}_x(2\beta_\ell)$ rotations. All qubits are measured in the computational basis, and each shot is a candidate solution.

On a device the cost layer is compiled into gates, and since all terms of (9.6) commute,

$$e^{-i\gamma\hat{\mathcal{H}}_C} = e^{-i\gamma c_0} \prod_{i<j} \hat{R}_{zz}(2\gamma J_{ij}) \prod_i \hat{R}_z(2\gamma h_i), \tag{9.12}$$

with one $\hat{R}_{zz}$ gate per edge for MaxCut and a global phase that has no effect on any measurement. The trained quantity is the expected cost

$$E(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle \boldsymbol{\gamma}, \boldsymbol{\beta} | \hat{\mathcal{H}}_C | \boldsymbol{\gamma}, \boldsymbol{\beta} \rangle = \sum_{\mathbf{b}} \left| \langle \mathbf{b} | \boldsymbol{\gamma}, \boldsymbol{\beta} \rangle \right|^2 C(\mathbf{b}), \tag{9.13}$$

the average cost of the measured bit strings. Since $\hat{\mathcal{H}}_C$ is diagonal, it is estimated from the same computational-basis shots that deliver the candidates, with a single measurement setting. Each QAOA angle is shared by several gates. For MaxCut, γ_ℓ enters the 12 gates $\hat{R}_{zz}(2\gamma_\ell J_{ij})$ with $c_g = 2J_{ij} = 1$, and β_ℓ enters N gates $\hat{R}_x(2\beta_\ell)$ with $c_g = 2$. By (4.8) the two derivatives of one layer need $2 \times (12 + 8) = 40$ circuits on a device, and a full gradient $40p$ circuits.

9.6 Training and sampling

At $p = 1$ the expected cost depends on two angles and can be drawn as a map. The eigenvalues of $\hat{\mathcal{H}}_C$ are integers, so $\hat{U}_C(\gamma + 2\pi) = \hat{U}_C(\gamma)$. A shift $\beta \to \beta + \pi/2$ multiplies the mixing layer by $(-i)^N \hat{\Pi}$, which flips every bit and leaves every cut unchanged, and reversing both angles complex-conjugates the state. The rectangle $0 \le \gamma \le \pi$, $-\pi/4 \le \beta \le \pi/4$ therefore contains every value of E (Figure 54). At $\gamma = 0$ or $\beta = 0$ the state keeps the uniform distribution and the expected cut equals the random value of six. The best depth-one state cuts 8.31 edges on average.

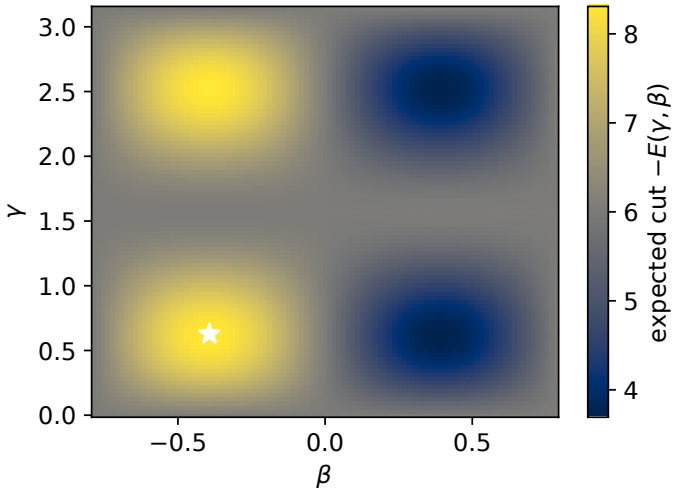


Figure 54: Expected cut $-E(\gamma, \beta)$ of the depth-one QAOA state (9.13) for the graph of Figure 52, on a grid of 101×101 angles, with the best grid point marked by a star. Exact state-vector values.

For $p > 1$ a grid is too expensive, since its size grows as the number of grid points per angle to the power $2p$. We minimise E with exact gradients and Adam from sixteen starting angle sets per depth, fifteen random and one linear ramp in which γ_ℓ grows and $|\beta_\ell|$ shrinks linearly with ℓ, and keep the best. We judge the trained state by the approximation ratio $r = -E/\mathrm{cut}_{\max}$, which compares the expected cut with the optimum and equals 0.6 for random guessing, by the probability P_{opt} of an optimal bit string and by M_{95}. Table 3 and Figure 55 show that each layer raises both measures. The probability of the optimum grows from 1/32 for random guessing to about one third at $p = 1$ and to 0.909 at $p = 4$, where the state puts almost all its weight on the maximum cut, and the number of shots needed to see an optimum falls from about a hundred to two. The landscape becomes harder to optimise with depth. Of the 16 starts, 16, 11, 3 and 1 reach the best expected cost within 10^{-3} at $p = 1$, 2, 3 and 4, so the $p = 4$ result rests on a single start.

In the deeper circuits (Figure 56) the trained cost angles γ_ℓ grow from the first layer to the last, and the mixing angles β_ℓ are negative and shrink in magnitude towards zero. The circuit starts with mostly mixing and ends with mostly cost evolution, the schedule of a slow interpolation from the mixing Hamiltonian to the cost Hamiltonian.

A device gives the distribution of Figure 55(b) only through shots, and Figure 57 shows how quickly an optimal bit string appears in a record of shots. We sample 1000 bit strings from the trained $p = 4$ state and the same number of uniformly random bit strings (Figure 58). The estimate of the expected cost from the shots, $\bar{C} = -9.870 \pm 0.016$, lies 2.5 standard errors from the exact value -9.830, a deviation that occurs in about one per cent of records, so the training signal is available

	expected cut $-E$	approximation ratio r	P_{opt}	M_{95}
random guessing	6.0	0.600	0.0312	95
$p = 1$	8.31	0.831	0.322	8
$p = 2$	9.25	0.925	0.522	5
$p = 3$	9.66	0.966	0.770	3
$p = 4$	9.83	0.983	0.909	2

Table 3: Trained QAOA states for MaxCut on the graph of Figure 52, with the maximum cut 10. Exact probabilities of the trained states. M_{95} is the number of shots for a 95 per cent chance of measuring an optimal bit string.

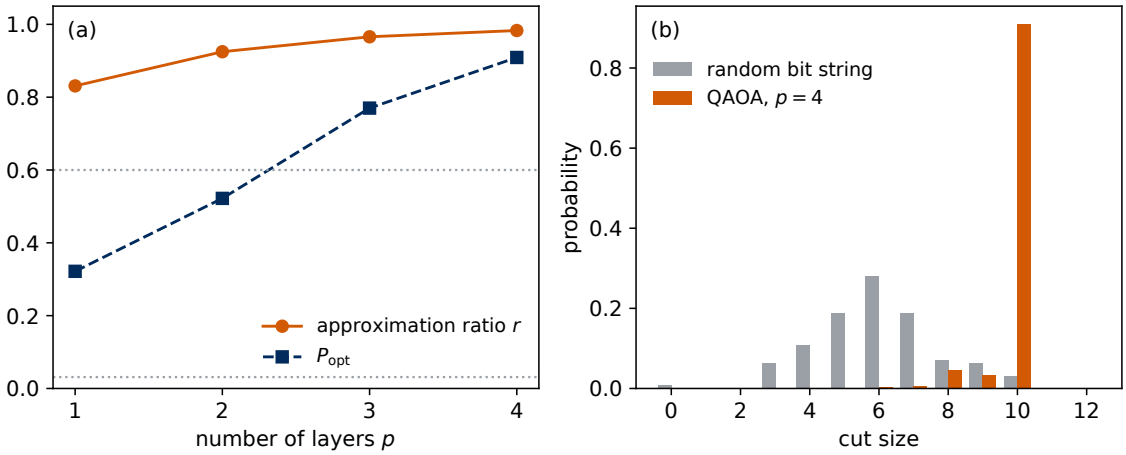


Figure 55: (a) Approximation ratio r (circles) and probability P_{opt} of an optimal bit string (squares) of the trained QAOA states against the number of layers p, with the random-guessing values 0.6 and 1/32 dotted. (b) Exact probability of each cut size for a uniformly random bit string (grey) and for the trained $p = 4$ state (green).

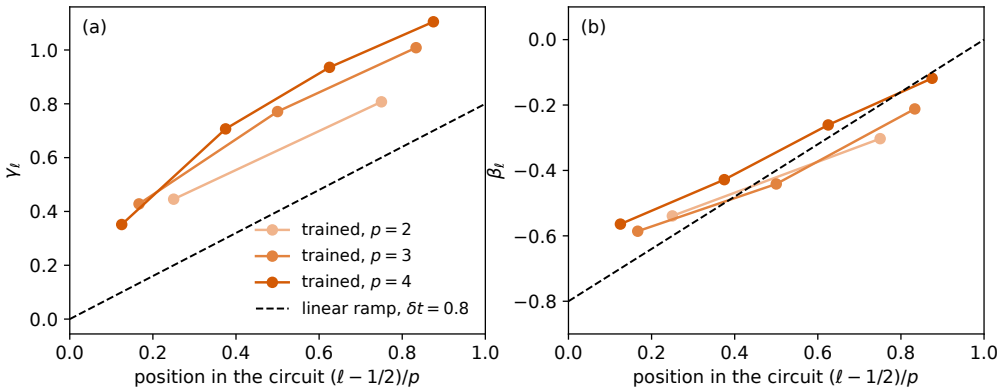


Figure 56: Trained QAOA angles for MaxCut, (a) γ_ℓ and (b) β_ℓ, against the position $(\ell - 1/2)/p$ of the layer in the circuit for $p = 2$, 3 and 4, compared with the linear annealing ramp (9.21) (black). The cost angles grow and the mixing angles shrink in magnitude along the circuit.

from the same records with the stated statistical error. An optimal bit string appears in QAOA shot 2, whereas the first optimal random bit string is shot 26. Both sources find the optimum within the 1000 shots, as expected for eight qubits. For applications the relevant quantity is the growth of the required number of shots with N, which is exponential for random guessing and depends for the QAOA on the problem class and on how the depth must grow with N.

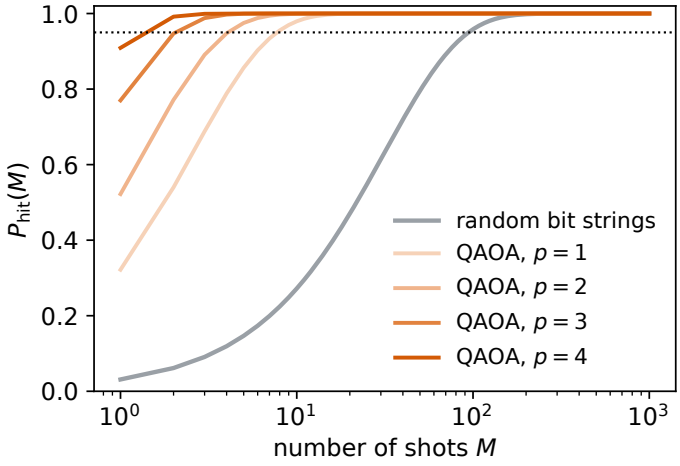


Figure 57: Probability $P_{\text{hit}}(M) = 1-(1-P_{\text{opt}})^M$ of measuring at least one optimal bit string in M shots (9.9), for random guessing and for the trained QAOA states, MaxCut on the graph of Figure 52. The dotted line marks 95 per cent.

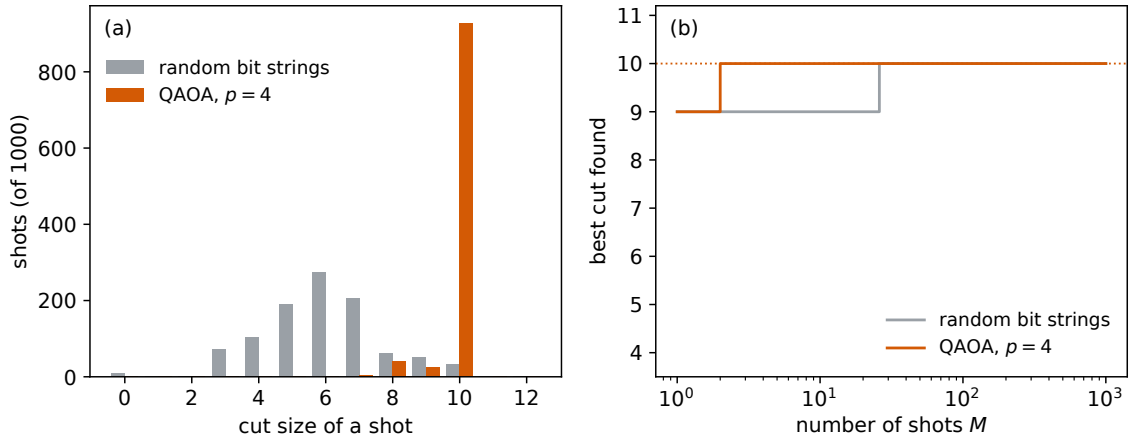


Figure 58: Sampling the trained $p = 4$ QAOA state. (a) Histogram of the cut sizes of 1000 measured bit strings for the QAOA state and for random bit strings. (b) Best cut among the first M shots, with the maximum cut dotted. Simulated measurement records.

9.7 Number partitioning

A list of positive integers $a_0, \ldots, a_{N-1}$ is to be divided into two groups whose sums are equal, or as close to equal as possible. Two movers sharing boxes of different weights, or two processors sharing jobs of different durations, face this problem. For the numbers $3, 1, 1, 2, 2, 1$ with total 10, the groups $\{3, 2\}$ and $\{1, 1, 2, 1\}$ both sum to 5, a perfect partition (Figure 59). Number partitioning is NP-complete in its decision form [49].

We label the group of number i by the bit b_i and its spin $z_i = 1 - 2b_i$. The difference between the two group sums is the residual $\sum_i a_i z_i$, and the cost is its square,

$$C(\mathbf{b}) = \Big(\sum_{i=0}^{N-1} a_i z_i \Big)^2 = \sum_i a_i^2 + 2 \sum_{i<j} a_i a_j \, z_i z_j, \tag{9.14}$$

where $z_i^2 = 1$ turns the diagonal terms into a constant. The cost Hamiltonian is therefore

$$\hat{\mathcal{H}}_C = \sum_{i<j} J_{ij} \, \hat{Z}_i \hat{Z}_j + \Big(\sum_i a_i^2 \Big) \hat{I}, \qquad J_{ij} = 2 a_i a_j, \tag{9.15}$$

without fields. In the binary variables, with the total $A = \sum_i a_i$, the residual is $A - 2\sum_i a_i x_i$, and

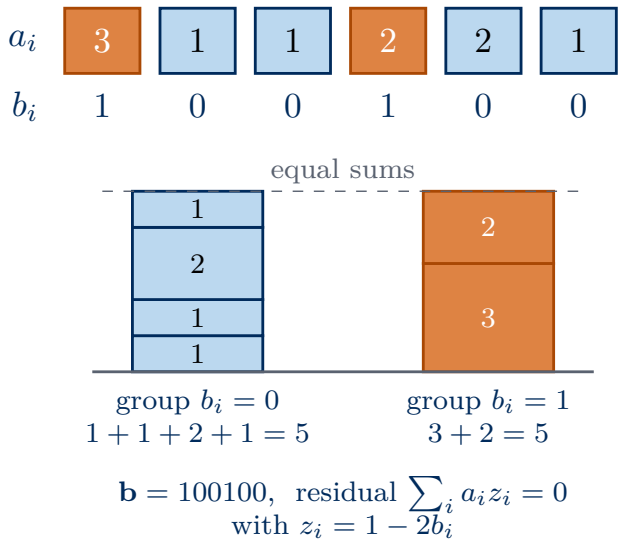


Figure 59: Number partitioning. The numbers $3, 1, 1, 2, 2, 1$ are split into two groups of equal sum 5, and the bit b_i labels the group of number i. The cost is the square of the difference of the group sums.

squaring it with $x_i^2 = x_i$ gives a QUBO plus a constant,

$$C = A^2 + \mathbf{x}^T Q\, \mathbf{x}, \qquad Q_{ij} = 4a_i a_j \ (i \neq j), \qquad Q_{ii} = 4a_i^2 - 4Aa_i, \tag{9.16}$$

and (9.5) applied to this matrix returns the couplings (9.15) and vanishing fields. Every pair of numbers is coupled, so the interaction graph is complete, with couplings of different strengths, unlike the sparse equal couplings of MaxCut. A perfect partition has the cost zero, so the approximation ratio is meaningless, and we judge a state by the probability P_{opt} of a perfect partition. Perfect partitions are often highly degenerate. For the instance $2, 1, 2, 1, 2, 1, 2, 1$ with total 12, a group of sum 6 contains two 2's and two 1's ($6 \times 6 = 36$ choices), three 2's (4 choices) or one 2 and four 1's (4 choices), so 44 of the 256 bit strings are perfect partitions and random guessing finds one with probability 0.172.

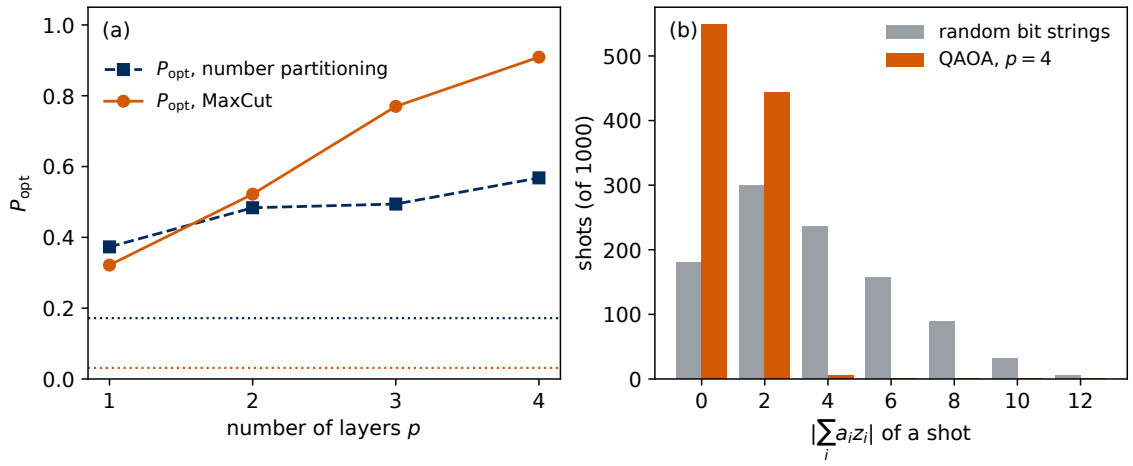


Figure 60: QAOA for number partitioning of $2, 1, 2, 1, 2, 1, 2, 1$. (a) Probability P_{opt} of a perfect partition against the number of layers, with the MaxCut values for comparison and the random-guessing probabilities of the two problems dotted. (b) Residuals $|\sum_i a_i z_i|$ of 1000 shots of the trained $p = 4$ state and of 1000 random bit strings. Exact probabilities in (a), simulated measurement records in (b).

Training the QAOA as for MaxCut raises the probability of a perfect partition from 0.172 to 0.568 at four layers (Figure 60). The gain over random guessing is much smaller than for MaxCut, because random guessing already succeeds often with 44 perfect bit strings, and beyond $p = 1$ the probability grows slowly with depth. The costs of this instance range up to 144, so the phases $\gamma C(\mathbf{b})$ of the cost

layer vary rapidly with γ, and the depth-one landscape changes on a scale of about 0.1 in γ with several separated low-cost regions. The trained state removes all residuals above 4 and puts almost all the weight that misses a perfect partition on the residual 2, the nearest imperfect split.

9.8 Maximum independent set

An independent set of a graph is a set of vertices of which no two are joined by an edge. In radio planning, for example, an edge joins two transmitter sites close enough to interfere, and the task is to switch on as many transmitters as possible without switching on two neighbours. The maximum independent set (MIS) problem asks for the largest such set. On a ring of six vertices the sets $\{0, 2, 4\}$ and $\{1, 3, 5\}$, the bit strings 101010 and 010101, are independent sets of the largest size three (Figure 61). On a ring of five vertices any three chosen vertices include two neighbours, so the maximum is two, because the odd cycle prevents alternation. A set is independent precisely when it is a clique, a set of mutually joined vertices, of the complement graph, and the clique problem is NP-complete [49], so the decision form of MIS is NP-complete too.

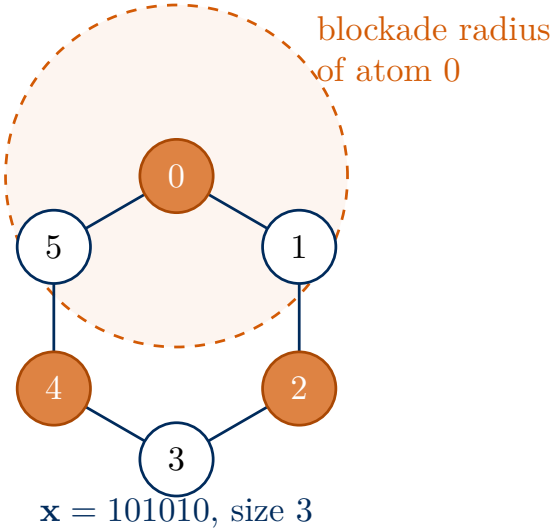


Figure 61: Maximum independent set on a ring of six vertices, 101010 of size three. The filled vertices form an independent set, no two of them joined by an edge, and the bit string marks the chosen vertices with ones.

A QUBO has no constraints, so we add the independence condition to the cost as a penalty,

$$C(\mathbf{x}) = -\sum_i x_i + B \sum_{(i,j)\in E} x_i x_j, \tag{9.17}$$

where the first term counts the chosen vertices and the second counts the violated edges with the penalty weight $B > 0$. For $B > 1$ every minimiser of (9.17) is an independent set. Removing one endpoint of a violated edge raises the first term by one and lowers the penalty by at least B, so it changes the cost by at most $1 - B < 0$, and a string with a violated edge cannot be optimal. On independent sets the penalty vanishes and the cost is minus the size, so the minimisers are the maximum independent sets. The QUBO matrix has $Q_{ii} = -1$ and $Q_{ij} = Q_{ji} = B/2$ on the edges, and (9.5) gives

$$J_{ij} = \frac{B}{4} \quad \text{on the edges,} \qquad h_i = \frac{2 - B d_i}{4}. \tag{9.18}$$

The fields are nonzero, because the cost favours $x_i = 1$ over $x_i = 0$ and is not invariant under flipping all bits.

Arrays of neutral atoms excited to Rydberg states realise the constraint without a penalty. Two atoms closer than the blockade radius cannot both be excited (Figure 62), because their strong interaction shifts the doubly excited state out of resonance with the laser. If the atoms sit at the vertices of a graph whose edges join atoms within the blockade radius, the laser drive reaches only

independent sets, with the excited atoms as the chosen vertices. Ref. [52] proposed to find maximum independent sets this way, and Ref. [53] carried it out with two-dimensional arrays of up to 289 atoms.

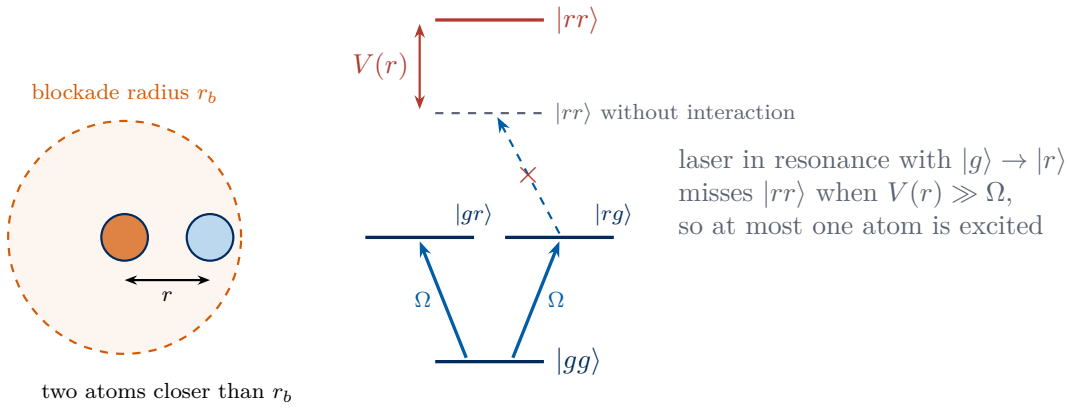


Figure 62: The Rydberg blockade. Two atoms closer than the blockade radius cannot both be excited, so atoms at the vertices of a graph whose edges join atoms within that radius reach only independent sets, with the excited atoms as the chosen vertices.

As an example we take the 2×4 grid, two rows of four vertices with edges between nearest neighbours, the blockade graph of eight atoms in two rows, with $B = 2$. Only 41 of the 256 bit strings are independent sets, including the empty set, and only two, the checkerboard patterns of size four, are maximum. Random guessing therefore rarely produces a valid answer, and the trained QAOA raises the probabilities of an independent set and of a maximum one by large factors (Figure 63), the latter from 2/256 to 0.456 at $p = 4$. Part of the weight remains on strings that violate the constraint, because the penalty enters only as an energy, and the training rewards large sets as well as feasibility. A measured bit string that violates the constraint is recognised classically at once and is discarded or repaired by removing one endpoint of each violated edge. Most shots of the trained $p = 4$ state are independent sets, and the largest group of them are maximum ones.

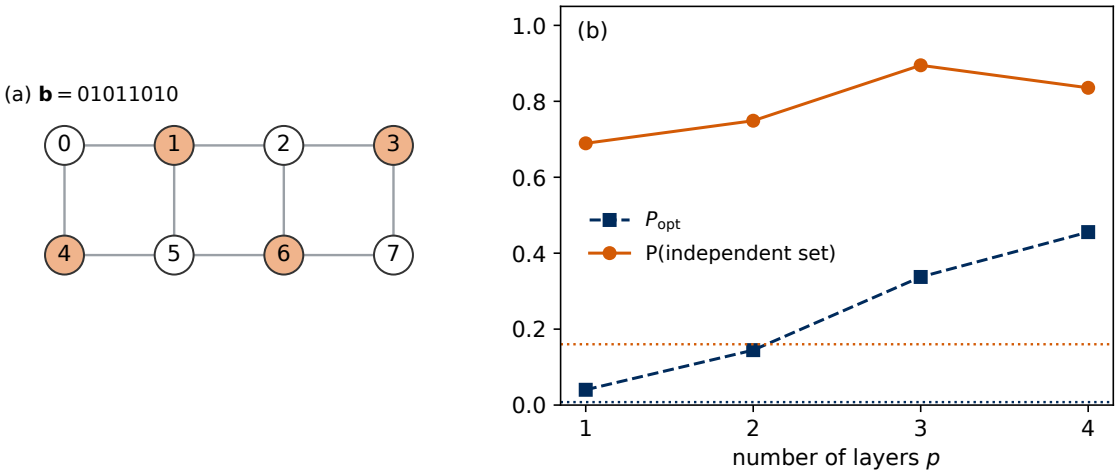


Figure 63: QAOA for the maximum independent set. (a) The 2×4 grid with one maximum independent set, the four orange vertices of a checkerboard pattern. (b) Probability P_{opt} of a maximum independent set (squares) and probability that a measured bit string is an independent set (circles) against the number of layers, with the random-guessing values dotted. Exact probabilities of the trained states.

9.9 Other problems in QUBO form

Many combinatorial problems take the QUBO form after their constraints are written as penalties, as for the independent set. Ref. [54] collects Ising formulations of many NP-complete problems, and

Table 4 lists a few common ones. In each penalty the weight B must exceed the largest gain that a violation can bring. Graph colouring and the travelling salesman need Nk and m^2 qubits, so the encoding of the constraints sets the size of the register.

problem	binary variables	cost to minimise
minimum vertex cover	$x_i = 1$ if vertex i is in the cover	$\sum_i x_i + B\sum_{(i,j)\in E}(1-x_i)(1-x_j)$
graph colouring with k colours	$x_{v,c} = 1$ if vertex v has colour c	$B\sum_v \big(1-\sum_c x_{v,c}\big)^2 + B\sum_{(u,v)\in E}\sum_c x_{u,c}x_{v,c}$
travelling salesman, m cities	$x_{c,t} = 1$ if city c is visited at step t	$\sum_{c,c'} d_{cc'}\sum_t x_{c,t}x_{c',t+1}$ plus penalties for one city per step and one step per city
knapsack with exact capacity W	$x_i = 1$ if item i is packed	$-\sum_i v_i x_i + B\big(\sum_i w_i x_i - W\big)^2$
portfolio selection of K assets	$x_i = 1$ if asset i is bought	$-\sum_i \mu_i x_i + q\sum_{i,j}\Sigma_{ij}x_i x_j + B\big(\sum_i x_i - K\big)^2$
Sherrington–Kirkpatrick spin glass	spins $z_i = \pm 1$	$\sum_{i<j} J_{ij} z_i z_j$ with random couplings J_{ij} [55]

Table 4: Further problems in QUBO or Ising form. Here $d_{cc'}$ are distances between cities, v_i and w_i the values and weights of items, μ_i the expected returns of assets, Σ_{ij} their covariance matrix, q a weight of the risk and B a penalty weight. A knapsack whose capacity is only an upper bound needs additional binary slack variables.

9.10 Relation to adiabatic evolution and annealing

The structure of the QAOA follows from adiabatic evolution under the interpolating Hamiltonian

$$\hat{\mathcal{H}}(s) = -(1-s)\sum_i \hat{X}_i + s\,\hat{\mathcal{H}}_C, \qquad 0 \le s \le 1, \tag{9.19}$$

whose ground state at $s = 0$ is $|+\rangle^{\otimes N}$ and at $s = 1$ the optimal bit string. If s is increased slowly enough from 0 to 1, the adiabatic theorem keeps the state in the instantaneous ground state, and the evolution ends in the optimum [56]. Quantum annealing follows the same schedule on dedicated hardware [57]. Dividing the evolution into p steps of duration δt at the values $s_\ell = (\ell - 1/2)/p$ and splitting each step into its two groups of mutually commuting terms gives

$$e^{-i\delta t\hat{\mathcal{H}}(s_\ell)} \approx e^{+i(1-s_\ell)\delta t\sum_i \hat{X}_i}\, e^{-is_\ell\delta t\hat{\mathcal{H}}_C}, \tag{9.20}$$

with an error of order δt^2. The product is a QAOA layer (9.10) with

$$\gamma_\ell = s_\ell\,\delta t, \qquad \beta_\ell = -(1-s_\ell)\,\delta t, \tag{9.21}$$

the linear ramp used as one of the starting points of the training. As $p \to \infty$ the QAOA therefore contains the discretised adiabatic evolution, which reaches the optimum [51]. The trained angles at small p show the same pattern, with γ_ℓ growing and $|\beta_\ell|$ shrinking along the circuit. At small depth the trained circuit need not follow the instantaneous ground state, since its angles are optimised only for the final expected cost.

10 From state ansätze to learning models

Parametrised circuits entered physics as ansätze for quantum states. In VQE and in the QAOA the circuit encodes a single state, the cost is the expectation value of a fixed Hamiltonian, and the trained angles describe that one state. The Born machine already used the circuit differently, since its cost

depended on a data set and its output was a distribution of samples. The same circuits were soon read as general machine-learning models.

In this reading (Figure 64) a data point $\mathbf{x}$ enters the circuit through its gate angles, for example as rotations $\hat{R}_y(x_j)$ on the qubits, and the trainable angles $\boldsymbol{\theta}$ act before, after or between these encoding gates. The prepared state $|\psi(\mathbf{x},\boldsymbol{\theta})\rangle$ then depends on the input as well as on the parameters. A loss summed over a data set, such as the squared difference between the measured output and a label for every training example, replaces the energy of one Hamiltonian as the cost. The output is an expectation value $\langle\hat{O}\rangle_{\mathbf{x},\boldsymbol{\theta}}$ read as a prediction, or a sample read as a generated data point. The gradients are obtained as in VQE, by automatic differentiation in simulation and by the parameter-shift rule on hardware, and barren plateaus limit deep and expressive circuits in the same way.

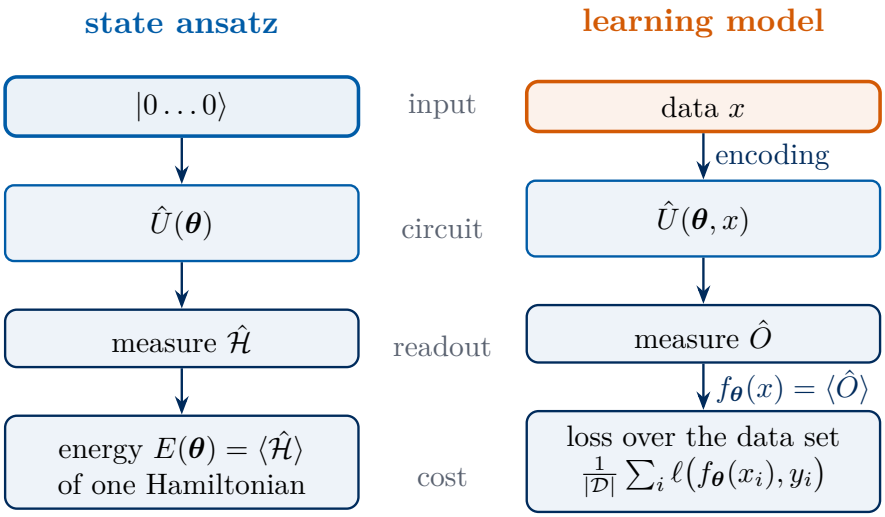


Figure 64: From state ansätze to learning models. In VQE and the QAOA the circuit prepares one state and the cost is the energy of a fixed Hamiltonian. As a learning model the circuit takes data through encoding gates, the cost is a loss over a data set, and the output is a prediction or a sample.

This reading was developed within a few years. Ref. [18] introduced quantum circuit learning, in which encoded inputs and trainable rotations fit functions and classify data, together with the parameter-shift gradient. Circuits as classifiers of labelled bit strings were proposed in Ref. [58], and low-depth classifier circuits acting on inputs encoded in the amplitudes of the register in Ref. [59]. The encoding was then recognised as a feature map into the Hilbert space of the register, whose inner products define a kernel for classical kernel methods [60, 61]. Finally, encoding the same data several times between trainable layers, called data re-uploading, makes even a single qubit a universal classifier [62]. In each case the ansatz is a family of functions of the data, and its quality is judged by its predictions on data that were not used in training.

11 Summary of the lecture

1 The many-body problem and its classical approximations

A state of N spins needs 2^N amplitudes, and exact methods are limited to a few dozen spins. Every trial state bounds the ground-state energy from above (1.4),

$$\langle\psi|\hat{\mathcal{H}}|\psi\rangle = \sum_m |c_m|^2 E_m \geq E_0,$$

and the Rayleigh–Ritz method minimises the energy over an ansatz. Mean field takes a product ansatz. For bosons in one mode the energy functional gives the Gross–Pitaevskii equation (1.14),

$$\Big[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) + g(N-1)|\phi(x)|^2\Big]\phi = \mu\,\phi,$$

a Hartree mean field derived with the field operator $\hat{\psi}(x) = \sum_i \phi_i(x)\hat{a}_i$ and $[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}$. For the TFIM (1.2) the site-factorised state $|\phi(\theta)\rangle^{\otimes N}$ has $E_{\mathrm{MF}} = -J(N-1)\cos^2\theta - hN\sin\theta$ (1.17), whose minimum obeys the self-consistent Weiss equation (1.19) and places the transition at $h_c^{\mathrm{MF}} = 2J(N-1)/N \to 2J$, twice the exact $h_c = J$. Product states carry no entanglement. Matrix product states bound the entanglement by $\log_2 \chi$ and describe area-law states, and Monte Carlo methods face the sign problem.

2 Parametrised circuits as quantum ansätze

A circuit of rotations prepares the trial state on the register (2.1),

$$|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|0\ldots0\rangle, \qquad \hat{R}_P(\theta) = e^{-i\theta\hat{P}/2} = \cos\tfrac{\theta}{2}\,\hat{I} - i\sin\tfrac{\theta}{2}\,\hat{P},$$

with angles that are pulse areas $\theta = \Omega t$ or interaction times $\theta = Jt$. The device prepares and measures, and a classical computer updates the angles (the VQE loop). The energy error bounds the infidelity through the gap (2.3), $E - E_0 \geq \Delta E\,(1-F)$.

3 The energy from bit strings

With $\hat{\mathcal{H}} = \sum_j c_j \hat{P}_j$, a Pauli string takes the value $(-1)^{\sum_{q\in S} b_q}$ after the rotation of its factors into $\hat{Z}$ by $\hat{H}$ or $\hat{H}\hat{S}^\dagger$. Strings sharing a basis on every qubit form one setting, two for the TFIM and three for the XXZ chain. The sample-mean estimate (3.5) has the standard error (3.7)

$$\sigma_{\tilde{E}} = \sqrt{\frac{1}{M}\sum_g \sigma_g^2},$$

so each tenfold reduction of the error costs a hundred times more shots.

4 Gradients

Along one angle the energy is a shifted cosine (4.4), $E(\theta_k) = a + b\cos(\theta_k - c)$, because every generator squares to the identity. Hence the exact parameter-shift rule (4.7) and, for an angle shared by gates with $\theta_g = c_g\varphi$, the chain rule (4.8),

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\boldsymbol{\theta} + \tfrac{\pi}{2}\mathbf{e}_k) - E(\boldsymbol{\theta} - \tfrac{\pi}{2}\mathbf{e}_k)}{2}, \qquad \frac{\partial E}{\partial \varphi} = \sum_g c_g \frac{\partial E}{\partial \theta_g}.$$

On a device the shift is a change of pulse length by $\pm\pi/(2\Omega)$, and one gradient costs $2n_{\mathrm{rot}}GM$ shots (4.9). In simulation reverse-mode automatic differentiation gives the same gradient at the cost of a few energy evaluations. Finite differences amplify shot noise by $1/\epsilon$, and SPSA uses two energies per step.

5 Circuit ansätze

An ansatz should be expressive, trainable, native to the hardware and symmetric like the Hamiltonian. The HEA (5.3) alternates $\hat{R}_z\hat{R}_y$ blocks with $\widehat{\mathrm{CZ}}$ chains and has $n = 2N(L+1)$ angles. The HVA (5.5) alternates evolutions under the terms of the TFIM from $|+\rangle^{\otimes N}$, has $n = 2L$ shared angles and conserves the parity $\hat{\Pi} = \prod_i \hat{X}_i$. The SPA applies the exchange gate (5.7) to a singlet product and conserves $\hat{Z}_{\mathrm{tot}}$. A brickwork of L layers with one gate of operator Schmidt rank two per layer across a cut obeys $S_{\mathrm{vN}} \leq \min(L, N/2)$ bits (5.10).

6 The variational quantum eigensolver for spin chains

The energy error is of second order in a small admixture ϵ of an excited state (6.2), whereas the entanglement entropy changes at first order (6.3), so the entropy error scales as $\sqrt{1-F}$ and exposes errors that the energy hides. In our runs (four restarts, 600 Adam steps) the structured ansätze reached better states with far fewer angles than the HEA at equal two-qubit gate count (Table 2). In the ordered TFIM a state within the ground doublet has (6.5)

$$E - E_0 = \frac{1 - \langle \hat{\Pi} \rangle}{2}\, \Delta E,$$

so the symmetry-broken branch reached by the HEA has $F = 1/2$, no entanglement and an energy error of only $\Delta E/2$, while the parity-conserving HVA reaches the cat state. Variational quantum deflation (6.6) adds the penalty $\lambda |\langle \tilde{\psi}_0 | \psi(\boldsymbol{\theta}) \rangle|^2$ with $\lambda > \Delta E$ and gives the first excited state. Across the phase diagrams of the TFIM and of the XXZ chain in a transverse field, small energy errors coexist with wrong states wherever an ansatz breaks a symmetry or misses entanglement, and the fidelity and the entropy reveal it.

7 Barren plateaus

The mean gradient over random angles vanishes and its variance measures the typical gradient. Behind a Haar-random unitary (7.3)

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \frac{\mathrm{Tr}(\hat{O}_0^2)}{2(d^2-1)}, \qquad d = 2^N,$$

which decays as 4^{-N} for the global cost and as 2^{-N} for sums of local Pauli strings. A local cost at fixed depth keeps a polynomial decay, because the derivative depends only on the gates in the light cone of the differentiated gate, and the HVA keeps large gradients because its gates generate a small set of unitaries. Resolving a gradient needs $M^\star = \sigma^2 / (2\,\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E])$ shots (7.6), about 2^N for a 2-design. Depolarising noise flattens the landscape exponentially in the depth.

8 Circuits as probability models

A continuous value x is mapped to the bin $k = \lfloor (x - x_{\min})/w \rfloor$ of width $w = (x_{\max} - x_{\min})/2^N$, and k in binary is the bit string, with b_0 the most significant bit. The Born machine (8.8) models $q_{\boldsymbol{\theta}}(\mathbf{b}) = |\langle \mathbf{b} | \hat{V}(\boldsymbol{\theta}) | 0 \rangle^{\otimes N}|^2$, and each measurement draws one sample. Its loss is the KL divergence to the data, equal to the negative log-likelihood up to the entropy of the data (8.12),

$$D_{\mathrm{KL}}(\tilde{p} \,\|\, q_{\boldsymbol{\theta}}) = -\frac{1}{|\mathcal{D}|} \sum_i \ln q_{\boldsymbol{\theta}}(\mathbf{b}^{(i)}) - H(\tilde{p}).$$

A product circuit gives independent bits (8.15) and at best the product of the marginals. Entangling layers create correlations, and deeper circuits eventually fit the fluctuations of a finite data set. Sampled models are judged by the total variation distance, and ancilla qubits turn the model into a mixture of Born distributions.

9 Combinatorial optimisation with QUBO and QAOA

A QUBO $C = \mathbf{x}^T Q \mathbf{x}$ becomes, with $x_i = (1 - z_i)/2$, the diagonal Ising Hamiltonian (9.6) with

$$J_{ij} = \frac{Q_{ij} + Q_{ji}}{4}, \qquad h_i = -\frac{1}{4} \sum_j (Q_{ij} + Q_{ji}),$$

whose basis states are the candidates and whose ground state is the optimum. MaxCut gives $\hat{\mathcal{H}}_C = \frac{1}{2} \sum_{(i,j) \in E} \hat{Z}_i \hat{Z}_j - |E|/2$, number partitioning all-to-all couplings $J_{ij} = 2 a_i a_j$, and the maximum

independent set a penalty $B > 1$ with fields $h_i = (2 - Bd_i)/4$. The QAOA (9.11) alternates $e^{-i\gamma_\ell \hat{\mathcal{H}}_C}$ and $e^{-i\beta_\ell \sum_i \hat{X}_i}$ from $|+\rangle^{\otimes N}$. The answer is the best sampled bit string, found with probability $1 - (1 - P_{\text{opt}})^M$ after M shots, and the expected cost from the same shots trains the angles. The trained angles resemble the Trotterised adiabatic schedule $\gamma_\ell = s_\ell \delta t$, $\beta_\ell = -(1 - s_\ell)\delta t$ (9.21).

10 From state ansätze to learning models

When data enter as gate angles and the cost becomes a loss over a data set, the parametrised circuit becomes a model for regression, classification, kernels and generation, with the same gradients and the same barren plateaus as the variational ansätze.

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